

Plug Puller Dial Reference

Every dial of the one-sided puller and the two-sided puller on its own page: what it moves, in a picture and a sentence, with the numbers the Customizer allows.

Model version 0.12.0

In every picture: black = the tool at its defaults; teal = the plug you measured; red dashed = what this dial moves.

The dial names are written exactly as the Customizer shows them. Use the bookmarks or the contents page to jump to a dial.

Contents

One-sided puller

Step 1 - Your Plug

- `plug_preset`: Plug preset
- `measure_plug_length`: Plug length
- `measure_plug_width_prong_end`: Plug width at the prong end
- `measure_plug_width_cord_end`: Plug width at the cord end
- `measure_plug_thickness_prong_end`: Plug thickness at the prong end
- `measure_plug_thickness_cord_end`: Plug thickness at the cord end
- `measure_cord_thickness`: Cord thickness
- `measure_wall_plate_style`: Outlet cover plate style
- `show_plug_preview`: Show the see-through plug

Step 2 - Size

- `size`: Hand size
- `measure_finger_width`: Finger width
- `measure_hand_width`: Hand width

Step 3 - Attachment

- `attachment`: Attachment
- `velcro_style`: Strap opening style
- `strap_width`: Strap width

Step 4 - Cord Hook

- `hook_hand`: Hook side

Advanced - Zip Tie Placement

- `zip_placement`: Zip-tie hole placement
- `zip_row_count`: Number of zip-tie rows
- `zip_pos_1`: Zip-tie pair 1 position
- `zip_pos_2`: Zip-tie pair 2 position
- `zip_pos_3`: Zip-tie pair 3 position
- `zip_edge_offset`: Zip-tie hole distance from the pocket wall

Advanced - Velcro Placement

- `velcro_placement`: Strap slot placement
- `velcro_pos`: Strap slot position

Advanced - Render Quality

- `quality`: Render quality

Custom Mode

- `reset_custom_to_medium`: Reset Custom to the Medium reference
- `custom_enable_auto_fit`: Auto-fit

Body Shape (Custom size only)

- `custom_puller_length`: Body length
- `custom_puller_bottom_width`: Body width at the widest point
- `custom_puller_bottom_corners`: Cord-end corner width
- `custom_puller_top_width`: Body width at the plug end
- `custom_puller_middle_width`: Body width at the middle
- `custom_puller_side_corner`: Side corner position
- `custom_body_thickness`: Body thickness
- `custom_body_round_bottom_only`: Round the cord half only

Plug Pocket (Custom size only)

- `custom_pocket_seat_diameter`: Pocket seat diameter
- `custom_pocket_width`: Pocket width
- `custom_pocket_depth`: Pocket depth
- `custom_pocket_dome_drop`: Pocket reference drop
- `custom_pocket_seat_floor`: Pocket seat floor thickness
- `custom_pocket_floor`: Plug recess floor thickness
- `custom_pocket_side_angle`: Pocket side taper

Finger Holes (Custom size only)

- `custom_enable_finger_holes`: Finger holes on or off
- `custom_finger_hole_diameter`: Finger hole diameter
- `custom_finger_hole_spacing`: Finger hole spacing
- `custom_finger_hole_y_position`: Finger hole position

T Hook (Custom size only)

- `custom_enable_t_hook`: Cord hook on or off
- `custom_t_hook_base_gap`: Hook slot width
- `custom_t_hook_length`: Hook length
- `custom_t_hook_holder_width`: Hook crossbar width
- `custom_t_hook_holder_length`: Hook crossbar height
- `custom_t_hook_gap_offset`: Hook slot inset
- `custom_t_hook_leg_offset`: Hook stem side offset
- `custom_t_hook_stem_offset`: Hook stem offset
- `custom_t_hook_catch_reach`: Hook catch reach
- `custom_t_hook_tip_drop`: Hook tip drop

Plug Wall Notch (Custom size only)

- `custom_enable_plug_wall_notch`: Wall notch on or off
- `custom_plug_wall_notch_width`: Wall notch width
- `custom_plug_wall_notch_height`: Wall notch depth
- `custom_plug_wall_notch_rounding`: Wall notch corner rounding

Zip Tie Holes (Custom size only)

- `custom_zip_tie_hole_diameter`: Zip-tie hole diameter
- `custom_zip_tie_height_spacing`: Zip-tie row spacing
- `custom_zip_tie_width_spacing`: Zip-tie column spacing
- `custom_zip_tie_distance_from_notch`: Zip-tie distance from the notch
- `custom_zip_tie_countersink`: Zip-tie countersink

Velcro / Wing Strap Holes (Custom size only)

- `custom_velcro_hole_length`: Strap slot length
- `custom_velcro_hole_width`: Strap slot width
- `custom_velcro_hole_x_center`: Strap slot distance from the centerline
- `custom_velcro_hole_y_center`: Strap slot position along the body
- `custom_velcro_hole_rotation`: Strap slot lean

Edge Rounding (Custom size only)

- `custom_body_side_rounding`: Body side rounding
- `custom_body_top_rounding`: Top edge rounding
- `custom_body_bottom_rounding`: Bottom edge rounding
- `custom_velcro_side_rounding`: Strap slot corner rounding
- `custom_velcro_top_bottom_rounding`: Strap opening edge rounding
- `custom_finger_hole_rounding`: Finger hole rim rounding
- `custom_t_hook_holder_side_rounding`: Hook crossbar corner rounding
- `custom_t_hook_gap_side_rounding`: Hook slot corner rounding
- `custom_t_hook_top_bottom_rounding`: Hook edge rounding

Two-sided puller

Step 1 - Your Plug

- `plug_preset`: Plug preset
- `measure_plug_length`: Plug length
- `measure_plug_width_prong_end`: Plug width at the prong end
- `measure_plug_width_cord_end`: Plug width at the cord end
- `measure_cord_thickness`: Cord thickness
- `plug_sides`: Plug sides
- `show_plug_preview`: Show the see-through plug

Step 2 - Size

- `size`: Hand size
- `measure_finger_width`: Finger width

Step 3 - Attachment

- `attachment`: Attachment
- `strap_width`: Strap width

Step 4 - Print Layout

- `print_layout`: Print layout

Advanced - Two-Sided Puller

- `plate_wall_boost`: Extra wall on the plate
- `plate_thickness`: Plate thickness
- `plate_grip_bite`: Grip bite
- `plate_cradle_depth`: Cradle depth
- `plate_grip_clearance`: Grip clearance
- `plate_cable_clearance`: Cord channel clearance
- `plate_finger_fit`: Finger hole fit
- `plate_finger_wall`: Finger lobe wall
- `plate_finger_inner_wall`: Wall beside the cord channel
- `plate_tooth_diameter`: Tooth size
- `plate_tooth_pitch`: Tooth spacing
- `plate_tooth_depth`: Tooth depth
- `plate_grip_zone_start`: Toothed zone start
- `plate_grip_zone_length`: Toothed zone length

- `plate_tip_flare`: Tip flare

- `plate_arm_tip_width`: Arm tip width
- `plate_edge_rounding`: Outer edge rounding
- `plate_strip_thickness`: Cable strip thickness
- `plate_zip_hole_diameter`: Zip-tie hole diameter
- `plate_zip_placement`: Zip-tie station placement
- `plate_zip_pos_1`: Zip-tie station 1 position
- `plate_zip_pos_2`: Zip-tie station 2 position
- `plate_zip_pos_3`: Zip-tie station 3 position
- `plate_slot_inner_wall`: Wall between the teeth and the slot
- `plate_velcro_slot_width`: Strap slot width
- `plate_velcro_slot_length`: Strap slot length

Advanced - Render Quality

- `quality`: Render quality

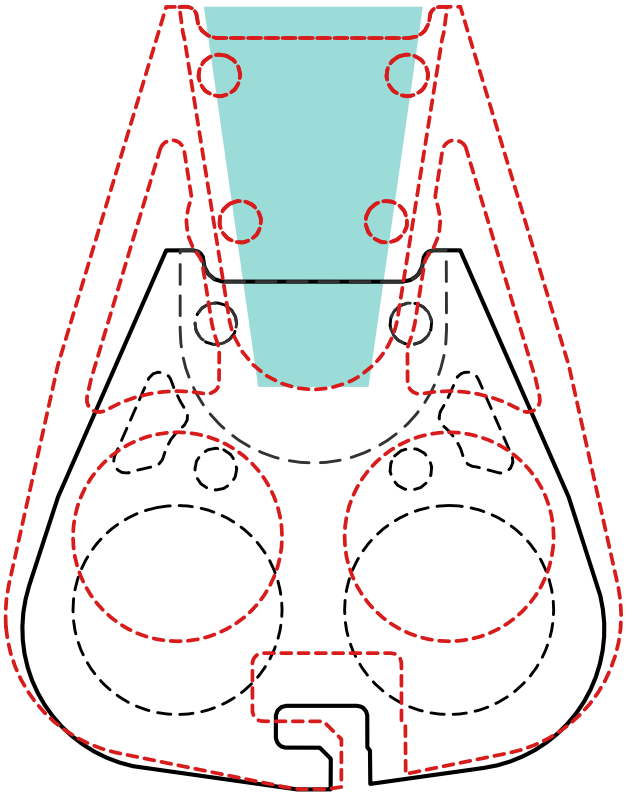
One-sided puller

plug_preset

One-sided puller · Step 1 - Your Plug

Plug preset

plug_preset
Measure my plug → Standard 3-prong plug - NEMA 5-15



black = the tool at its defaults;
teal = the plug you measured;
red dashed = what this dial moves

Fills in all six plug numbers from a measured reference plug, so the pocket, the wall notch, the cord hook and the zip-tie and wing positions all take that plug's shape at once.

Default	Range	Step	Unit
Measure my plug	5 options	—	—

Options: Measure my plug, Flat 2-prong lamp plug - NEMA 1-15, Standard 3-prong plug - NEMA 5-15, Heavy-duty extension cord - NEMA 5-15, Wide 2-prong appliance plug - NEMA 1-15.

Before: Measure my plug. After: Standard 3-prong plug - NEMA 5-15.

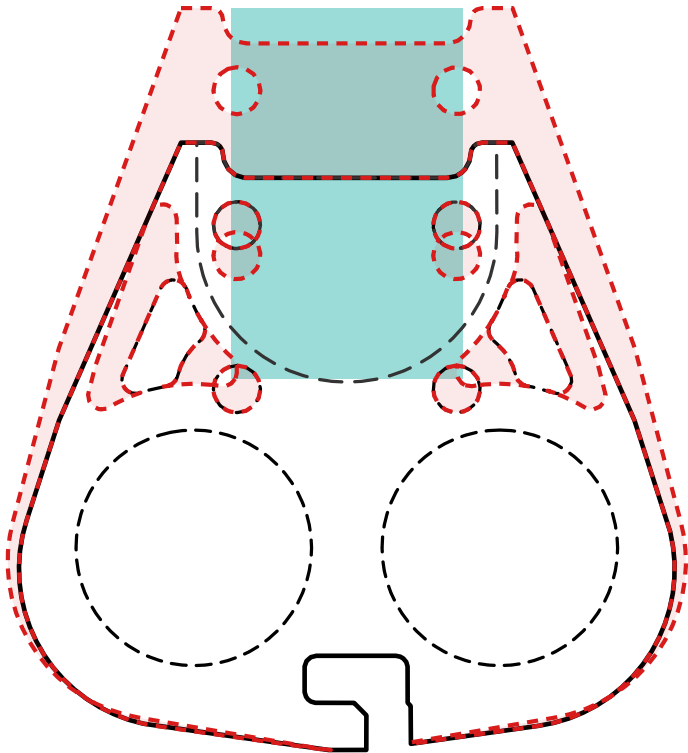
Moves: body edge, pocket, seat, wall notch, wing openings, zip-tie holes.

measure_plug_length

One-sided puller · Step 1 - Your Plug

Plug length

measure_plug_length
25.5 → 40



black = the tool at its defaults;
teal = the plug you measured;
red dashed = what this dial moves

Runs the pocket farther toward the finger holes and stretches the whole body to keep the finger holes and zip-tie rows clear of it.

Default	Range	Step	Unit
25.5	12 to 85	0.5	mm

Before: 25.5. After: 40.

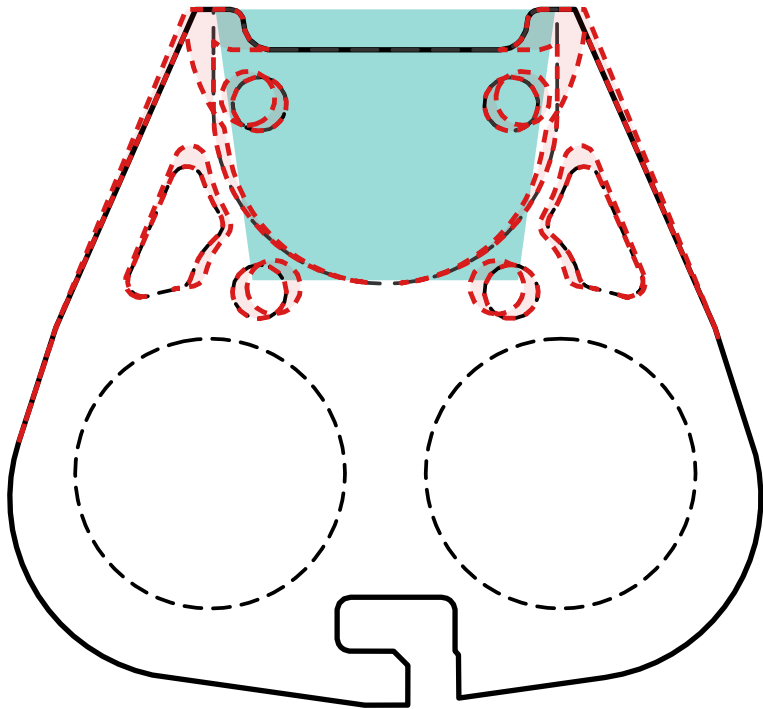
Moves: body edge, seat, wall notch, wing openings, zip-tie holes.

measure_plug_width_prong_end

One-sided puller · Step 1 - Your Plug

Plug width at the prong end

measure_plug_width_prong_end
25 → 32



black = the tool at its defaults;
teal = the plug you measured;
red dashed = what this dial moves

Widens or narrows the pocket, its seat and the wall notch at the plug end, and moves the zip-tie holes and wing openings with them.

Default	Range	Step	Unit
25	8 to 38	0.5	mm

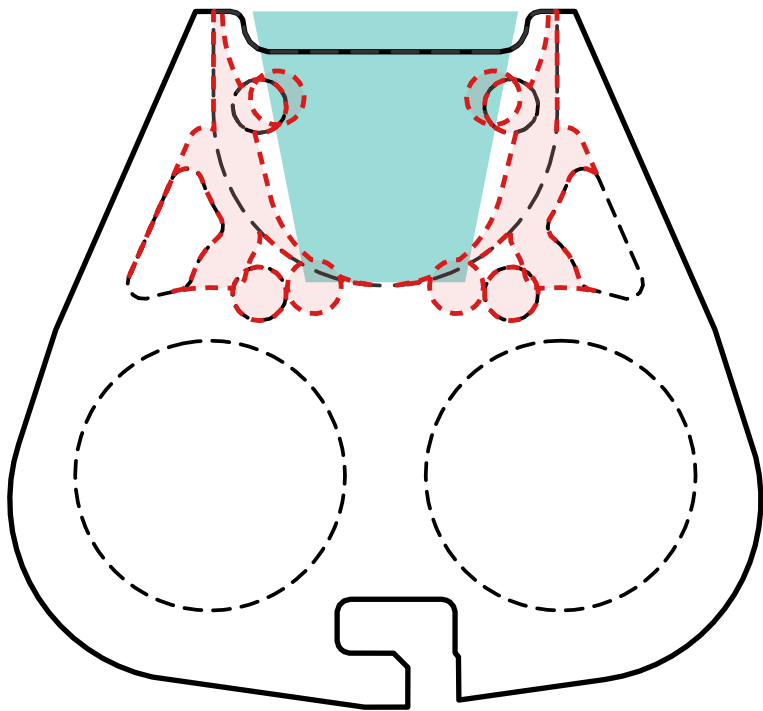
Before: 25. After: 32.

Moves: seat, wall notch, wing openings, zip-tie holes.

measure_plug_width_cord_end

One-sided puller · Step 1 - Your Plug
Plug width at the cord end

measure_plug_width_cord_end
25 → 15



black = the tool at its defaults;
teal = the plug you measured;
red dashed = what this dial moves

Tilts the pocket's side walls to match the plug's taper, and the zip-tie holes and wing openings lean in with them.

Default	Range	Step	Unit
25	8 to 38	0.5	mm

Before: 25. After: 15.

Moves: pocket, wing openings.

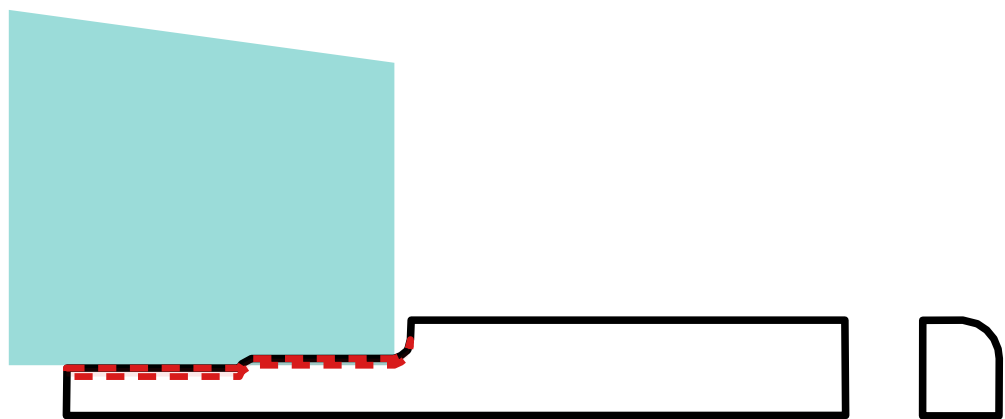
measure_plug_thickness_prong_end

One-sided puller · Step 1 - Your Plug

Plug thickness at the prong end

measure_plug_thickness_prong_end

20 → 23.5
(section-x: 0)



black = the tool at its defaults;
teal = the plug you measured;
red dashed = what this dial moves

Deepens or shallows the two pocket floors, the round seat and the plug recess, to suit the fatter of the two thickness numbers.

Default	Range	Step	Unit
20	4 to 40	0.5	mm

Before: 20. After: 23.5. Section at x = 0 mm.

The fatter of the two thickness numbers sets the floors, so lowering one alone changes nothing; 24 mm or more sends you to the two-sided puller.

Moves: pocket, seat.

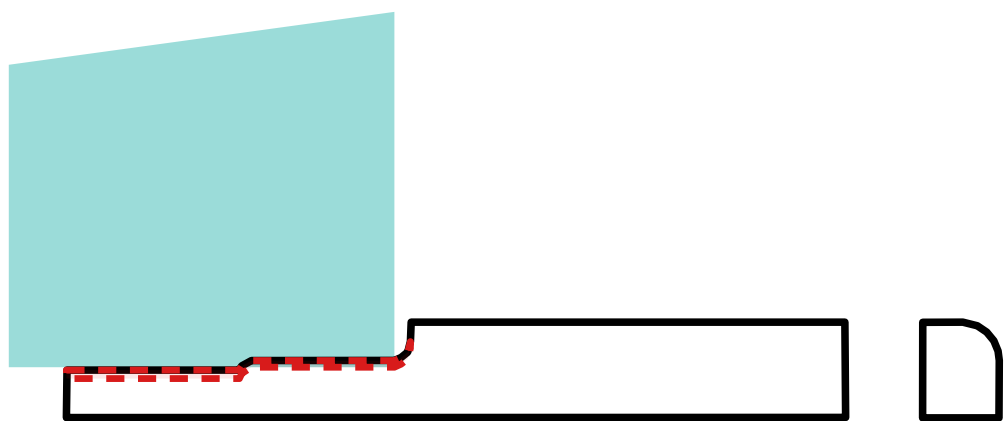
measure_plug_thickness_cord_end

One-sided puller · Step 1 - Your Plug

Plug thickness at the cord end

measure_plug_thickness_cord_end

20 → 23.5
(section-x: 0)



black = the tool at its defaults;
teal = the plug you measured;
red dashed = what this dial moves

Deepens or shallows the two pocket floors, the round seat and the plug recess, when this end is the fatter end of the plug.

Default	Range	Step	Unit
20	4 to 40	0.5	mm

Before: 20. After: 23.5. Section at x = 0 mm.

The fatter of the two thickness numbers sets the floors, so lowering one alone changes nothing; 24 mm or more sends you to the two-sided puller.

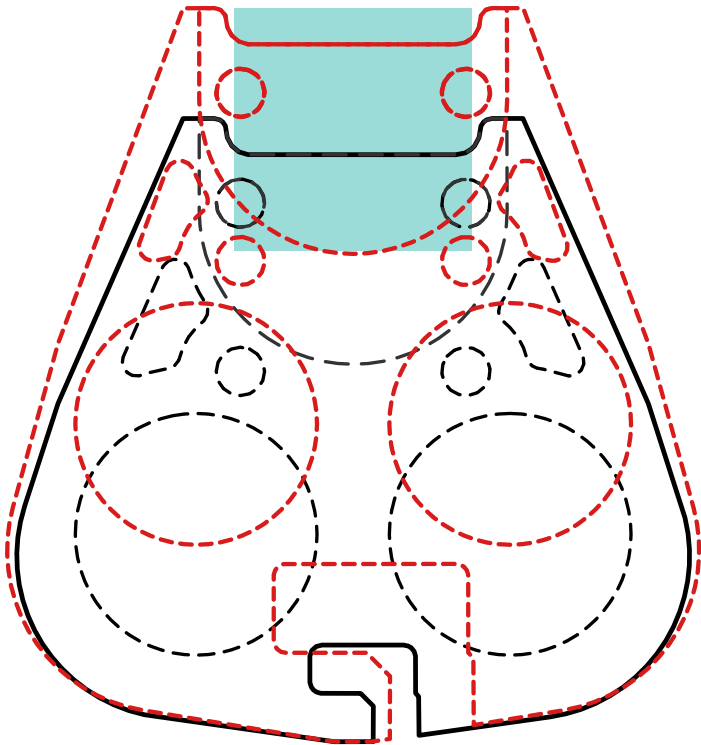
Moves: pocket, seat.

measure_cord_thickness

One-sided puller · Step 1 - Your Plug

Cord thickness

measure_cord_thickness
4 → 8



black = the tool at its defaults;
teal = the plug you measured;
red dashed = what this dial moves

Widens the cord hook's slot and crossbar to fit the cord, and moves the finger holes and the whole body up to make room.

Default	Range	Step	Unit
4	1.5 to 9	0.5	mm

Before: 4. After: 8.

Moves: body edge, finger holes, pocket, wall notch, wing openings, zip-tie holes.

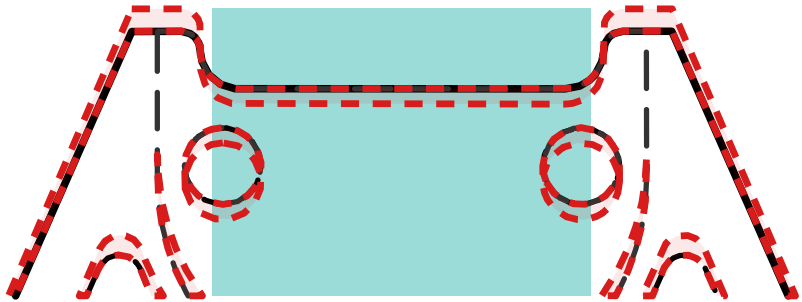
measure_wall_plate_style

One-sided puller · Step 1 - Your Plug

Outlet cover plate style

measure_wall_plate_style

Standard flat plate → Oversized / Jumbo



black = the tool at its defaults;
teal = the plug you measured;
red dashed = what this dial moves

Cuts the wall notch at the plug end deeper or shallower to suit the cover plate, and the zip-tie rows shift down with it.

Default	Range	Step	Unit
Standard flat plate	4 options	—	—

Options: Standard flat plate, Rocker / Decora, Oversized / Jumbo, No plate / flush.

Before: Standard flat plate. After: Oversized / Jumbo.

Moves: body edge, pocket, wall notch, wing openings, zip-tie holes.

show_plug_preview

One-sided puller · Step 1 - Your Plug

Show the see-through plug

show_plug_preview
This dial changes no shape.
Changes no shape: the see-through plug is a preview aid and is never exported.

Shows or hides the see-through plug in the preview and changes nothing in the printed tool.

Default	Range	Step	Unit
true	true / false	—	—

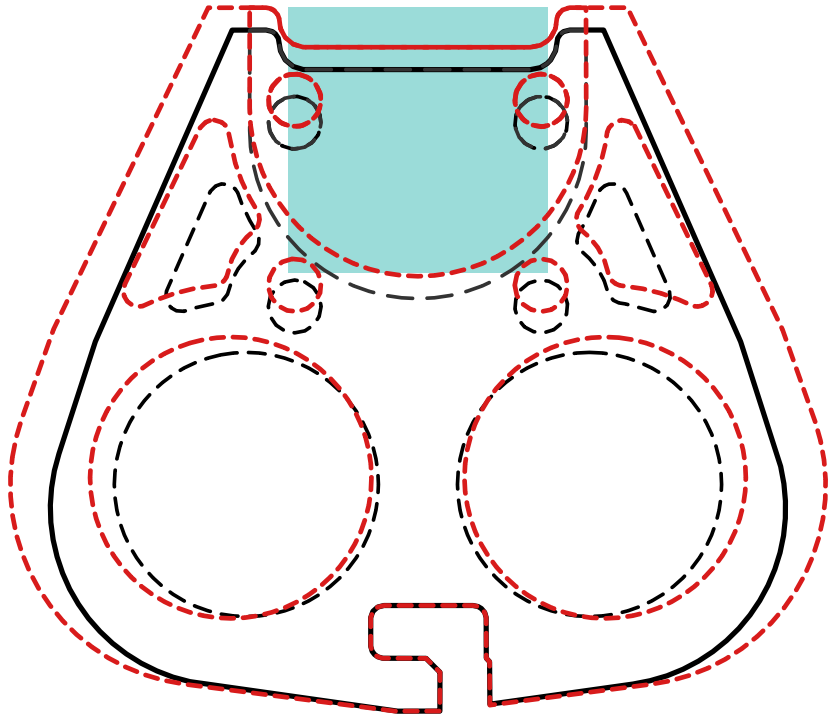
This dial changes no shape. Changes no shape: the see-through plug is a preview aid and is never exported.

size

One-sided puller · Step 2 - Size

Hand size

size
Medium → Large



black = the tool at its defaults;
teal = the plug you measured;
red dashed = what this dial moves

Scales the whole body, both finger holes, the wing openings and the zip-tie grid to the chosen hand size.

Default	Range	Step	Unit
Medium	5 options	—	—

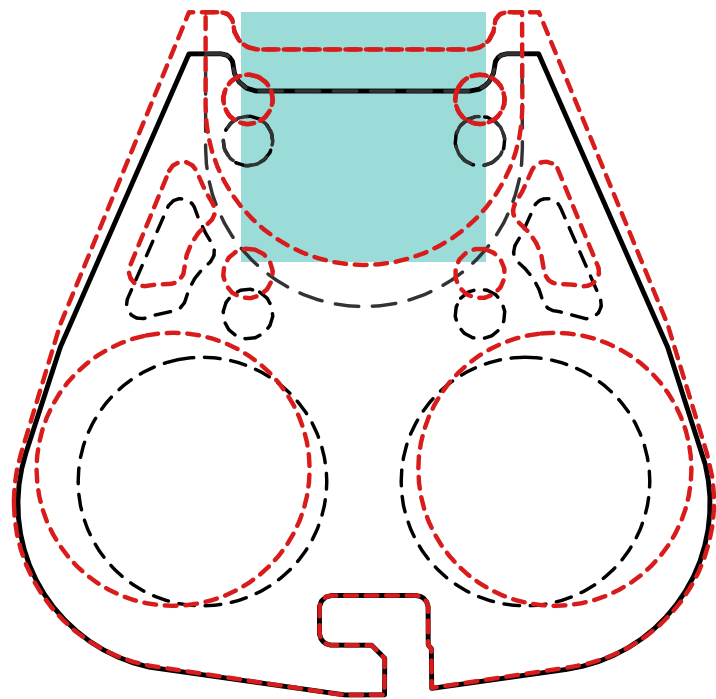
Options: Small, Medium, Large, Measure my hand, Custom.
Before: Medium. After: Large.
Moves: body edge, finger holes, pocket, wing openings, zip-tie holes.

measure_finger_width

One-sided puller · Step 2 - Size

Finger width

measure_finger_width
20 → 26
(size: Measure my hand)



black = the tool at its defaults;
teal = the plug you measured;
red dashed = what this dial moves

Widens both finger holes, spaces them farther apart and pushes them and the body up to keep the walls printable.

Default	Range	Step	Unit
20	14 to 32	0.5	mm

Before: 20. After: 26. Context: size = Measure my hand.

Only acts when size is Measure my hand.

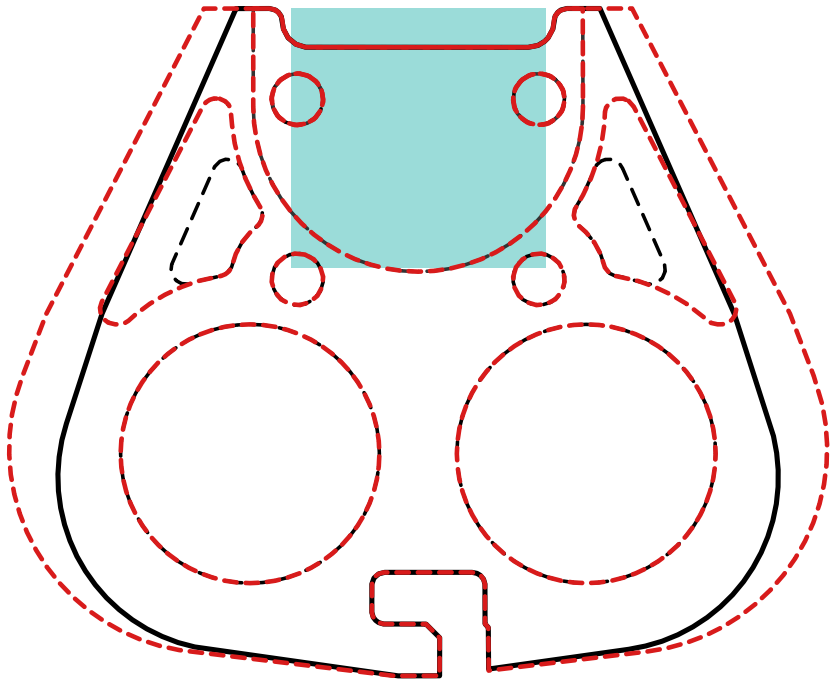
Moves: finger holes, pocket, wall notch, wing openings.

measure_hand_width

One-sided puller · Step 2 - Size

Hand width

measure_hand_width
85 → 100
(size: Measure my hand)



black = the tool at its defaults;
teal = the plug you measured;
red dashed = what this dial moves

Widens the whole body outline and thickens the slab in step with the hand.

Default	Range	Step	Unit
85	60 to 110	1	mm

Before: 85. After: 100. Context: size = Measure my hand.

Only acts when size is Measure my hand.

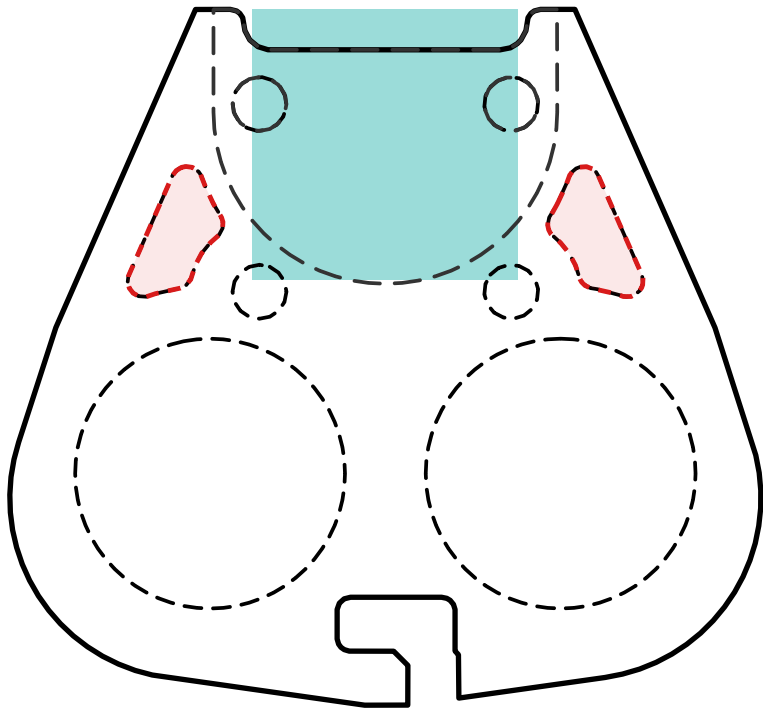
Moves: wing openings.

attachment

One-sided puller · Step 3 - Attachment

Attachment

attachment
Zip ties + Velcro → Zip ties



black = the tool at its defaults;
teal = the plug you measured;
red dashed = what this dial moves

Adds or removes the zip-tie holes and the wing or slot openings for the strap.

Default	Range	Step	Unit
Zip ties + Velcro	4 options	—	—

Options: Zip ties, Velcro strap, Zip ties + Velcro, None.

Before: Zip ties + Velcro. After: Zip ties.

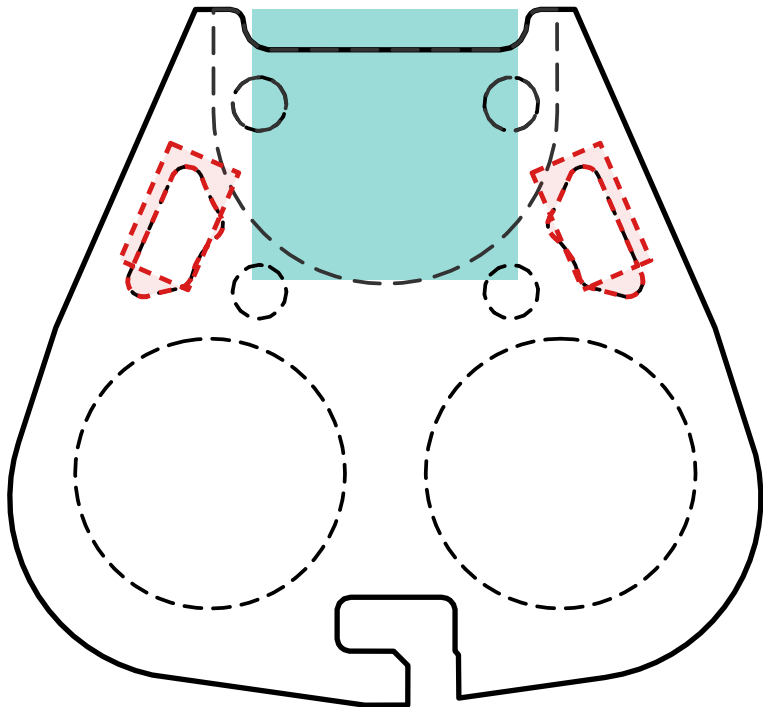
Moves: wing openings.

velcro_style

One-sided puller · Step 3 - Attachment

Strap opening style

velcro_style
Wing → Classic slot



black = the tool at its defaults;
teal = the plug you measured;
red dashed = what this dial moves

Swaps the curved wing openings for a pair of plain rectangular slots that lean along the body's sides.

Default	Range	Step	Unit
Wing	2 options	—	—

Options: Wing, Classic slot.

Before: Wing. After: Classic slot.

Moves: wing openings.

Can trip: WING OPENING SMALLER THAN STRAP WIDTH.

strap_width

One-sided puller · Step 3 - Attachment

Strap width

strap_width

This dial changes no shape.

Changes no shape: it only sets the width the W-14 check compares the wing opening against.

Sets the strap width the wing opening is checked against and changes no shape.

Default	Range	Step	Unit
15	10 to 25	1	mm

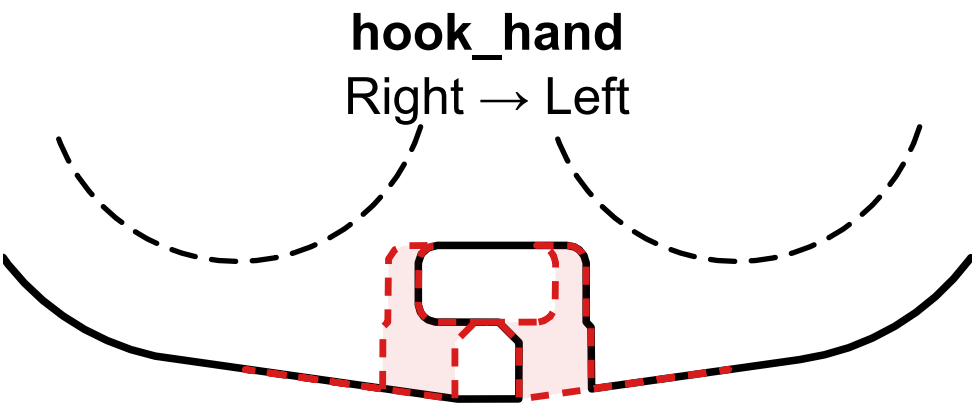
This dial changes no shape. Changes no shape: it only sets the width the W-14 check compares the wing opening against.

Can trip: WING OPENING SMALLER THAN STRAP WIDTH; WING WEB COLLAPSED - NO ROOM FOR STRAP.

hook_hand

One-sided puller · Step 4 - Cord Hook

Hook side



black = the tool at its defaults;
teal = the plug you measured;
red dashed = what this dial moves

Mirrors the cord hook at the cord end so its catch faces the other side, and nothing else moves.

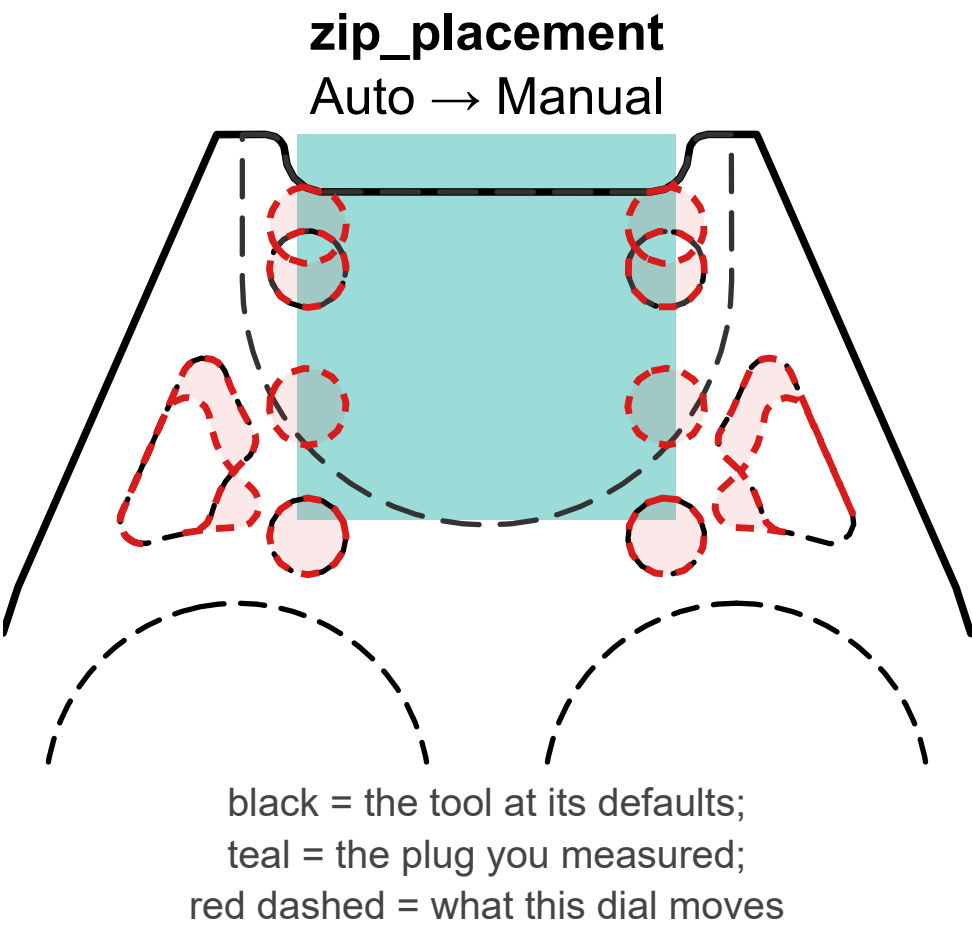
Default	Range	Step	Unit
Right	2 options	—	—

Options: Right, Left.
Before: Right. After: Left.
Moves: hook.

zip_placement

One-sided puller · Advanced - Zip Tie Placement

Zip-tie hole placement



Switches the zip-tie pairs from the automatic rows to the three position dials, which slide each pair along the plug's side.

Default	Range	Step	Unit
Auto	2 options	—	—

Options: Auto, Manual.

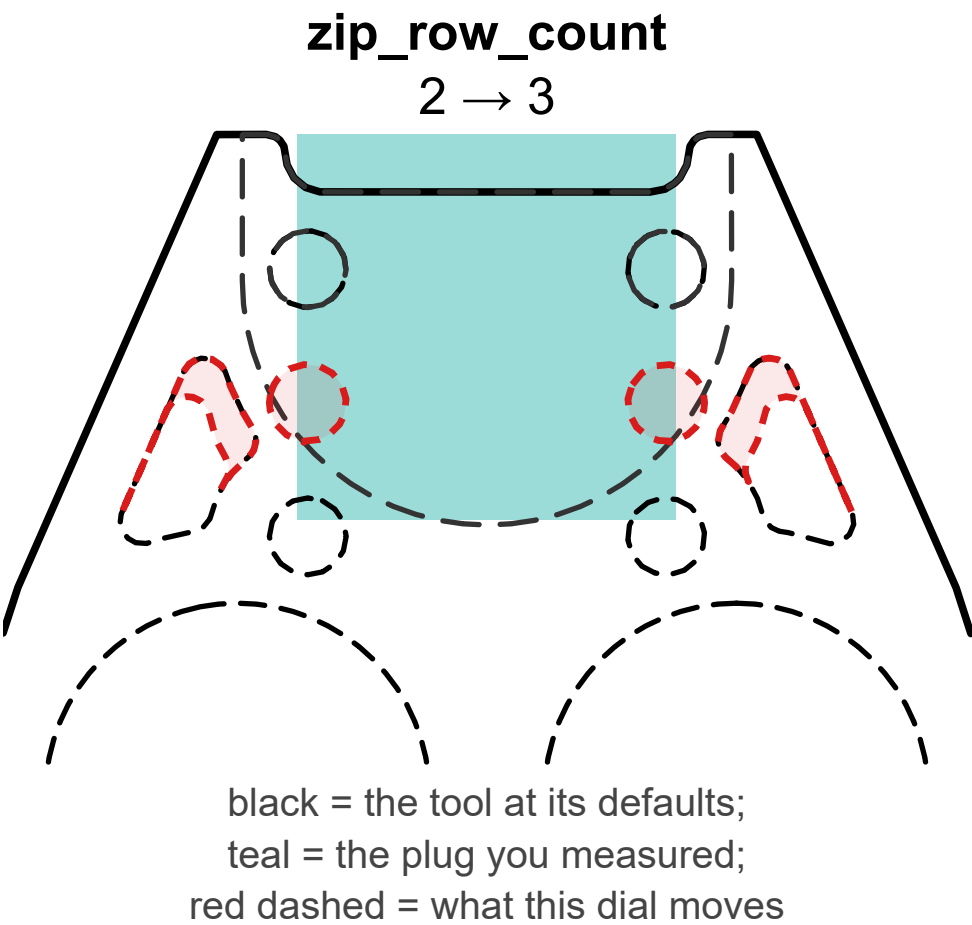
Before: Auto. After: Manual.

Moves: seat, wall notch, wing openings, zip-tie holes.

zip_row_count

One-sided puller · Advanced - Zip Tie Placement

Number of zip-tie rows



Adds or removes a pair of zip-tie holes, and the rows squeeze together when the body has no room for another.

Default	Range	Step	Unit
2	1 to 3	1	—

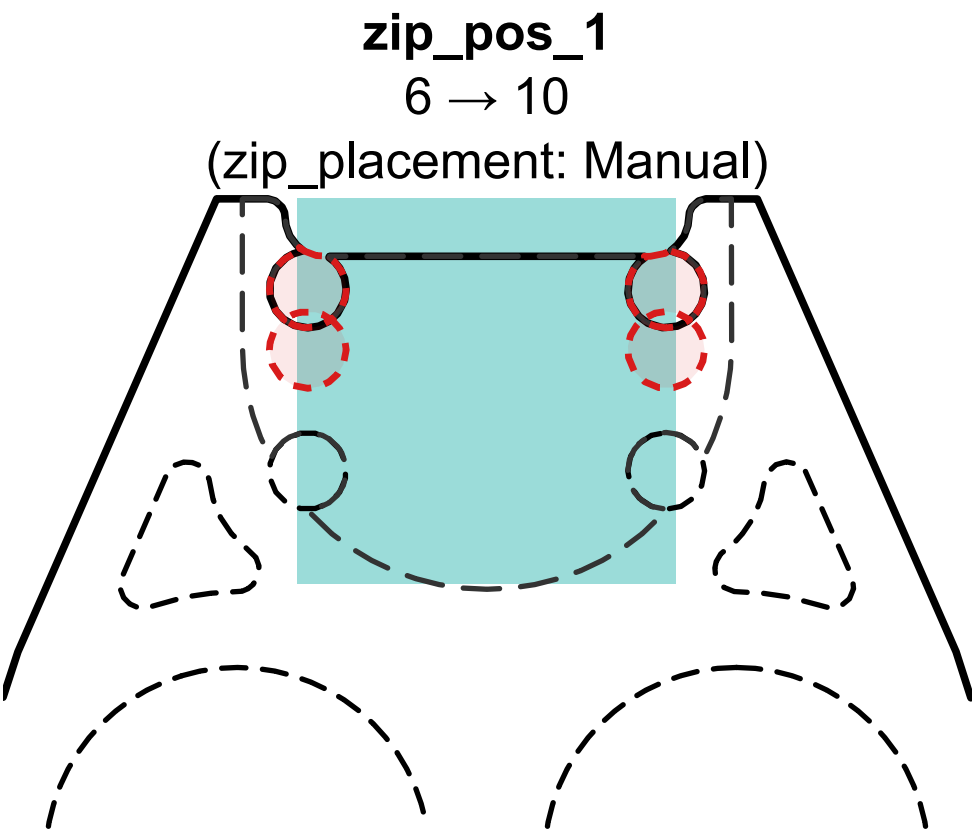
Before: 2. After: 3.

Moves: wing openings, zip-tie holes.

zip_pos_1

One-sided puller · Advanced - Zip Tie Placement

Zip-tie pair 1 position



black = the tool at its defaults;
teal = the plug you measured;
red dashed = what this dial moves

Slides the first pair of zip-tie holes along the plug's side, measured from the plug end.

Default	Range	Step	Unit
6	0 to 55	0.5	mm

Before: 6. After: 10. Context: zip_placement = Manual.

Only acts when zip_placement is Manual.

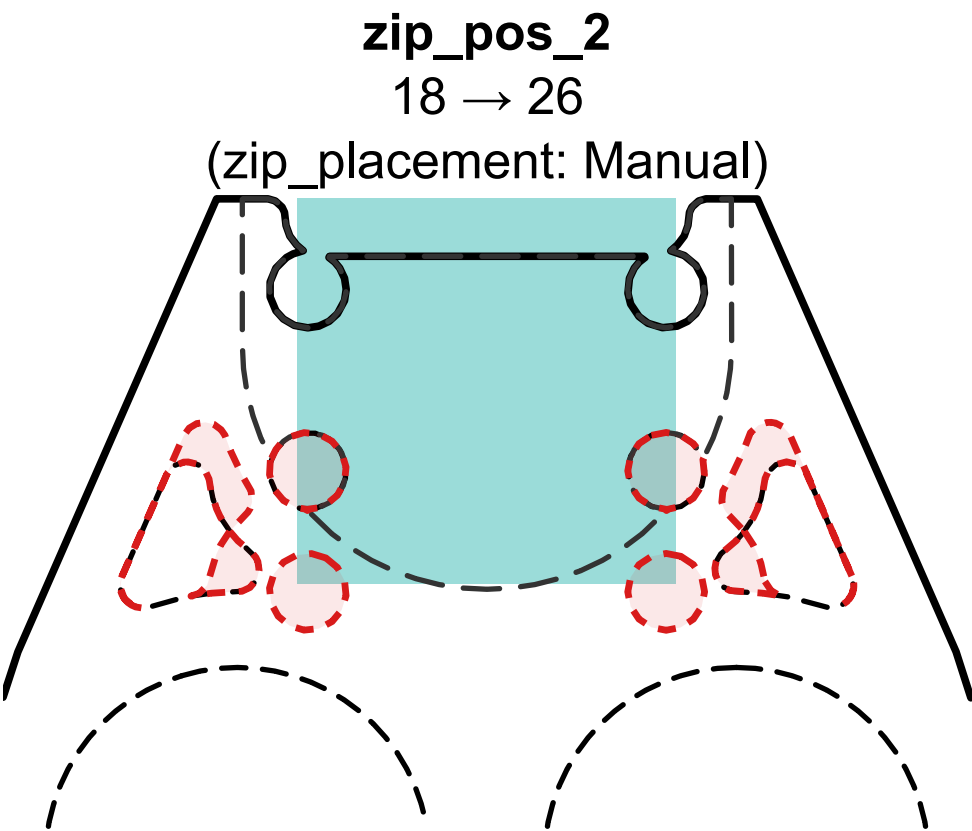
Moves: seat, wall notch, zip-tie holes.

Can trip: ZIP TIE HOLES HIT FINGER HOLES; ZIP TIE ROWS OVERLAP EACH OTHER; ZIP TIE HOLES HIT VELCRO SLOTS.

zip_pos_2

One-sided puller · Advanced - Zip Tie Placement

Zip-tie pair 2 position



black = the tool at its defaults;
teal = the plug you measured;
red dashed = what this dial moves

Slides the second pair of zip-tie holes along the plug's side, measured from the plug end.

Default	Range	Step	Unit
18	0 to 55	0.5	mm

Before: 18. After: 26. Context: zip_placement = Manual.

Only acts when zip_placement is Manual.

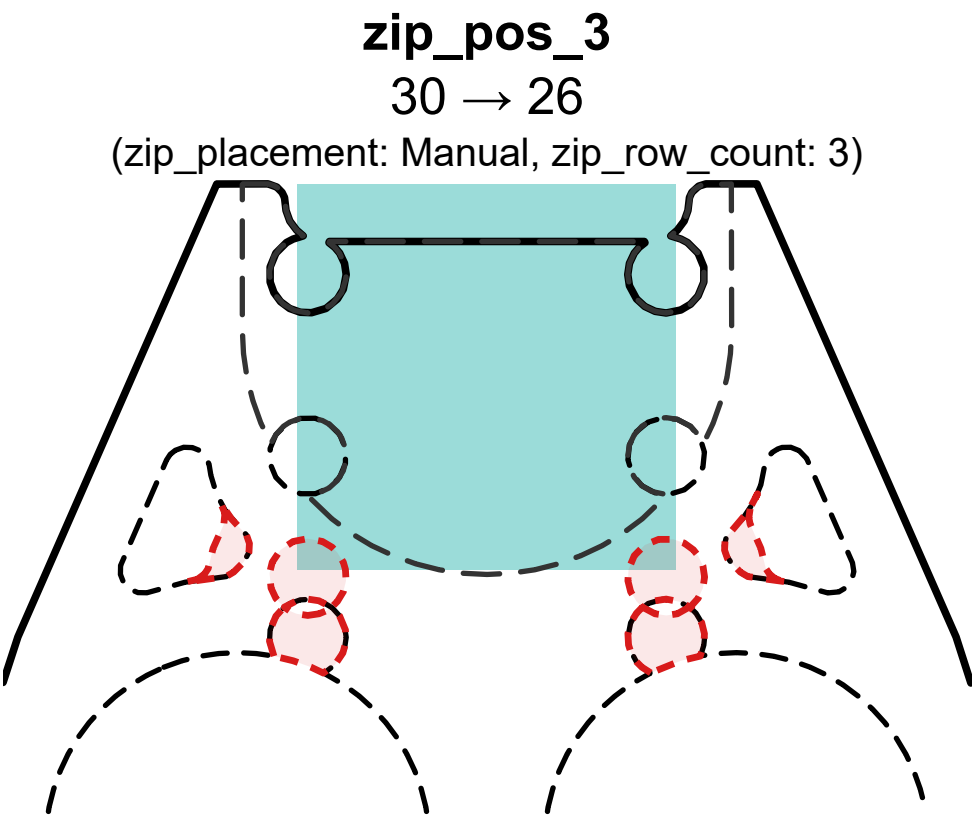
Moves: wing openings, zip-tie holes.

Can trip: ZIP TIE HOLES HIT FINGER HOLES; ZIP TIE ROWS OVERLAP EACH OTHER; ZIP TIE HOLES HIT VELCRO SLOTS.

zip_pos_3

One-sided puller · Advanced - Zip Tie Placement

Zip-tie pair 3 position



black = the tool at its defaults;
teal = the plug you measured;
red dashed = what this dial moves

Slides the third pair of zip-tie holes along the plug's side, measured from the plug end.

Default	Range	Step	Unit
30	0 to 55	0.5	mm

Before: 30. After: 26. Context: zip_placement = Manual, zip_row_count = 3.

Only acts when zip_placement is Manual and zip_row_count is 3.

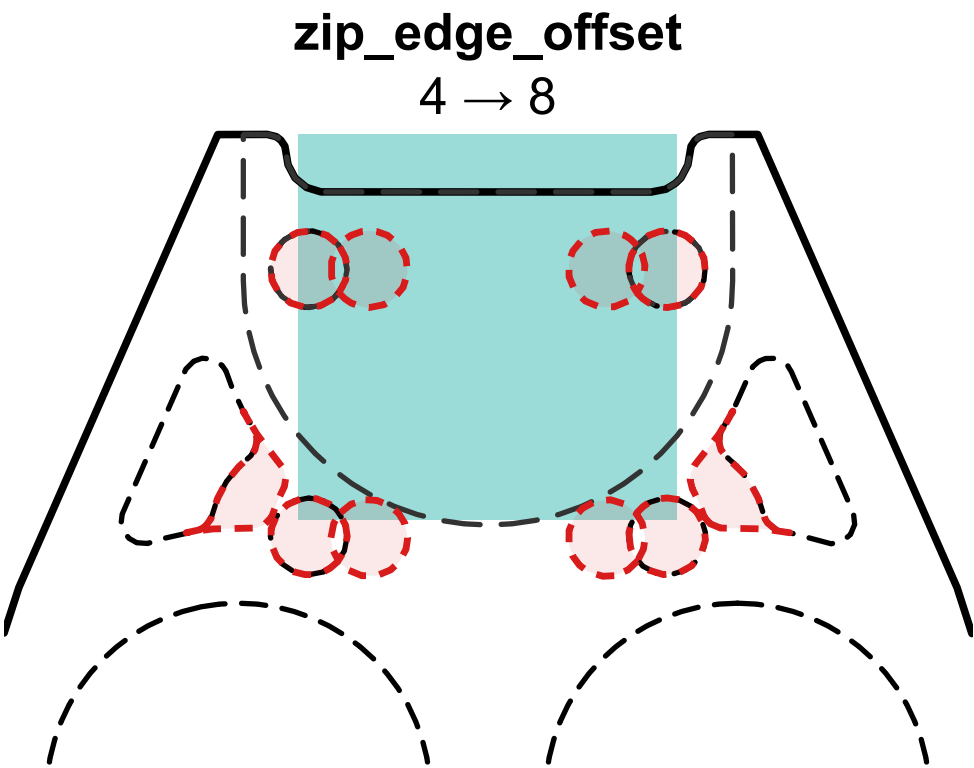
Moves: wing openings, zip-tie holes.

Can trip: ZIP TIE HOLES HIT FINGER HOLES; ZIP TIE ROWS OVERLAP EACH OTHER; ZIP TIE HOLES HIT VELCRO SLOTS.

zip_edge_offset

One-sided puller · Advanced - Zip Tie Placement

Zip-tie hole distance from the pocket wall



black = the tool at its defaults;
teal = the plug you measured;
red dashed = what this dial moves

Moves every zip-tie hole farther from or closer to the pocket wall, and the wing openings reshape around them.

Default	Range	Step	Unit
4	2.5 to 12	0.25	mm

Before: 4. After: 8.

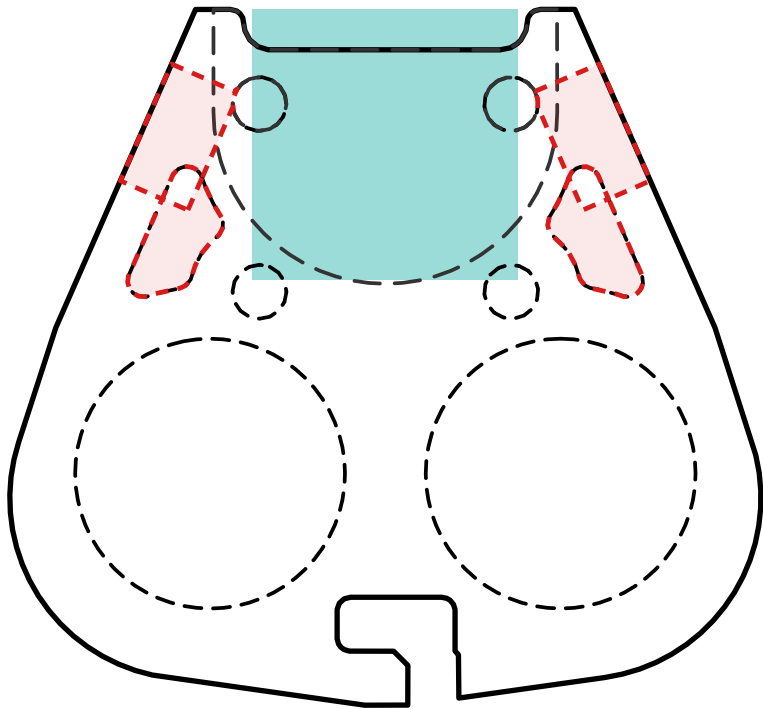
Moves: wing openings, zip-tie holes.

velcro_placement

One-sided puller · Advanced - Velcro Placement

Strap slot placement

velcro_placement
Auto → Manual



black = the tool at its defaults;
teal = the plug you measured;
red dashed = what this dial moves

Switches the strap openings to a pair of plain slots that the position dial slides along the plug's side, whatever the opening style.

Default	Range	Step	Unit
Auto	2 options	—	—

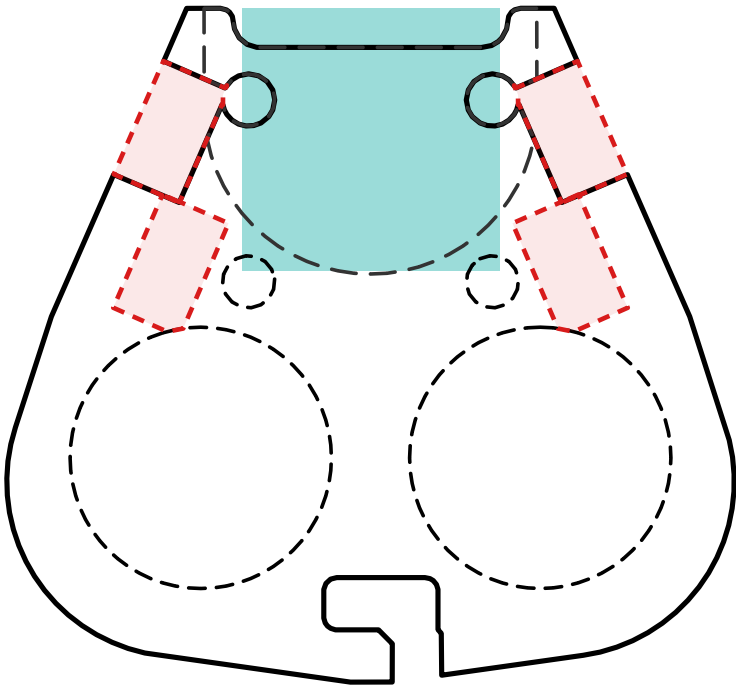
Options: Auto, Manual.
Before: Auto. After: Manual.
Moves: wing openings.

velcro_pos

One-sided puller · Advanced - Velcro Placement

Strap slot position

velcro_pos
12 → 25
(velcro_style: Classic slot, velcro_placement: Manual)



black = the tool at its defaults;
teal = the plug you measured;
red dashed = what this dial moves

Slides the pair of strap slots along the plug's side, measured from the plug end.

Default	Range	Step	Unit
12	0 to 55	0.5	mm

Before: 12. After: 25. Context: velcro_style = Classic slot, velcro_placement = Manual.

Only acts when velcro_placement is Manual.

Moves: classic slots.

Can trip: ZIP TIE HOLES HIT VELCRO SLOTS.

quality

One-sided puller · Advanced - Render Quality

Render quality

quality

This dial changes no shape.

Changes no shape: it only sets the number of segments in every circle and arc.

Sets how many flat segments draw each curve and changes no dimension.

Default	Range	Step	Unit
64	24 to 128	8	segments

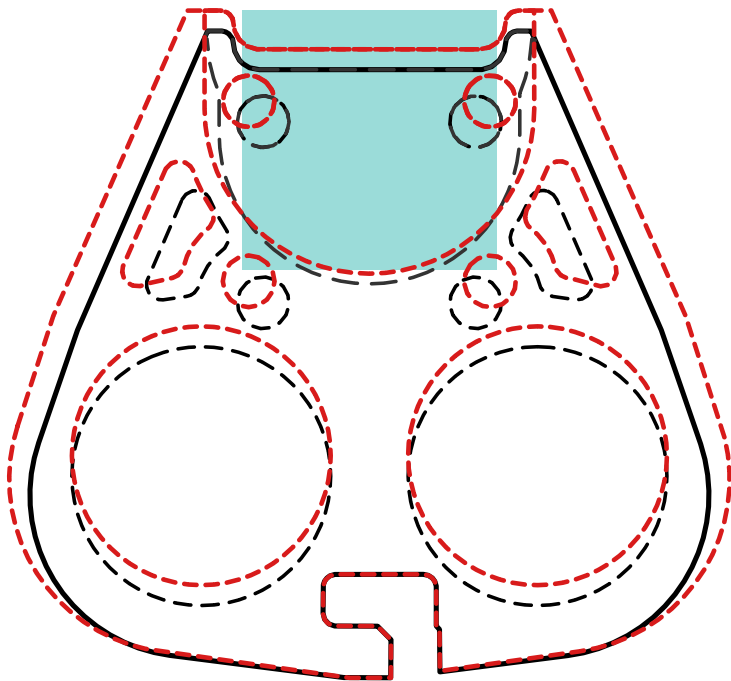
This dial changes no shape. Changes no shape: it only sets the number of segments in every circle and arc.

reset_custom_to_medium

One-sided puller · Custom Mode

Reset Custom to the Medium reference

reset_custom_to_medium
false → true
(size: Custom)



black = the tool at its defaults;
teal = the plug you measured;
red dashed = what this dial moves

Swaps every Custom slider for the Medium reference values for one render, so the whole tool snaps back to the Medium shape.

Default	Range	Step	Unit
false	true / false	—	—

Before: false. After: true. Context: size = Custom.

Only acts when size is Custom.

Moves: finger holes, wall notch, wing openings, zip-tie holes.

custom_enable_auto_fit

One-sided puller · Custom Mode

Auto-fit

custom_enable_auto_fit

This dial changes no shape.

Changes no shape at the Custom defaults: it only clamps sliders that leave their safe range, and every clamp is reported in the console.

Lets the tool clamp any Custom slider that would push a feature past the body or into a neighbor, and changes nothing while every slider is inside its safe range.

Default	Range	Step	Unit
true	true / false	—	—

This dial changes no shape. Changes no shape at the Custom defaults: it only clamps sliders that leave their safe range, and every clamp is reported in the console.

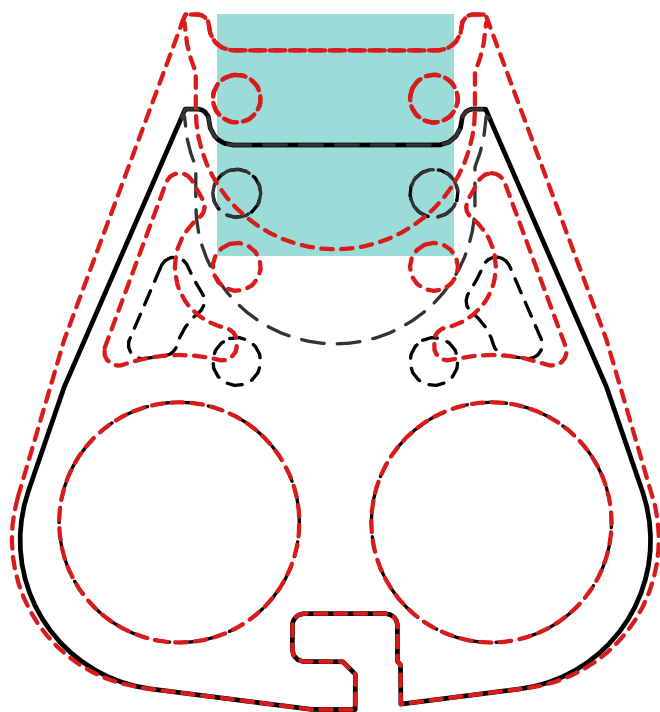
Can trip: ZIP TIE HOLES HIT FINGER HOLES; FINGER HOLES OUTSIDE BODY; POCKET SEAT WIDER THAN TOP EDGE; POCKET WIDER THAN BODY; WALL NOTCH WIDER THAN TOP EDGE.

custom_puller_length

One-sided puller · Body Shape (Custom size only)

Body length

custom_puller_length
63.5 → 73.5
(size: Custom)



black = the tool at its defaults;
teal = the plug you measured;
red dashed = what this dial moves

Stretches the whole body from the cord end to the plug end.

Default	Range	Step	Unit
63.5	50 to 120	0.5	mm

Before: 63.5. After: 73.5. Context: size = Custom.

Only acts when size is Custom.

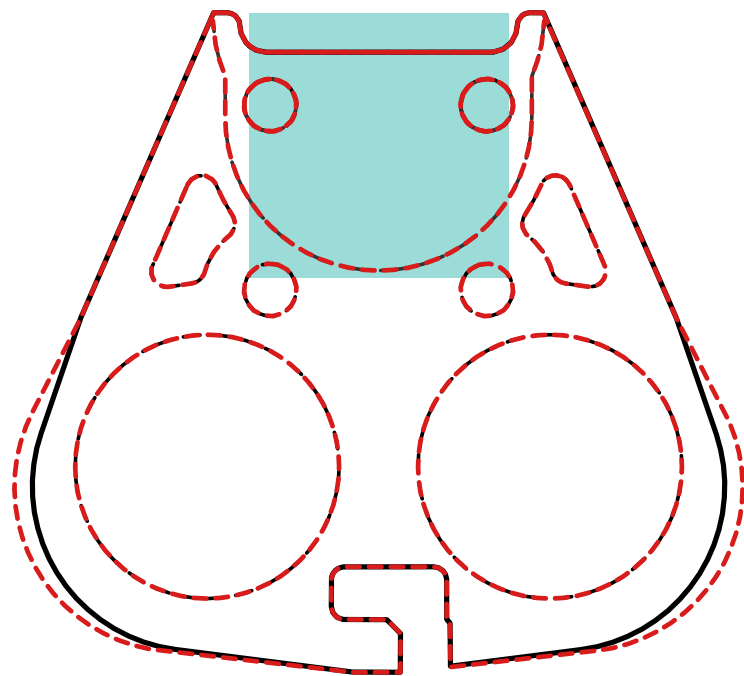
Moves: body edge, pocket, seat, wall notch, zip-tie holes.

custom_puller_bottom_width

One-sided puller · Body Shape (Custom size only)

Body width at the widest point

custom_puller_bottom_width
77.6 → 87.6
(size: Custom)



black = the tool at its defaults;
teal = the plug you measured;
red dashed = what this dial moves

Widens the body at its widest point, just above the cord end, before the side rounding is applied.

Default	Range	Step	Unit
77.6	50 to 120	0.05	mm

Before: 77.6. After: 87.6. Context: size = Custom.

Only acts when size is Custom.

Moves: body edge.

custom_puller_bottom_corners

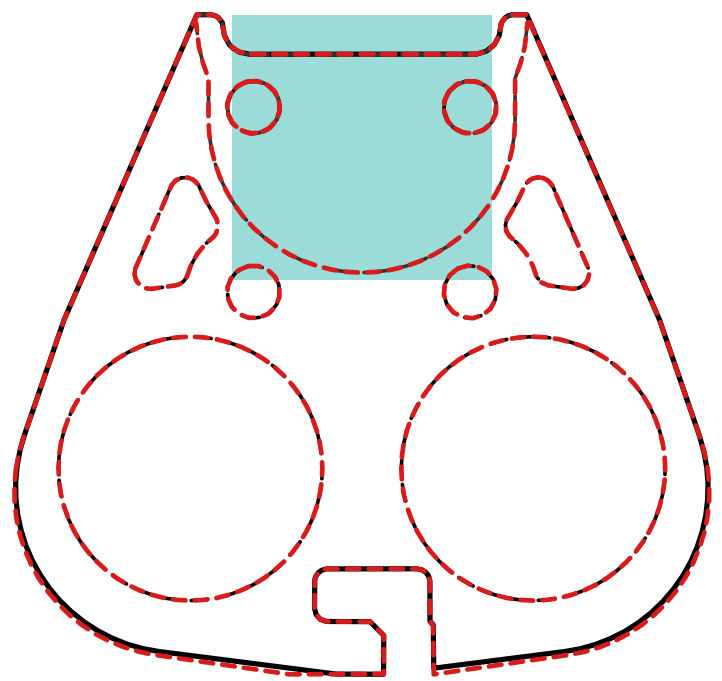
One-sided puller · Body Shape (Custom size only)

Cord-end corner width

custom_puller_bottom_corners

3 → 13

(size: Custom)



black = the tool at its defaults;
teal = the plug you measured;
red dashed = what this dial moves

Widens the flat at the cord end where the two lower corners of the body outline sit.

Default	Range	Step	Unit
3	3 to 80	0.25	mm

Before: 3. After: 13. Context: size = Custom.

Only acts when size is Custom.

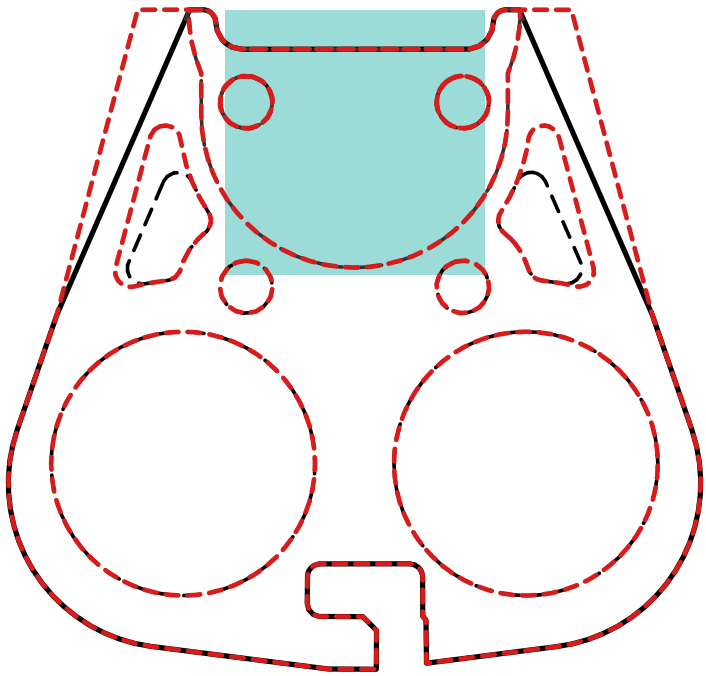
Moves: body edge.

custom_puller_top_width

One-sided puller · Body Shape (Custom size only)

Body width at the plug end

custom_puller_top_width
31.75 → 41.75
(size: Custom)



black = the tool at its defaults;
teal = the plug you measured;
red dashed = what this dial moves

Widens the plug end of the body, where the wall notch and the pocket seat sit.

Default	Range	Step	Unit
31.75	25 to 80	0.25	mm

Before: 31.75. After: 41.75. Context: size = Custom.

Only acts when size is Custom.

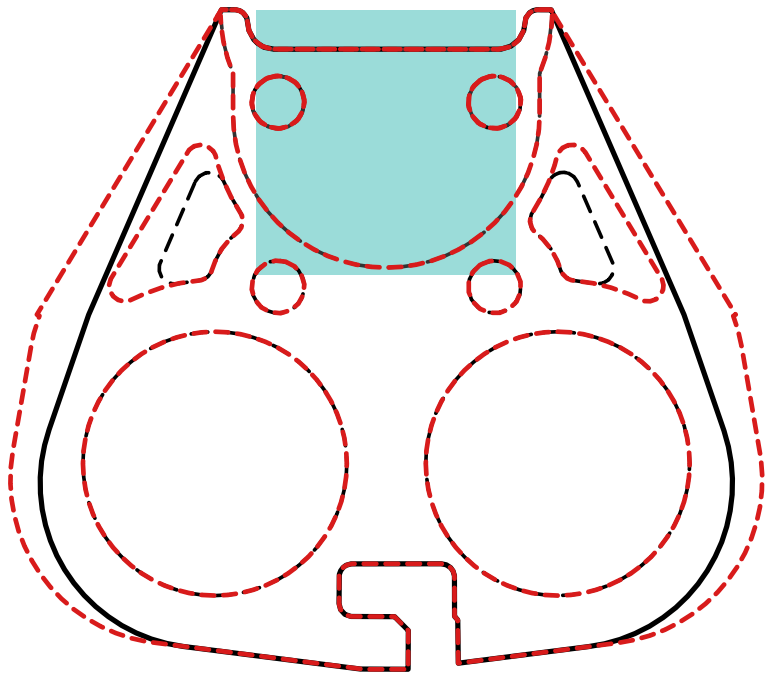
Moves: body edge, wing openings.

custom_puller_middle_width

One-sided puller · Body Shape (Custom size only)

Body width at the middle

custom_puller_middle_width
57.35 → 67.35
(size: Custom)



black = the tool at its defaults;
teal = the plug you measured;
red dashed = what this dial moves

Widens the body at the middle control point, bulging or straightening the side edges between the widest point and the plug end.

Default	Range	Step	Unit
57.35	25 to 120	0.05	mm

Before: 57.35. After: 67.35. Context: size = Custom.

Only acts when size is Custom.

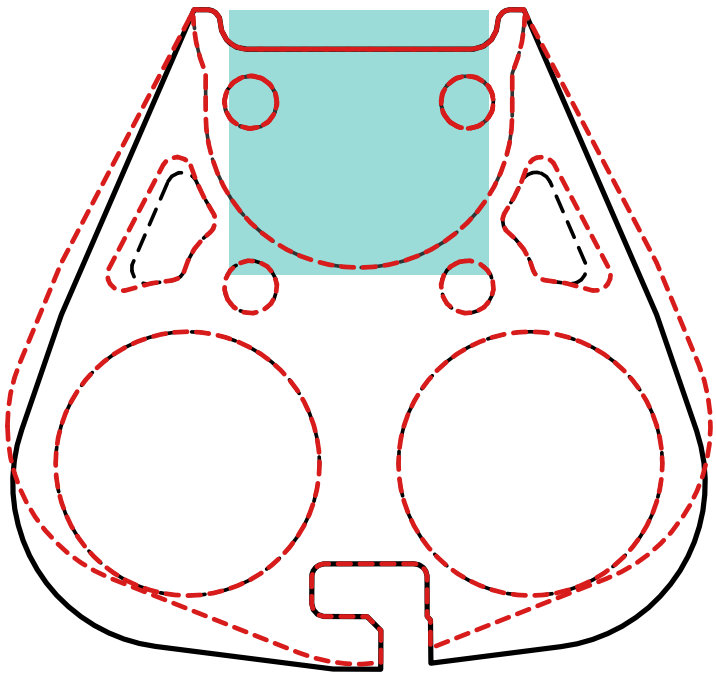
Moves: body edge, wing openings.

custom_puller_side_corner

One-sided puller · Body Shape (Custom size only)

Side corner position

custom_puller_side_corner
4.65 → 14.65
(size: Custom)



black = the tool at its defaults;
teal = the plug you measured;
red dashed = what this dial moves

Moves the two widest corners of the body outline up toward the plug end or down toward the cord end.

Default	Range	Step	Unit
4.65	3 to 35	0.05	mm

Before: 4.65. After: 14.65. Context: size = Custom.

Only acts when size is Custom.

Moves: body edge, wing openings.

custom_body_thickness

One-sided puller · Body Shape (Custom size only)

Body thickness

custom_body_thickness
6.35 → 9
(size: Custom, section-y: 35)



black = the tool at its defaults;
teal = the plug you measured;
red dashed = what this dial moves

Thickens the whole slab, so every hole and the pocket get deeper walls.

Default	Range	Step	Unit
6.35	4 to 15	0.05	mm

Before: 6.35. After: 9. Context: size = Custom. Section at y = 35 mm.

Only acts when size is Custom.

Moves: body edge.

custom_body_round_bottom_only

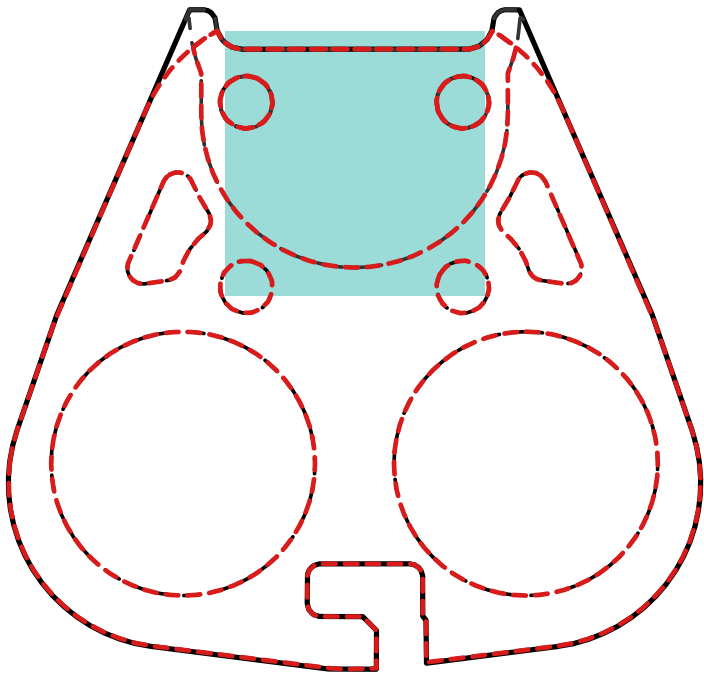
One-sided puller · Body Shape (Custom size only)

Round the cord half only

custom_body_round_bottom_only

true → false

(size: Custom)



black = the tool at its defaults;
teal = the plug you measured;
red dashed = what this dial moves

Rounds the whole outline instead of only the cord half, so the plug end loses its crisp corners.

Default	Range	Step	Unit
true	true / false	—	—

Before: true. After: false. Context: size = Custom.

Only acts when size is Custom.

Moves: body edge, hook, seat, wall notch.

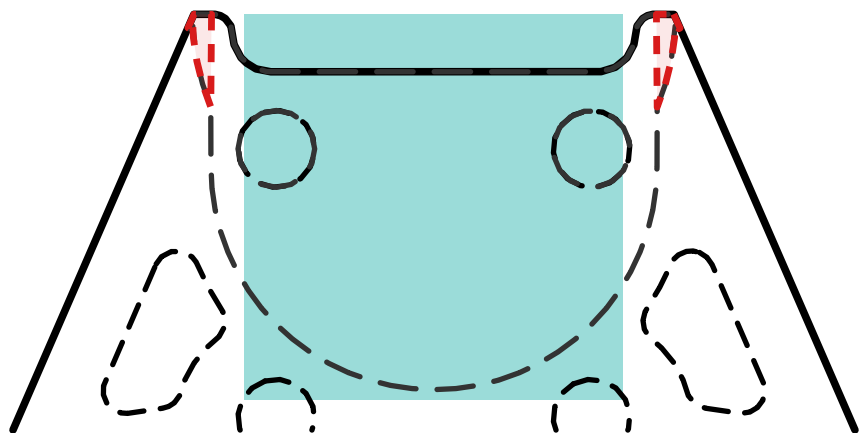
custom_pocket_seat_diameter

One-sided puller · Plug Pocket (Custom size only)

Pocket seat diameter

custom_pocket_seat_diameter

31.75 → 28
(size: Custom)



black = the tool at its defaults;
teal = the plug you measured;
red dashed = what this dial moves

Widens the round seat cut into the plug end of the pocket.

Default	Range	Step	Unit
31.75	10 to 45	0.05	mm

Before: 31.75. After: 28. Context: size = Custom.

Only acts when size is Custom.

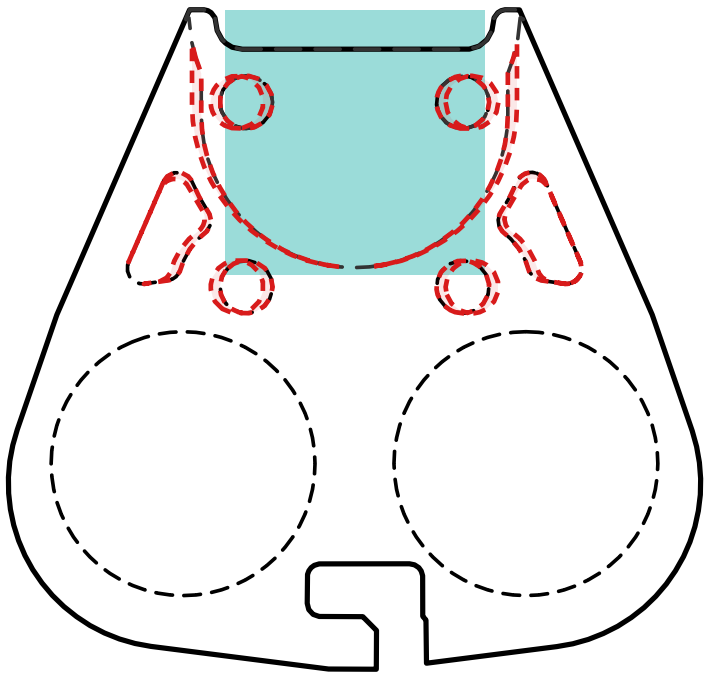
Moves: seat.

custom_pocket_width

One-sided puller · Plug Pocket (Custom size only)

Pocket width

custom_pocket_width
28.85 → 34
(size: Custom)



black = the tool at its defaults;
teal = the plug you measured;
red dashed = what this dial moves

Widens the plug recess, and the zip-tie holes and wing openings move outward with its wall.

Default	Range	Step	Unit
28.85	10 to 45	0.05	mm

Before: 28.85. After: 34. Context: size = Custom.

Only acts when size is Custom.

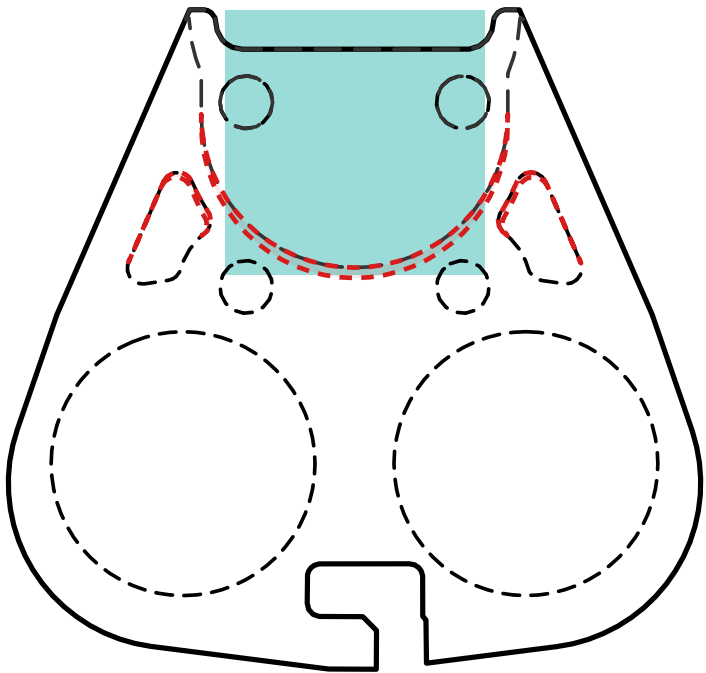
Moves: pocket, seat, wing openings, zip-tie holes.

custom_pocket_depth

One-sided puller · Plug Pocket (Custom size only)

Pocket depth

custom_pocket_depth
24.5 → 34
(size: Custom)



black = the tool at its defaults;
teal = the plug you measured;
red dashed = what this dial moves

Runs the plug recess farther from the plug end toward the finger holes.

Default	Range	Step	Unit
24.5	5 to 60	0.5	mm

Before: 24.5. After: 34. Context: size = Custom.

Only acts when size is Custom.

Moves: pocket, wing openings.

custom_pocket_dome_drop

One-sided puller · Plug Pocket (Custom size only)

Pocket reference drop

custom_pocket_dome_drop
This dial changes no shape.
Changes no shape: the recess is a rounded-nose shape whose width at the plug end is the pocket width and whose walls follow the side taper, so this reference point cancels out; measured with straight and with tapered walls.

Changes no shape of the pocket, because the recess outline is built from the plug width at the plug end and the side taper, and this reference point cancels out of it.

Default	Range	Step	Unit
2.15	0 to 8	0.05	mm

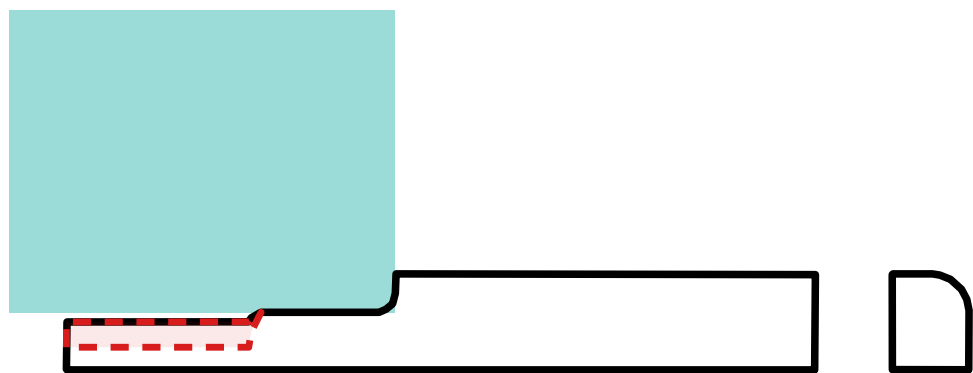
This dial changes no shape. Changes no shape: the recess is a rounded-nose shape whose width at the plug end is the pocket width and whose walls follow the side taper, so this reference point cancels out; measured with straight and with tapered walls.

custom_pocket_seat_floor

One-sided puller · Plug Pocket (Custom size only)

Pocket seat floor thickness

custom_pocket_seat_floor
3.175 → 1.5
(size: Custom, section-x: 0)



black = the tool at its defaults;
teal = the plug you measured;
red dashed = what this dial moves

Thins or thickens the floor left under the round seat, so the seat is cut deeper or shallower.

Default	Range	Step	Unit
3.175	0 to 8	0.025	mm

Before: 3.175. After: 1.5. Context: size = Custom. Section at x = 0 mm.

Only acts when size is Custom.

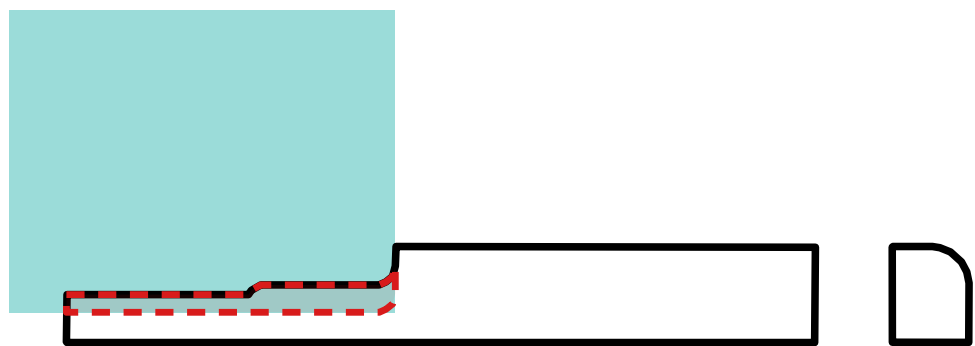
Moves: seat.

custom_pocket_floor

One-sided puller · Plug Pocket (Custom size only)

Plug recess floor thickness

custom_pocket_floor
3.81 → 2
(size: Custom, section-x: 0)



black = the tool at its defaults;
teal = the plug you measured;
red dashed = what this dial moves

Thins or thickens the floor left under the plug recess, so the recess is cut deeper or shallower.

Default	Range	Step	Unit
3.81	0 to 12	0.025	mm

Before: 3.81. After: 2. Context: size = Custom. Section at x = 0 mm.

Only acts when size is Custom.

Moves: pocket, seat.

custom_pocket_side_angle

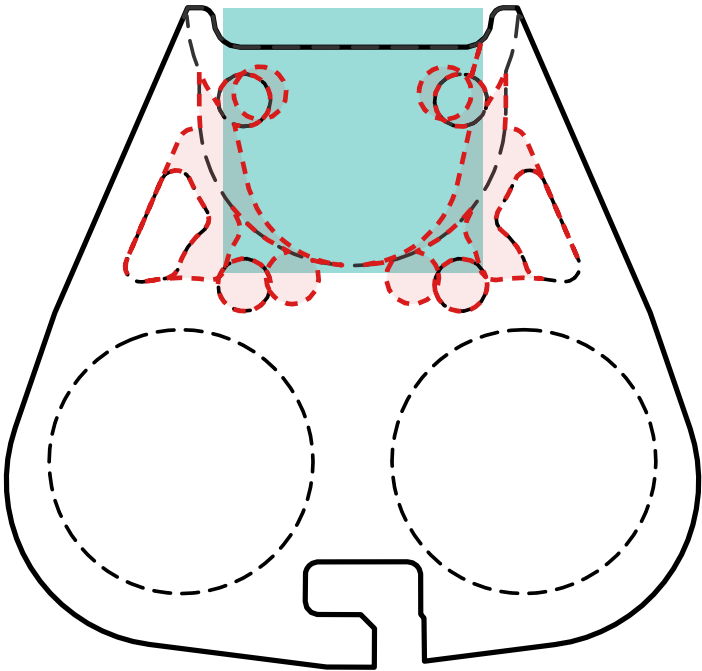
One-sided puller · Plug Pocket (Custom size only)

Pocket side taper

custom_pocket_side_angle

0 → 10

(size: Custom)



black = the tool at its defaults;
teal = the plug you measured;
red dashed = what this dial moves

Tilts the pocket's side walls inward toward the cord end, and the zip-tie holes and strap slots lean in along the same line.

Default	Range	Step	Unit
0	-15 to 25	0.5	mm

Before: 0. After: 10. Context: size = Custom.

Only acts when size is Custom.

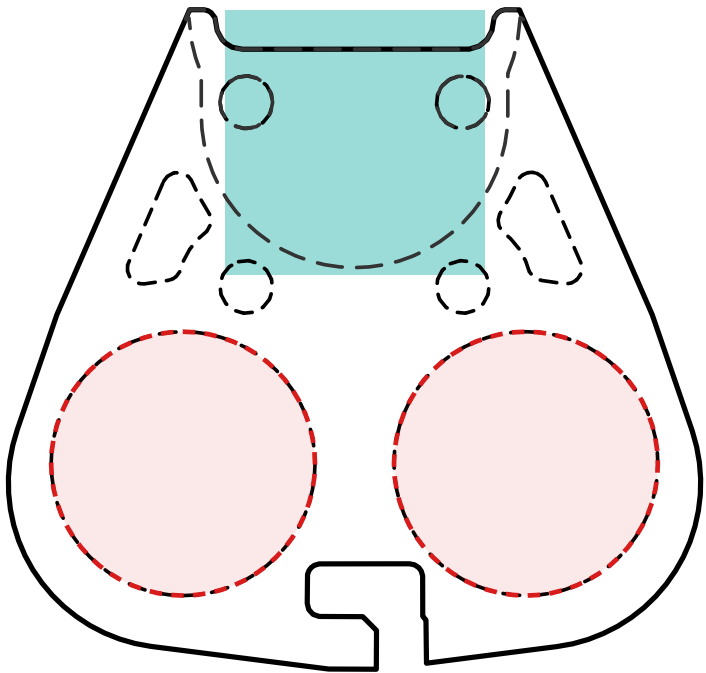
Moves: pocket, wing openings.

custom_enable_finger_holes

One-sided puller · Finger Holes (Custom size only)

Finger holes on or off

custom_enable_finger_holes
true → false
(size: Custom)



black = the tool at its defaults;
teal = the plug you measured;
red dashed = what this dial moves

Cuts or leaves out both finger holes.

Default	Range	Step	Unit
true	true / false	—	—

Before: true. After: false. Context: size = Custom.

Only acts when size is Custom.

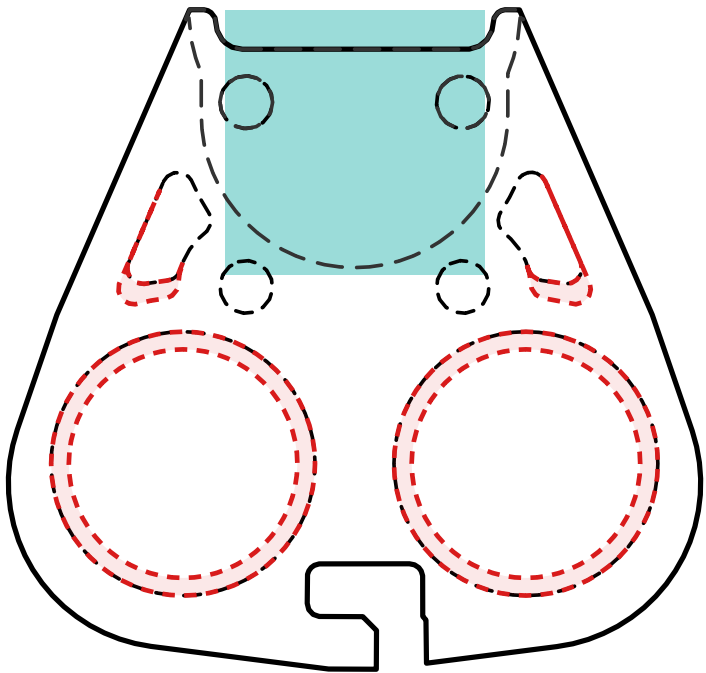
Moves: body edge, finger holes.

custom_finger_hole_diameter

One-sided puller · Finger Holes (Custom size only)

Finger hole diameter

custom_finger_hole_diameter
25.4 → 22
(size: Custom)



black = the tool at its defaults;
teal = the plug you measured;
red dashed = what this dial moves

Widens both finger holes.

Default	Range	Step	Unit
25.4	15 to 40	0.1	mm

Before: 25.4. After: 22. Context: size = Custom.

Only acts when size is Custom.

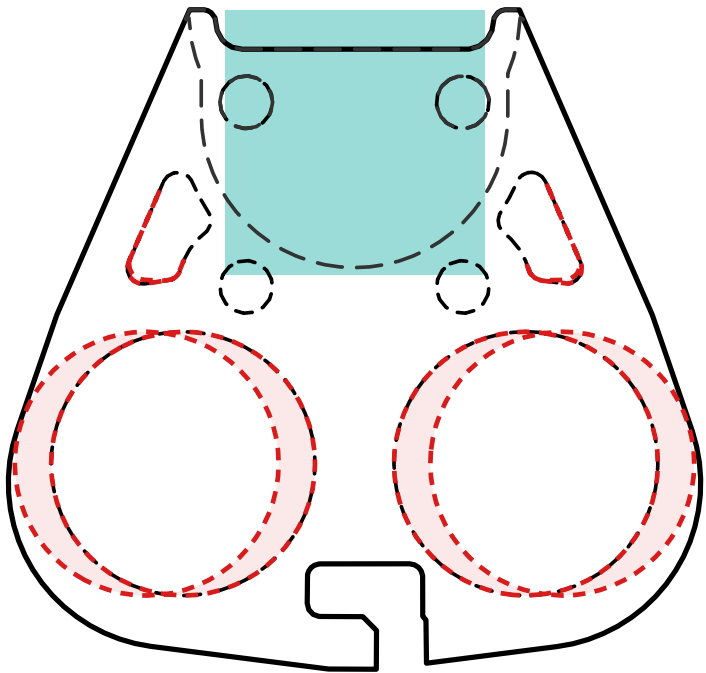
Moves: finger holes, wing openings.

custom_finger_hole_spacing

One-sided puller · Finger Holes (Custom size only)

Finger hole spacing

custom_finger_hole_spacing
33 → 40
(size: Custom)



black = the tool at its defaults;
teal = the plug you measured;
red dashed = what this dial moves

Moves the two finger holes farther apart or closer together.

Default	Range	Step	Unit
33	20 to 50	0.5	mm

Before: 33. After: 40. Context: size = Custom.

Only acts when size is Custom.

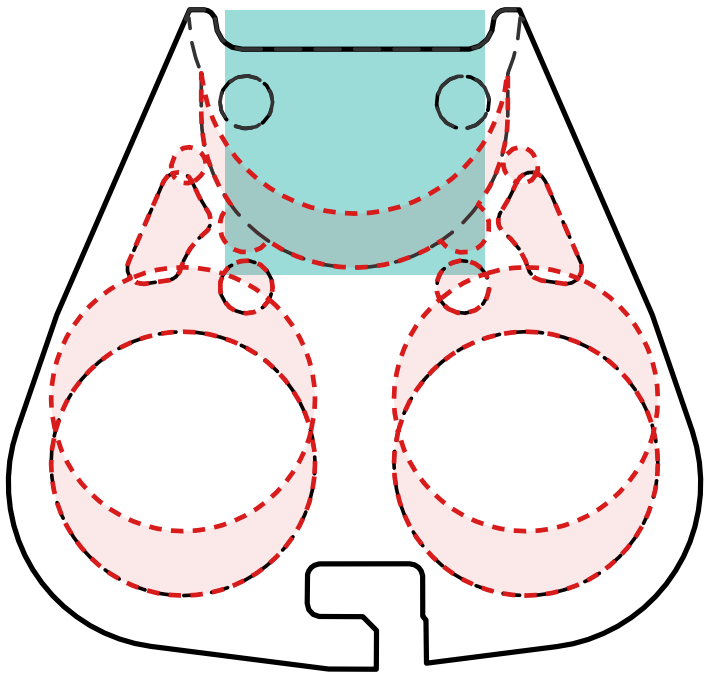
Moves: finger holes, wing openings.

custom_finger_hole_y_position

One-sided puller · Finger Holes (Custom size only)

Finger hole position

custom_finger_hole_y_position
19.8 → 26
(size: Custom)



black = the tool at its defaults;
teal = the plug you measured;
red dashed = what this dial moves

Slides both finger holes up toward the pocket or down toward the cord hook.

Default	Range	Step	Unit
19.8	10 to 50	0.1	mm

Before: 19.8. After: 26. Context: size = Custom.

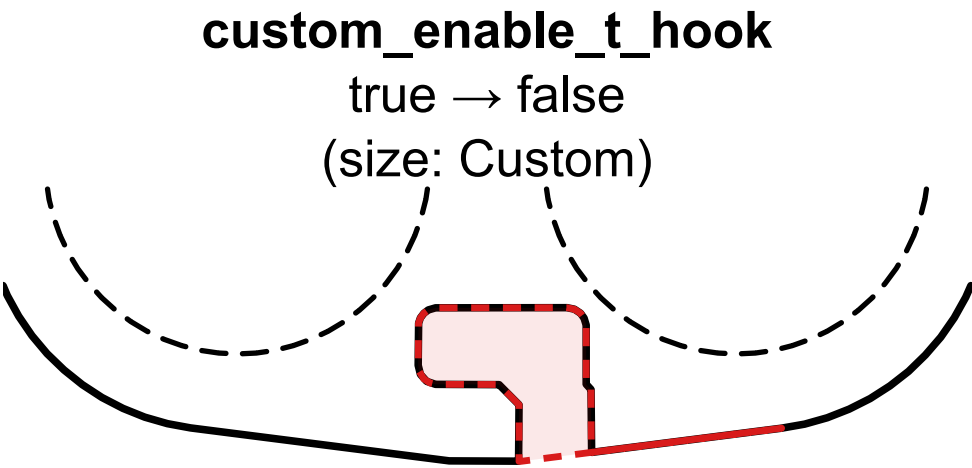
Only acts when size is Custom.

Moves: finger holes, pocket.

custom_enable_t_hook

One-sided puller · T Hook (Custom size only)

Cord hook on or off



black = the tool at its defaults;
teal = the plug you measured;
red dashed = what this dial moves

Cuts or leaves out the cord hook at the cord end.

Default	Range	Step	Unit
true	true / false	—	—

Before: true. After: false. Context: size = Custom.

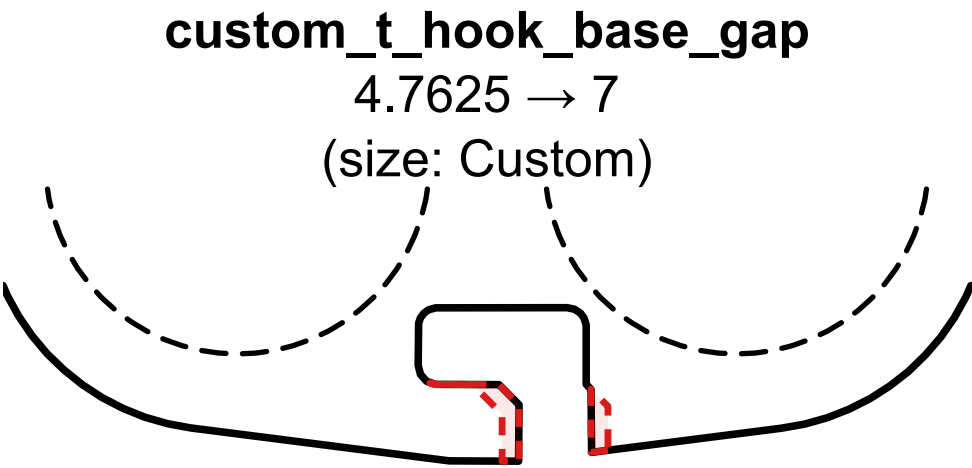
Only acts when size is Custom.

Moves: hook.

custom_t_hook_base_gap

One-sided puller · T Hook (Custom size only)

Hook slot width



black = the tool at its defaults;
teal = the plug you measured;
red dashed = what this dial moves

Widens the slot the cord slides into at the cord end.

Default	Range	Step	Unit
4.7625	3 to 10	0.05	mm

Before: 4.7625. After: 7. Context: size = Custom.

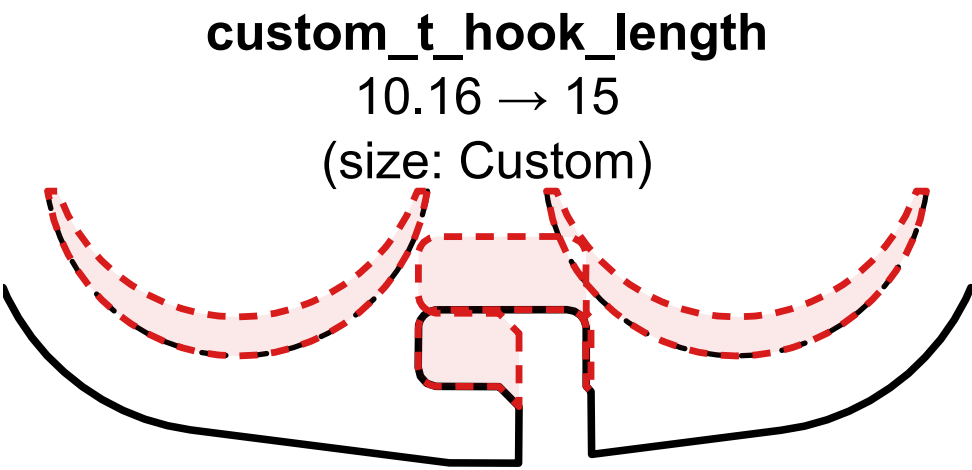
Only acts when size is Custom.

Moves: hook.

custom_t_hook_length

One-sided puller · T Hook (Custom size only)

Hook length



black = the tool at its defaults;
teal = the plug you measured;
red dashed = what this dial moves

Runs the cord hook farther into the body from the cord end.

Default	Range	Step	Unit
10.16	6 to 20	0.05	mm

Before: 10.16. After: 15. Context: size = Custom.

Only acts when size is Custom.

Moves: finger holes, hook, pocket, wing openings, zip-tie holes.

custom_t_hook_holder_width

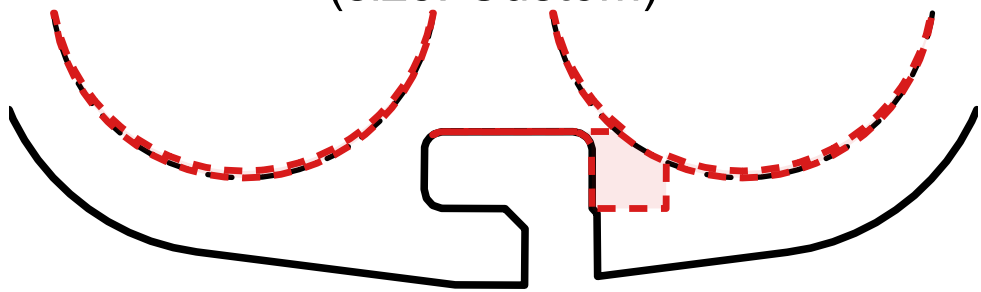
One-sided puller · T Hook (Custom size only)

Hook crossbar width

custom_t_hook_holder_width

11.1125 → 16

(size: Custom)



black = the tool at its defaults;
teal = the plug you measured;
red dashed = what this dial moves

Widens the crossbar the cord hooks under at the top of the slot.

Default	Range	Step	Unit
11.1125	8 to 25	0.05	mm

Before: 11.1125. After: 16. Context: size = Custom.

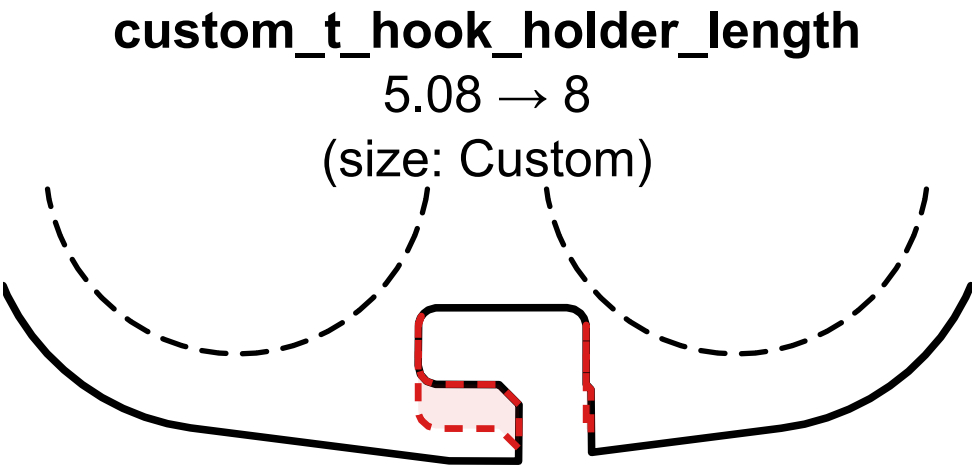
Only acts when size is Custom.

Moves: finger holes, hook, wing openings, zip-tie holes.

custom_t_hook_holder_length

One-sided puller · T Hook (Custom size only)

Hook crossbar height



black = the tool at its defaults;
teal = the plug you measured;
red dashed = what this dial moves

Makes the crossbar opening taller or shorter along the body.

Default	Range	Step	Unit
5.08	2 to 15	0.05	mm

Before: 5.08. After: 8. Context: size = Custom.

Only acts when size is Custom.

Moves: hook.

custom_t_hook_gap_offset

One-sided puller · T Hook (Custom size only)

Hook slot inset

custom_t_hook_gap_offset
This dial changes no shape.
Changes no shape: the J-hook cord catch is drawn without this dial; it only appears in the Cutouts Only 2D debug overlay.

Changes no shape of the printed tool, because the J-hook cord catch does not read this dial.

Default	Range	Step	Unit
0	0 to 8	0.5	mm

This dial changes no shape. Changes no shape: the J-hook cord catch is drawn without this dial; it only appears in the Cutouts Only 2D debug overlay.

custom_t_hook_leg_offset

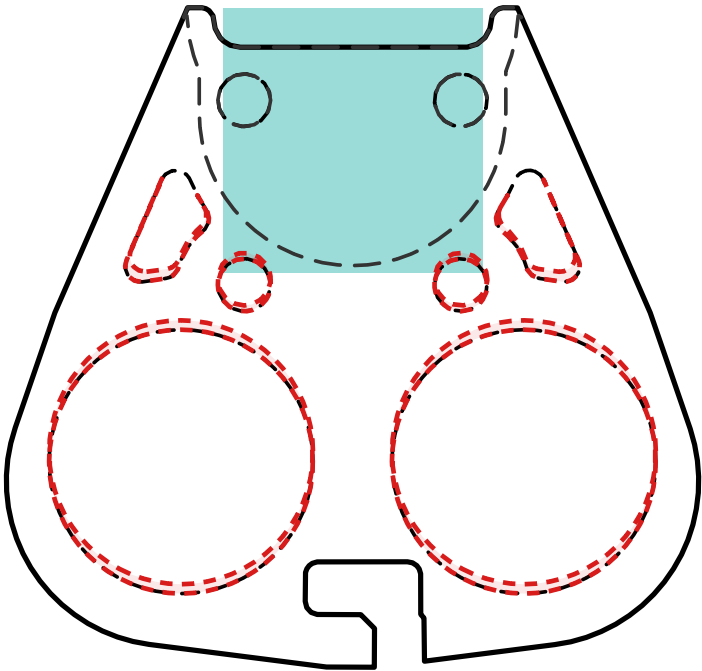
One-sided puller · T Hook (Custom size only)

Hook stem side offset

custom_t_hook_leg_offset

0 → 3

(size: Custom)



black = the tool at its defaults;
teal = the plug you measured;
red dashed = what this dial moves

Moves nothing on the hook itself, and only pushes the finger holes up when the auto-fit keep-out around the hook grows with it.

Default	Range	Step	Unit
0	-5 to 5	0.5	mm

Before: 0. After: 3. Context: size = Custom.

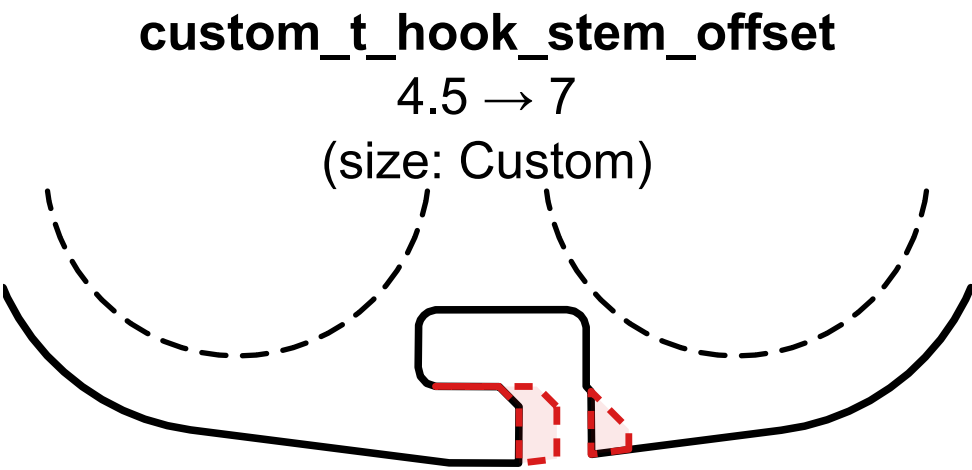
Only acts when size is Custom; the J-hook cord catch is drawn without this dial, and only the auto-fit clamp on the finger holes reads it.

Moves: finger holes, wing openings, zip-tie holes.

custom_t_hook_stem_offset

One-sided puller · T Hook (Custom size only)

Hook stem offset



black = the tool at its defaults;
teal = the plug you measured;
red dashed = what this dial moves

Shifts the cord slot sideways along the crossbar, making the J of the hook deeper or shallower.

Default	Range	Step	Unit
4.5	0 to 8	0.05	mm

Before: 4.5. After: 7. Context: size = Custom.

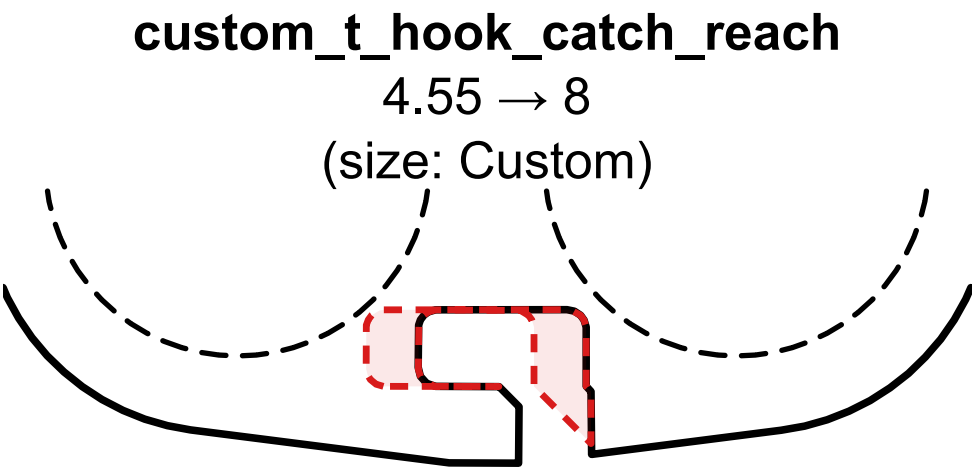
Only acts when size is Custom.

Moves: hook.

custom_t_hook_catch_reach

One-sided puller · T Hook (Custom size only)

Hook catch reach



black = the tool at its defaults;
teal = the plug you measured;
red dashed = what this dial moves

Extends the crossbar past the slot on the catch side, lengthening the lip the cord hooks under.

Default	Range	Step	Unit
4.55	0 to 10	0.05	mm

Before: 4.55. After: 8. Context: size = Custom.

Only acts when size is Custom.

Moves: hook.

custom_t_hook_tip_drop

One-sided puller · T Hook (Custom size only)

Hook tip drop

custom_t_hook_tip_drop

This dial changes no shape.

Only acts when size is Custom. Changes no shape: the hook slot's mouth already opens through the cord end (the body ends at $Y = 0$), so sagging the mouth farther below that edge cuts only air; measured at 1.98, 4 and 0.

Sags the slot's mouth farther below the cord end so a hooked cord cannot back out.

Default	Range	Step	Unit
1.98	0 to 5	0.05	mm

This dial changes no shape. Only acts when size is Custom. Changes no shape: the hook slot's mouth already opens through the cord end (the body ends at $Y = 0$), so sagging the mouth farther below that edge cuts only air; measured at 1.98, 4 and 0.

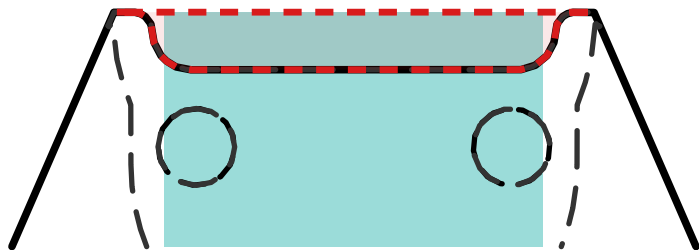
custom_enable_plug_wall_notch

One-sided puller · Plug Wall Notch (Custom size only)

Wall notch on or off

custom_enable_plug_wall_notch

true → false
(size: Custom)



black = the tool at its defaults;
teal = the plug you measured;
red dashed = what this dial moves

Cuts or leaves out the notch at the plug end that straddles the outlet's cover plate.

Default	Range	Step	Unit
true	true / false	—	—

Before: true. After: false. Context: size = Custom.

Only acts when size is Custom.

Moves: wall notch.

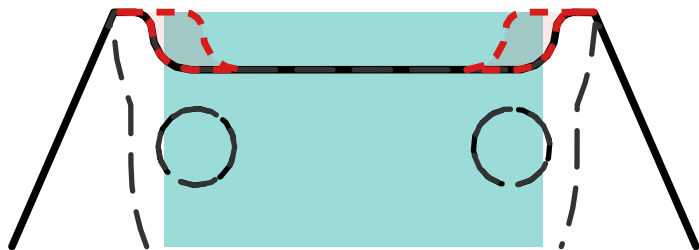
custom_plug_wall_notch_width

One-sided puller · Plug Wall Notch (Custom size only)

Wall notch width

custom_plug_wall_notch_width

26.67 → 20
(size: Custom)



black = the tool at its defaults;
teal = the plug you measured;
red dashed = what this dial moves

Widens or narrows the notch at the plug end.

Default	Range	Step	Unit
26.67	5 to 40	0.01	mm

Before: 26.67. After: 20. Context: size = Custom.

Only acts when size is Custom.

Moves: wall notch.

custom_plug_wall_notch_height

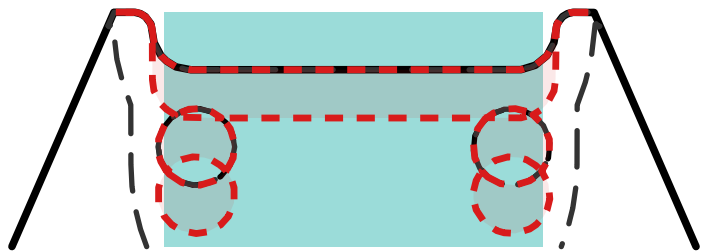
One-sided puller · Plug Wall Notch (Custom size only)

Wall notch depth

custom_plug_wall_notch_height

3.81 → 7

(size: Custom)



black = the tool at its defaults;
teal = the plug you measured;
red dashed = what this dial moves

Cuts the notch deeper or shallower into the plug end.

Default	Range	Step	Unit
3.81	0 to 10	0.05	mm

Before: 3.81. After: 7. Context: size = Custom.

Only acts when size is Custom.

Moves: wall notch, wing openings, zip-tie holes.

custom_plug_wall_notch_rounding

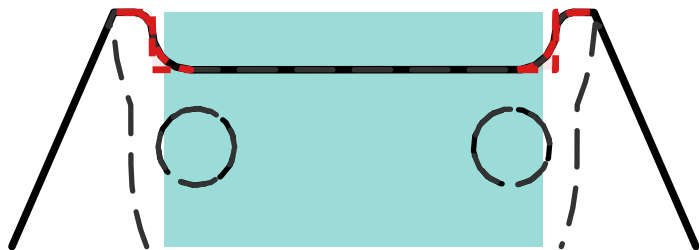
One-sided puller · Plug Wall Notch (Custom size only)

Wall notch corner rounding

custom_plug_wall_notch_rounding

2.54 → 0

(size: Custom)



black = the tool at its defaults;
teal = the plug you measured;
red dashed = what this dial moves

Rounds or squares the notch's two inner corners.

Default	Range	Step	Unit
2.54	0 to 5	0.01	mm

Before: 2.54. After: 0. Context: size = Custom.

Only acts when size is Custom.

Moves: wall notch.

custom_zip_tie_hole_diameter

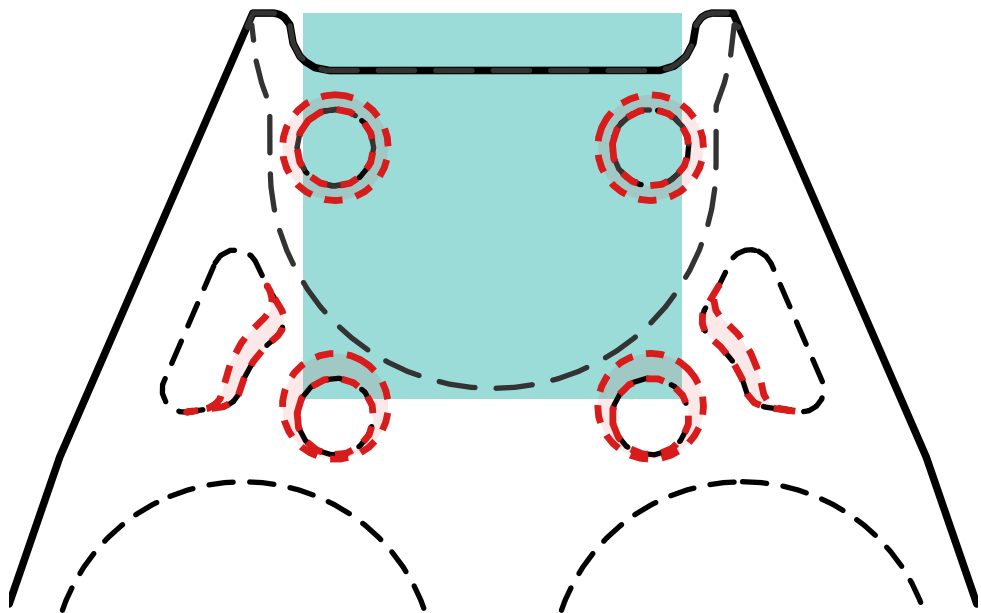
One-sided puller · Zip Tie Holes (Custom size only)

Zip-tie hole diameter

custom_zip_tie_hole_diameter

5.08 → 7

(size: Custom)



black = the tool at its defaults;
teal = the plug you measured;
red dashed = what this dial moves

Widens every zip-tie hole.

Default	Range	Step	Unit
5.08	2 to 8	0.02	mm

Before: 5.08. After: 7. Context: size = Custom.

Only acts when size is Custom.

Moves: wing openings, zip-tie holes.

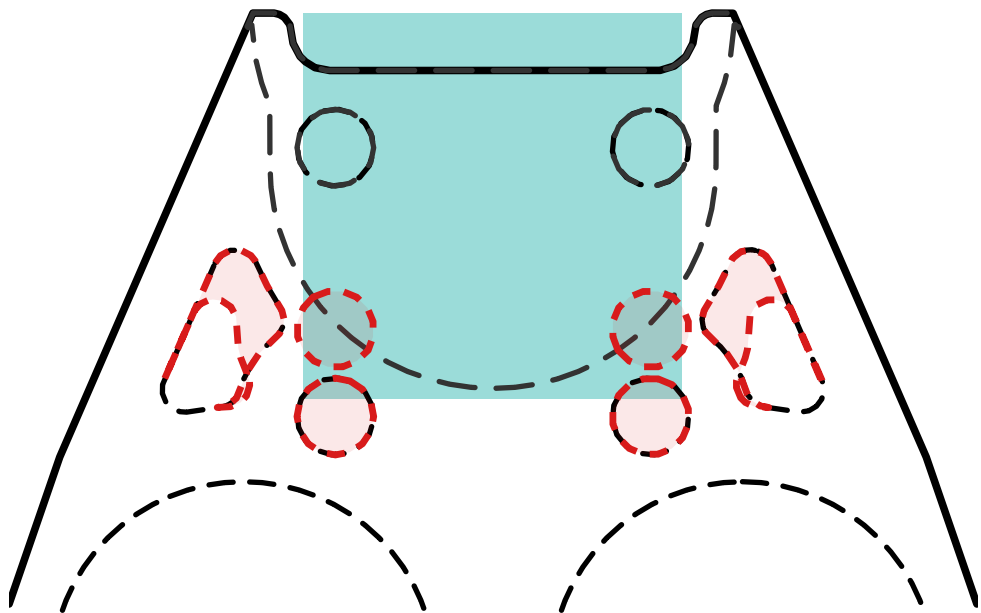
custom_zip_tie_height_spacing

One-sided puller · Zip Tie Holes (Custom size only)

Zip-tie row spacing

custom_zip_tie_height_spacing

17.78 → 12
(size: Custom)



black = the tool at its defaults;
teal = the plug you measured;
red dashed = what this dial moves

Moves the second row of zip-tie holes closer to or farther from the first.

Default	Range	Step	Unit
17.78	5 to 30	0.02	mm

Before: 17.78. After: 12. Context: size = Custom.

Only acts when size is Custom.

Moves: wing openings, zip-tie holes.

custom_zip_tie_width_spacing

One-sided puller · Zip Tie Holes (Custom size only)

Zip-tie column spacing

custom_zip_tie_width_spacing

This dial changes no shape.

Changes no shape of its own: the zip-tie columns sit zip_edge_offset inside the pocket wall; this dial only feeds the auto-fit finger keep-out, which trimmed the row spacing by 0.05 mm at the Custom defaults.

Moves no zip-tie hole, because the two columns follow the pocket wall, and only feeds the auto-fit keep-out that can trim the row spacing by a hair.

Default	Range	Step	Unit
17.7	10 to 50	0.02	mm

This dial changes no shape. Changes no shape of its own: the zip-tie columns sit zip_edge_offset inside the pocket wall; this dial only feeds the auto-fit finger keep-out, which trimmed the row spacing by 0.05 mm at the Custom defaults.

custom_zip_tie_distance_from_notch

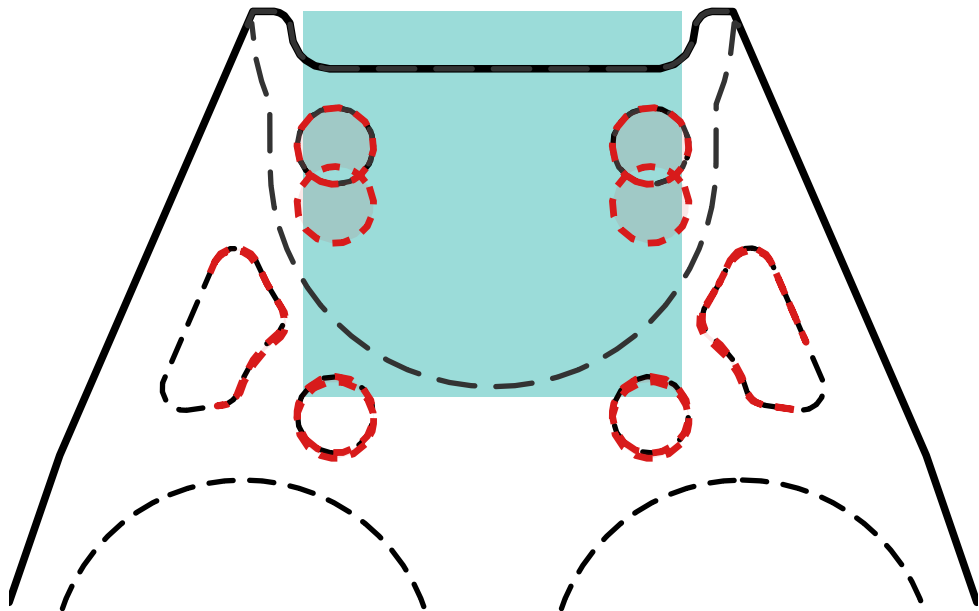
One-sided puller · Zip Tie Holes (Custom size only)

Zip-tie distance from the notch

custom_zip_tie_distance_from_notch

5.1 → 9

(size: Custom)



black = the tool at its defaults;
teal = the plug you measured;
red dashed = what this dial moves

Moves the first row of zip-tie holes farther from the wall notch, and the second row follows.

Default	Range	Step	Unit
5.1	1 to 15	0.05	mm

Before: 5.1. After: 9. Context: size = Custom.

Only acts when size is Custom.

Moves: wing openings, zip-tie holes.

custom_zip_tie_countersink

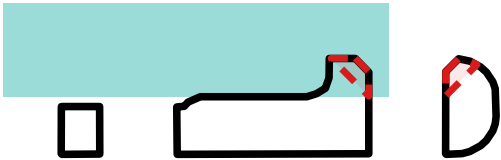
One-sided puller · Zip Tie Holes (Custom size only)

Zip-tie countersink

custom_zip_tie_countersink

0.9 → 2.5

(size: Custom, section-x: 10.4)



black = the tool at its defaults;
teal = the plug you measured;
red dashed = what this dial moves

Flares the top of the zip-tie holes wider so the tie head sits flush.

Default	Range	Step	Unit
0.9	0 to 3	0.05	mm

Before: 0.9. After: 2.5. Context: size = Custom. Section at x = 10.4 mm.

Only acts when size is Custom.

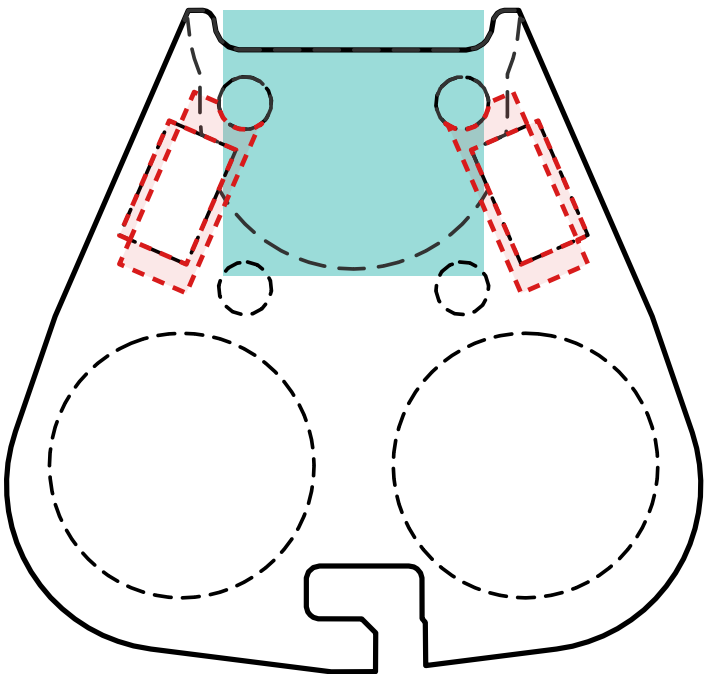
Moves: pocket, zip-tie holes.

custom_velcro_hole_length

One-sided puller · Velcro / Wing Strap Holes (Custom size only)

Strap slot length

custom_velcro_hole_length
12 → 18
(size: Custom, velcro_style: Classic slot)



black = the tool at its defaults;
teal = the plug you measured;
red dashed = what this dial moves

Lengthens each plain strap slot along its lean.

Default	Range	Step	Unit
12	6 to 20	0.5	mm

Before: 12. After: 18. Context: size = Custom, velcro_style = Classic slot.

Only acts when size is Custom and velcro_style is Classic slot.

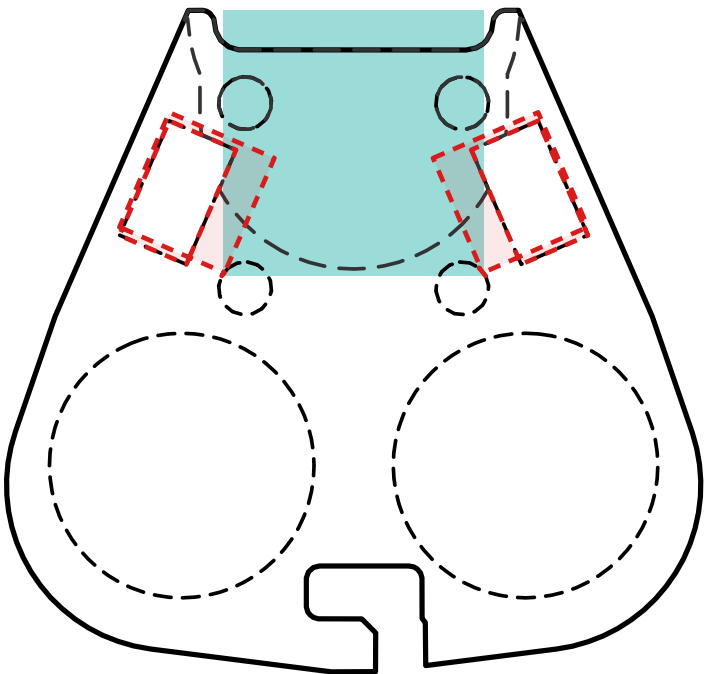
Moves: classic slots.

custom_velcro_hole_width

One-sided puller · Velcro / Wing Strap Holes (Custom size only)

Strap slot width

custom_velcro_hole_width
7 → 11
(size: Custom, velcro_style: Classic slot)



black = the tool at its defaults;
teal = the plug you measured;
red dashed = what this dial moves

Widens each plain strap slot.

Default	Range	Step	Unit
7	3 to 14	0.5	mm

Before: 7. After: 11. Context: size = Custom, velcro_style = Classic slot.

Only acts when size is Custom and velcro_style is Classic slot.

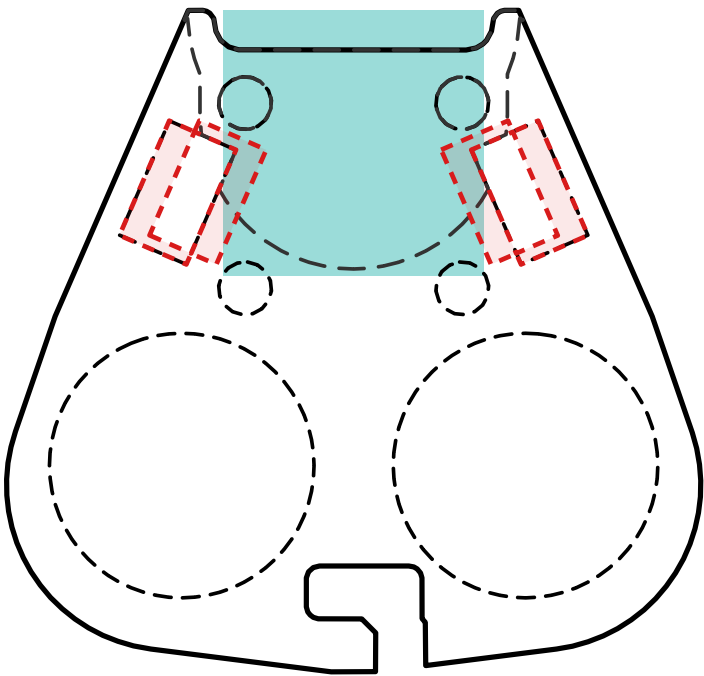
Moves: classic slots.

custom_velcro_hole_x_center

One-sided puller · Velcro / Wing Strap Holes (Custom size only)

Strap slot distance from the centerline

custom_velcro_hole_x_center
19.4 → 14
(size: Custom, velcro_style: Classic slot)



black = the tool at its defaults;
teal = the plug you measured;
red dashed = what this dial moves

Moves the two plain strap slots closer to or farther from the centerline.

Default	Range	Step	Unit
19.4	5 to 35	0.05	mm

Before: 19.4. After: 14. Context: size = Custom, velcro_style = Classic slot.

Only acts when size is Custom and velcro_style is Classic slot.

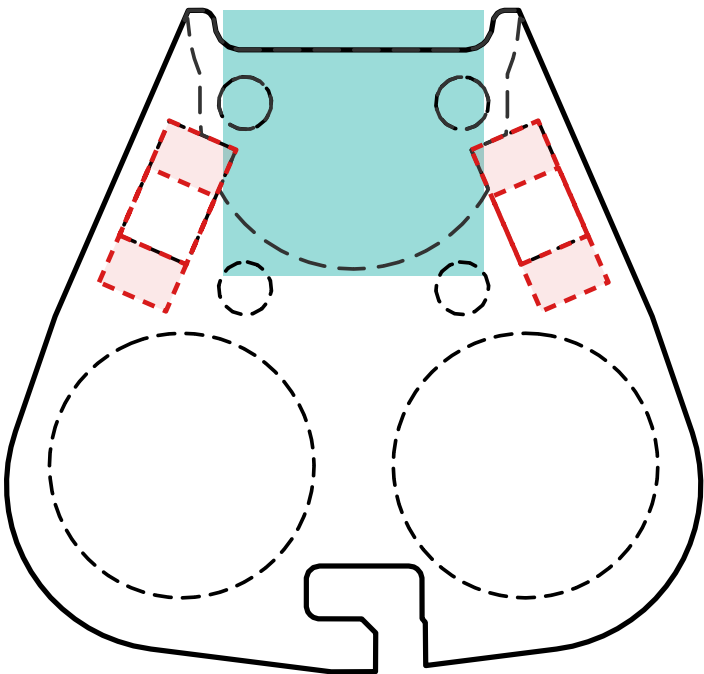
Moves: classic slots.

custom_velcro_hole_y_center

One-sided puller · Velcro / Wing Strap Holes (Custom size only)

Strap slot position along the body

custom_velcro_hole_y_center
46 → 38
(size: Custom, velcro_style: Classic slot)



black = the tool at its defaults;
teal = the plug you measured;
red dashed = what this dial moves

Slides the two plain strap slots toward the cord end or the plug end.

Default	Range	Step	Unit
46	30 to 80	0.5	mm

Before: 46. After: 38. Context: size = Custom, velcro_style = Classic slot.

Only acts when size is Custom and velcro_style is Classic slot.

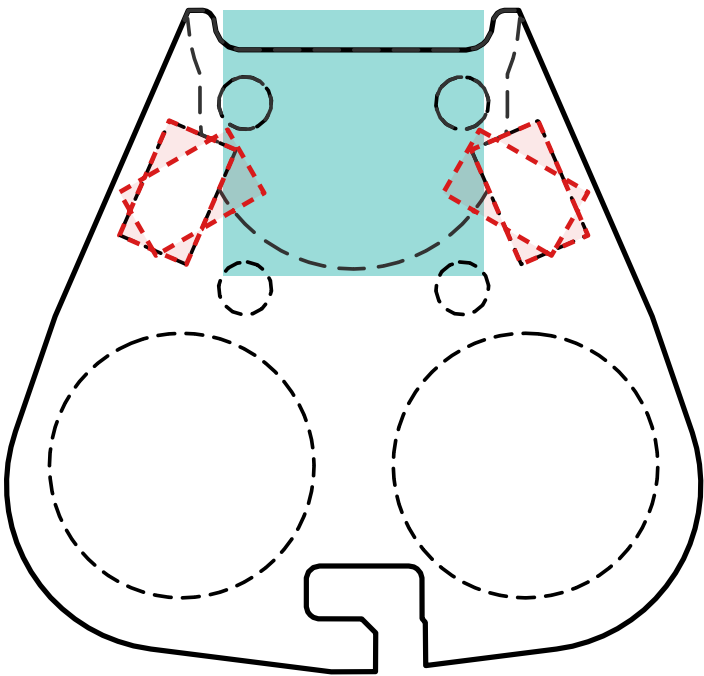
Moves: classic slots.

custom_velcro_hole_rotation

One-sided puller · Velcro / Wing Strap Holes (Custom size only)

Strap slot lean

custom_velcro_hole_rotation
23.5 → 60
(size: Custom, velcro_style: Classic slot)



black = the tool at its defaults;
teal = the plug you measured;
red dashed = what this dial moves

Leans the two plain strap slots more or less steeply, mirrored on the two sides.

Default	Range	Step	Unit
23.5	0 to 180	0.5	deg

Before: 23.5. After: 60. Context: size = Custom, velcro_style = Classic slot.

Only acts when size is Custom and velcro_style is Classic slot.

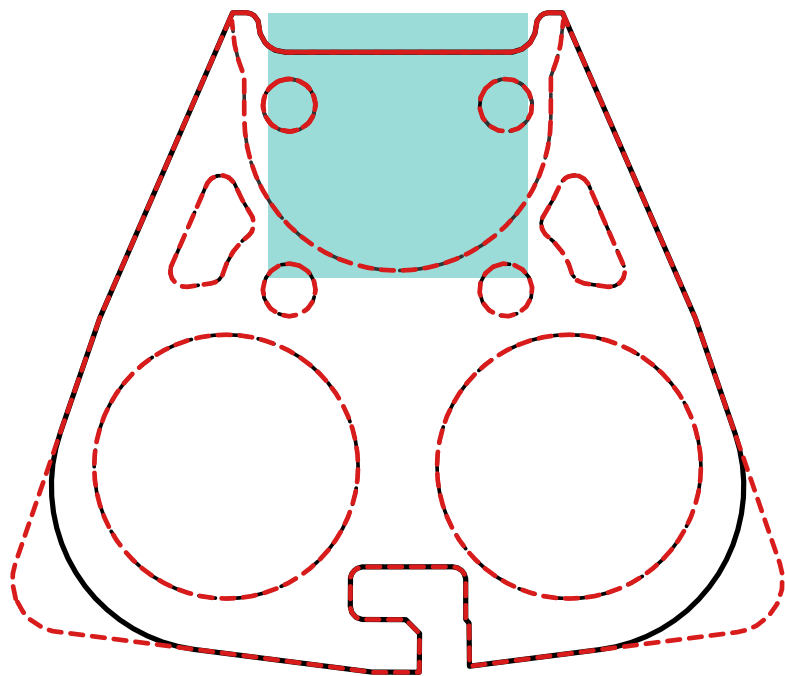
Moves: classic slots.

custom_body_side_rounding

One-sided puller · Edge Rounding (Custom size only)

Body side rounding

custom_body_side_rounding
15.85 → 5
(size: Custom)



black = the tool at its defaults;
teal = the plug you measured;
red dashed = what this dial moves

Rounds the body outline's corners more or less, down to the plain octagon at 0.

Default	Range	Step	Unit
15.85	0 to 30	0.05	mm

Before: 15.85. After: 5. Context: size = Custom.

Only acts when size is Custom.

Moves: body edge.

custom_body_top_rounding

One-sided puller · Edge Rounding (Custom size only)

Top edge rounding

custom_body_top_rounding
2.54 → 0
(size: Custom, section-y: 35)



black = the tool at its defaults;
teal = the plug you measured;
red dashed = what this dial moves

Rounds or squares the edge where the body's top face meets its sides.

Default	Range	Step	Unit
2.54	0 to 5	0.01	mm

Before: 2.54. After: 0. Context: size = Custom. Section at y = 35 mm.

Only acts when size is Custom.

Moves: body edge.

custom_body_bottom_rounding

One-sided puller · Edge Rounding (Custom size only)

Bottom edge rounding

custom_body_bottom_rounding
0 → 2
(size: Custom, section-y: 35)



black = the tool at its defaults;
teal = the plug you measured;
red dashed = what this dial moves

Rounds or squares the edge where the body's bottom face meets its sides.

Default	Range	Step	Unit
0	0 to 5	0.1	mm

Before: 0. After: 2. Context: size = Custom. Section at y = 35 mm.

Only acts when size is Custom.

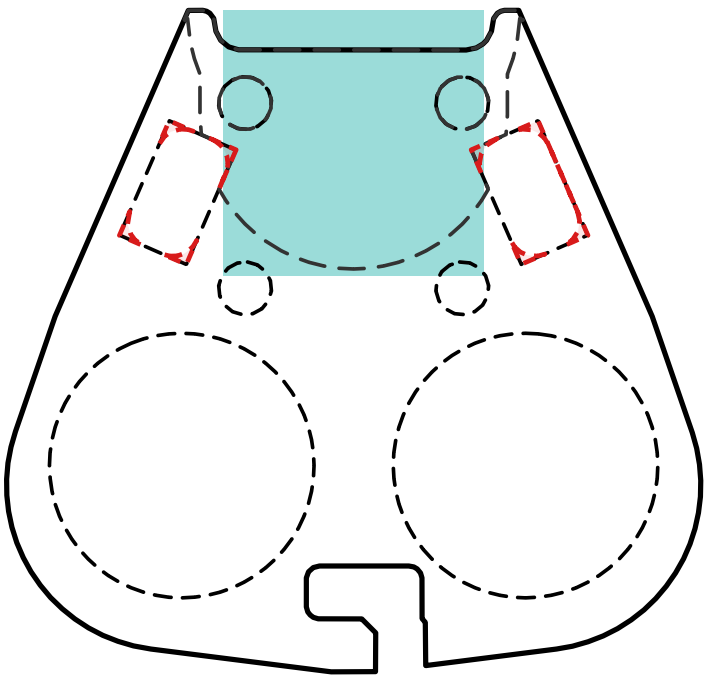
Moves: body edge.

custom_velcro_side_rounding

One-sided puller · Edge Rounding (Custom size only)

Strap slot corner rounding

custom_velcro_side_rounding
0 → 2.5
(size: Custom, velcro_style: Classic slot)



black = the tool at its defaults;
teal = the plug you measured;
red dashed = what this dial moves

Rounds the four corners of each plain strap slot.

Default	Range	Step	Unit
0	0 to 3	0.5	mm

Before: 0. After: 2.5. Context: size = Custom, velcro_style = Classic slot.

Only acts when size is Custom and velcro_style is Classic slot.

Moves: classic slots.

custom_velcro_top_bottom_rounding

One-sided puller · Edge Rounding (Custom size only)

Strap opening edge rounding

custom_velcro_top_bottom_rounding

0 → 1.5

(size: Custom, section-y: 46)



black = the tool at its defaults;
teal = the plug you measured;
red dashed = what this dial moves

Flares the top and bottom edges of the strap openings so the strap does not chafe.

Default	Range	Step	Unit
0	0 to 3	0.1	mm

Before: 0. After: 1.5. Context: size = Custom. Section at y = 46 mm.

Only acts when size is Custom.

Moves: pocket, wing openings.

custom_finger_hole_rounding

One-sided puller · Edge Rounding (Custom size only)

Finger hole rim rounding

custom_finger_hole_rounding

2.5 → 0

(size: Custom, section-y: 19.8)



black = the tool at its defaults;
teal = the plug you measured;
red dashed = what this dial moves

Rounds or squares the rims of both finger holes on both faces.

Default	Range	Step	Unit
2.5	0 to 5	0.1	mm

Before: 2.5. After: 0. Context: size = Custom. Section at y = 19.8 mm.

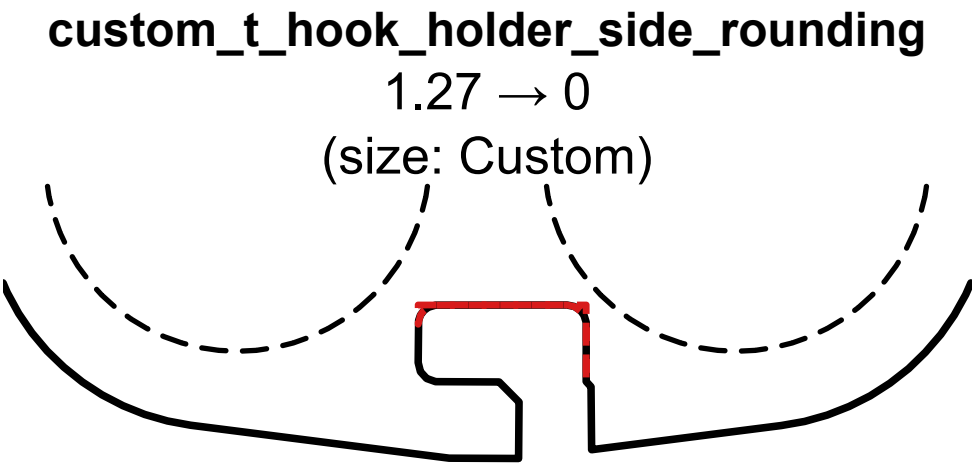
Only acts when size is Custom.

Moves: finger holes.

custom_t_hook_holder_side_rounding

One-sided puller · Edge Rounding (Custom size only)

Hook crossbar corner rounding



black = the tool at its defaults;
teal = the plug you measured;
red dashed = what this dial moves

Rounds or squares the top corners of the hook's crossbar.

Default	Range	Step	Unit
1.27	0 to 3	0.01	mm

Before: 1.27. After: 0. Context: size = Custom.

Only acts when size is Custom.

Moves: hook.

custom_t_hook_gap_side_rounding

One-sided puller · Edge Rounding (Custom size only)

Hook slot corner rounding

custom_t_hook_gap_side_rounding

This dial changes no shape.

Changes no shape: the J-hook cord catch is drawn without this dial; it only appears in the Cutouts Only 2D debug overlay.

Changes no shape of the printed tool, because the J-hook cord catch draws its own corners.

Default	Range	Step	Unit
0	0 to 3	0.1	mm

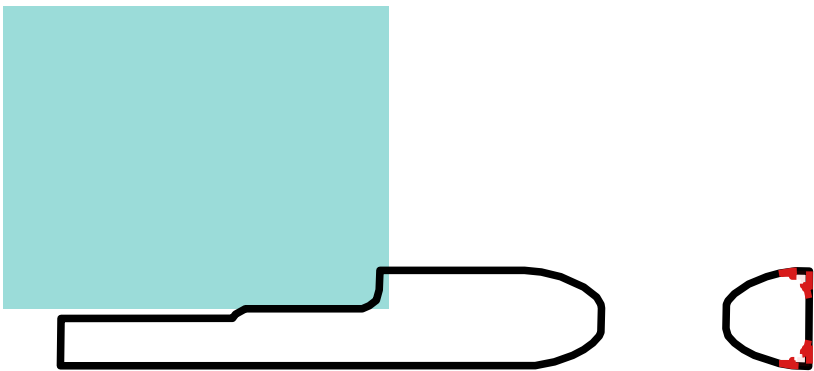
This dial changes no shape. Changes no shape: the J-hook cord catch is drawn without this dial; it only appears in the Cutouts Only 2D debug overlay.

custom_t_hook_top_bottom_rounding

One-sided puller · Edge Rounding (Custom size only)

Hook edge rounding

custom_t_hook_top_bottom_rounding
0 → 2
(size: Custom, section-x: 4.5)



black = the tool at its defaults;
teal = the plug you measured;
red dashed = what this dial moves

Flares the top and bottom edges of the cord hook's slot and crossbar.

Default	Range	Step	Unit
0	0 to 3	0.1	mm

Before: 0. After: 2. Context: size = Custom. Section at x = 4.5 mm.

Only acts when size is Custom.

Moves: hook.

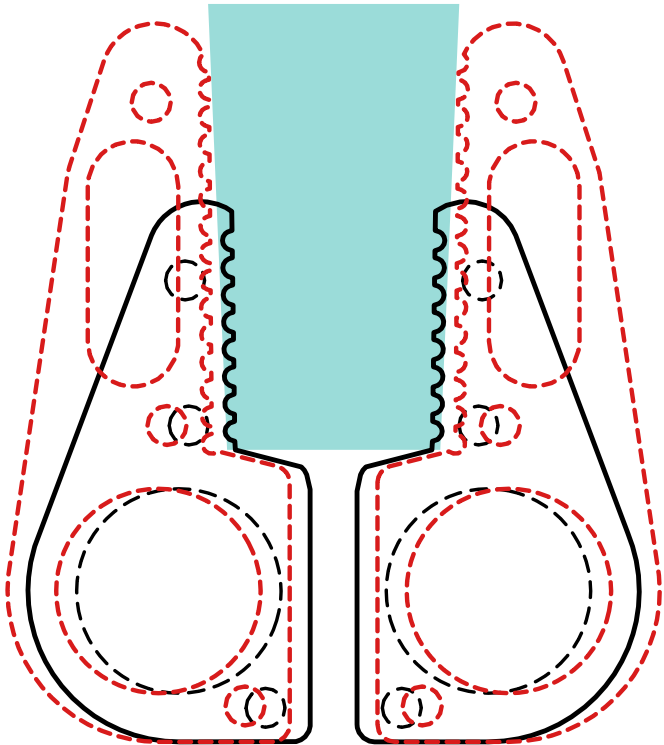
plug_preset

Two-sided puller · Step 1 - Your Plug

Plug preset

plug_preset

Measure my plug → Heavy-duty extension cord - NEMA 5-15



black = the tool at its defaults;
teal = the plug you measured;
red dashed = what this dial moves

Fills in the plug's length, both widths and the cord from the measured heavy-duty cord plug, so the arms, the grip gap, the cord channel and the plate length all take that plug's shape at once.

Default	Range	Step	Unit
Measure my plug	5 options	—	—

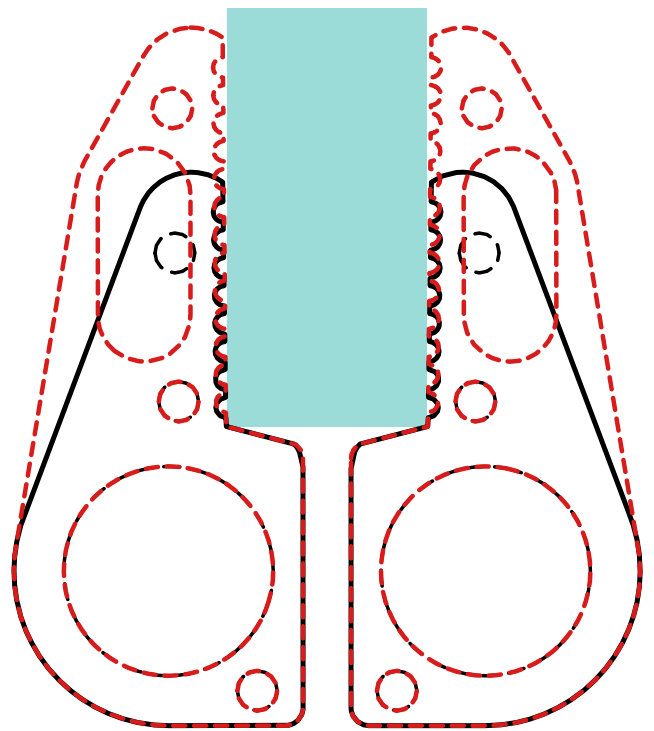
Options: Measure my plug, Heavy-duty extension cord - NEMA 5-15, USB-C laptop tip, Flat 2-prong lamp plug - NEMA 1-15, Standard 3-prong plug - NEMA 5-15.
Before: Measure my plug. After: Heavy-duty extension cord - NEMA 5-15.
Moves: arms, finger lobes, plate edge, zip stations.

measure_plug_length

Two-sided puller · Step 1 - Your Plug

Plug length

measure_plug_length
25.5 → 40



black = the tool at its defaults;
teal = the plug you measured;
red dashed = what this dial moves

Lengthens both arms so the teeth cover the whole plug body, and moves the zip-tie stations and the strap slot with them.

Default	Range	Step	Unit
25.5	12 to 85	0.5	mm

Before: 25.5. After: 40.

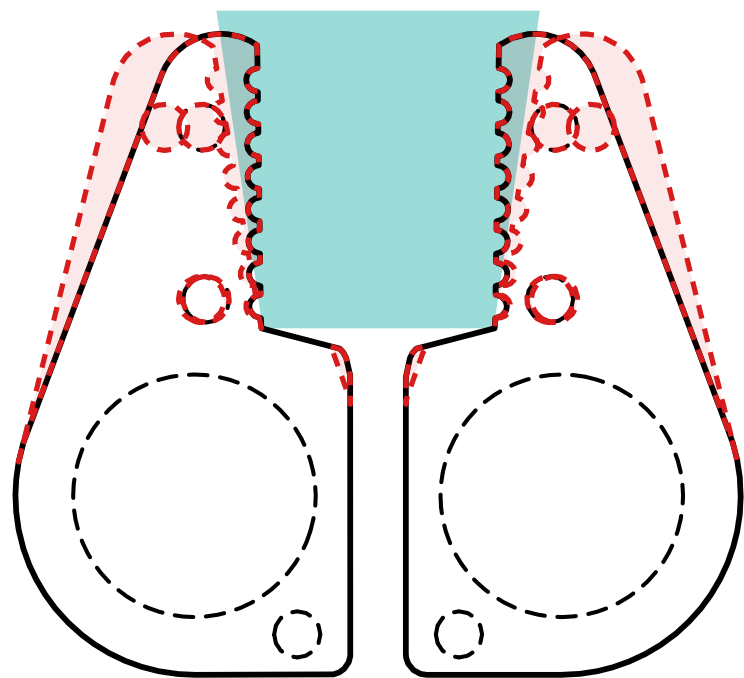
Moves: arms.

measure_plug_width_prong_end

Two-sided puller · Step 1 - Your Plug

Plug width at the prong end

measure_plug_width_prong_end
20 → 28



black = the tool at its defaults;
teal = the plug you measured;
red dashed = what this dial moves

Opens or closes the gap between the arms at their tips, where the plug's prong end sits.

Default	Range	Step	Unit
20	5 to 40	0.5	mm

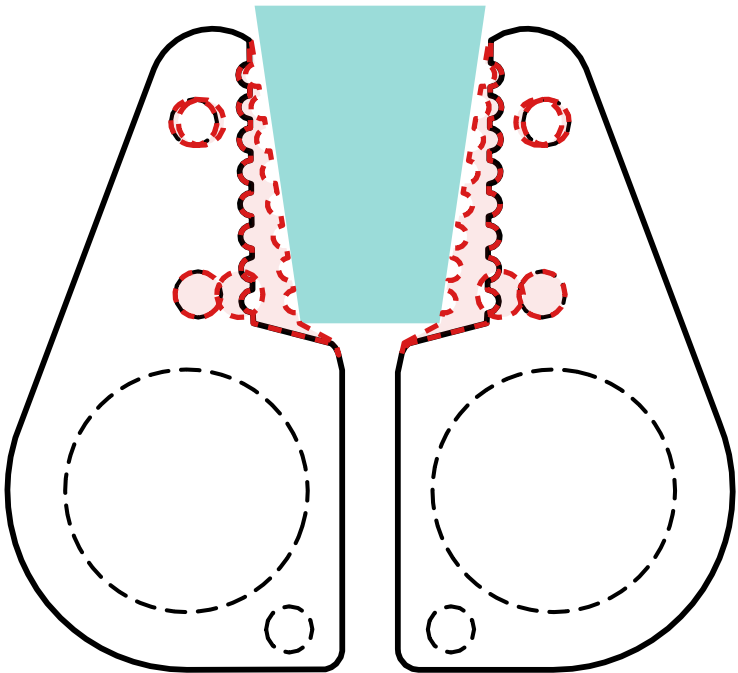
Before: 20. After: 28.

Moves: arms, cord channel, finger lobes, zip stations.

measure_plug_width_cord_end

Two-sided puller · Step 1 - Your Plug
Plug width at the cord end

measure_plug_width_cord_end
20 → 12



black = the tool at its defaults;
teal = the plug you measured;
red dashed = what this dial moves

Opens or closes the gap between the arms at the plug's back end, so the arms taper to match the plug.

Default	Range	Step	Unit
20	5 to 40	0.5	mm

Before: 20. After: 12.

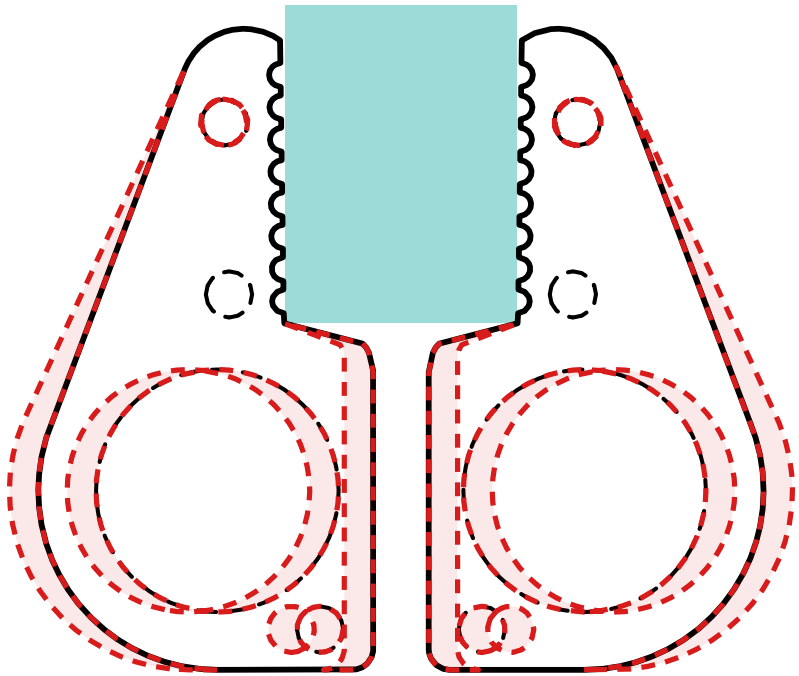
Moves: arms, teeth, zip stations.

measure_cord_thickness

Two-sided puller · Step 1 - Your Plug

Cord thickness

measure_cord_thickness
4 → 9



black = the tool at its defaults;
teal = the plug you measured;
red dashed = what this dial moves

Widens the cord channel between the finger holes, and the finger lobes move outward with it.

Default	Range	Step	Unit
4	1.5 to 12	0.5	mm

Before: 4. After: 9.

Moves: arms, cord channel, finger lobes, zip stations.

plug_sides

Two-sided puller · Step 1 - Your Plug

Plug sides

plug_sides
Rounded sides → Flat sides
(section-y: 45)



black = the tool at its defaults;
teal = the plug you measured;
red dashed = what this dial moves

Removes the sloped cradle from the arms' gripping edges so the teeth bite straight along a flat-sided plug.

Default	Range	Step	Unit
Rounded sides	2 options	—	—

Options: Rounded sides, Flat sides.
Before: Rounded sides. After: Flat sides. Section at y = 45 mm.
Moves: arms, teeth.

show_plug_preview

Two-sided puller · Step 1 - Your Plug

Show the see-through plug

show_plug_preview

This dial changes no shape.

Changes no shape: the see-through plug is a preview aid and is never exported.

Shows or hides the see-through plug in the preview and changes nothing in the printed plates.

Default	Range	Step	Unit
true	true / false	—	—

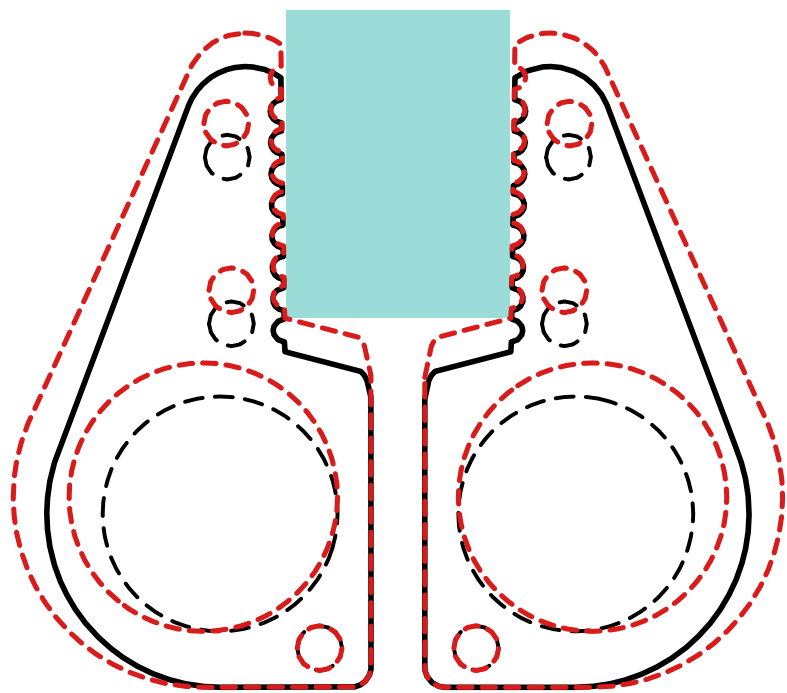
This dial changes no shape. Changes no shape: the see-through plug is a preview aid and is never exported.

size

Two-sided puller · Step 2 - Size

Hand size

size
Medium → Large



black = the tool at its defaults;
teal = the plug you measured;
red dashed = what this dial moves

Widens both finger holes and their lobes, so the whole plate grows around them.

Default	Range	Step	Unit
Medium	4 options	—	—

Options: Small, Medium, Large, Measure my hand.

Before: Medium. After: Large.

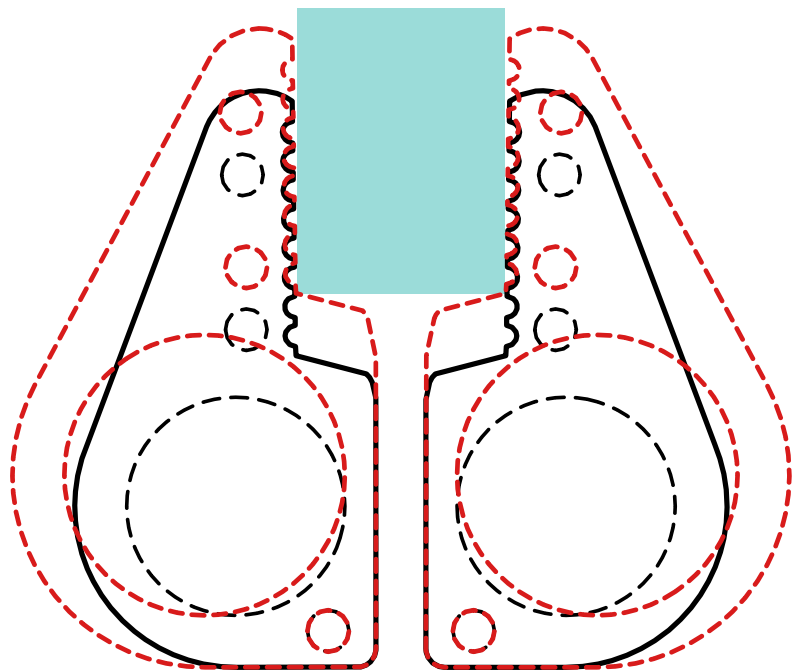
Moves: arms, finger lobes, zip stations.

measure_finger_width

Two-sided puller · Step 2 - Size

Finger width

measure_finger_width
20 → 26
(size: Measure my hand)



black = the tool at its defaults;
teal = the plug you measured;
red dashed = what this dial moves

Widens both finger holes and their lobes.

Default	Range	Step	Unit
20	14 to 32	0.5	mm

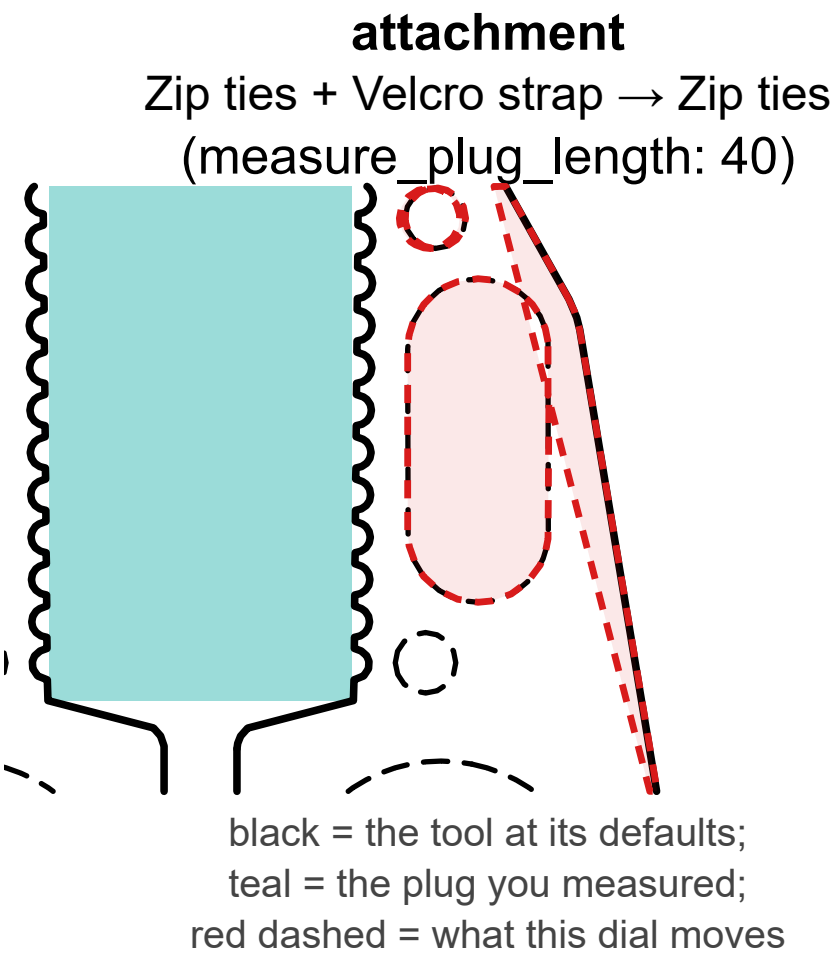
Before: 20. After: 26. Context: size = Measure my hand.

Only acts when size is Measure my hand.

Moves: arms, finger lobes, plate edge, zip stations.

attachment

Two-sided puller · Step 3 - Attachment
Attachment



Removes the strap slot from each arm, while the zip-tie stations stay because they hold the plates together.

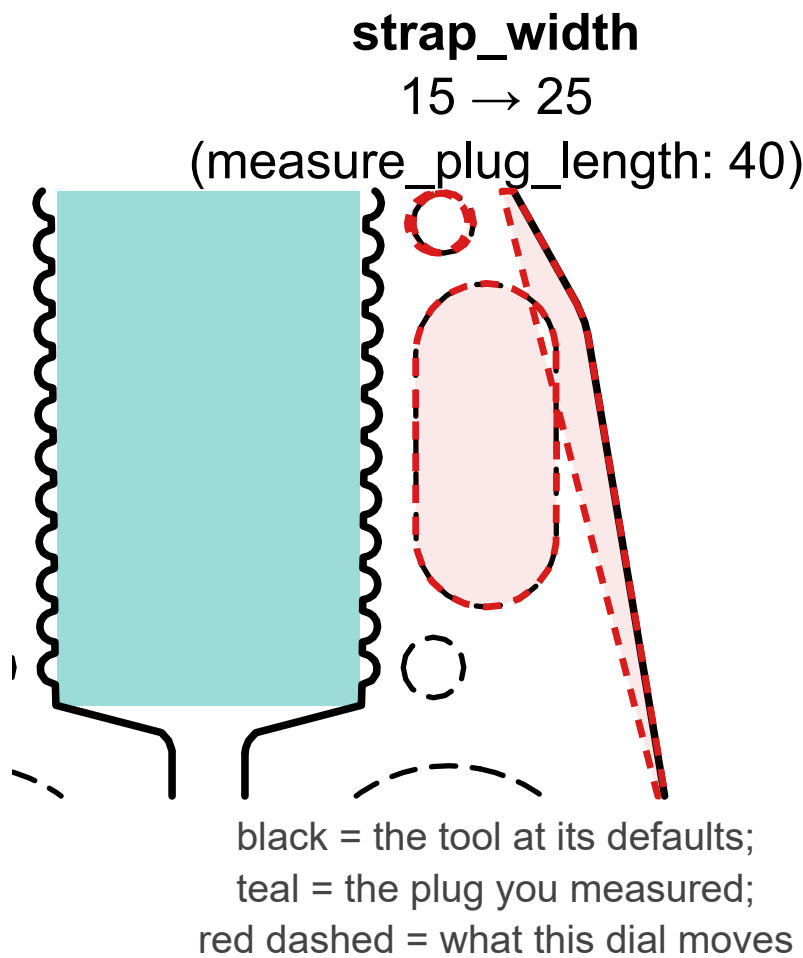
Default	Range	Step	Unit
Zip ties + Velcro strap	2 options	—	—

Options: Zip ties + Velcro strap, Zip ties.
Before: Zip ties + Velcro strap. After: Zip ties. Context: measure_plug_length = 40.
Drawn on a 40 mm plug: on the default 25.5 mm plug the arms are too short for a strap slot.
Moves: arms, strap slot, zip stations.

strap_width

Two-sided puller · Step 3 - Attachment

Strap width



Sets the strap the arm slot must clear, and when the slot's window between the zip-tie stations is shorter than the strap plus 1.5 mm the slot is left out.

Default	Range	Step	Unit
15	10 to 25	1	mm

Before: 15. After: 25. Context: measure_plug_length = 40.

Drawn on a 40 mm plug: on the default 25.5 mm plug the arms are too short for a strap slot.

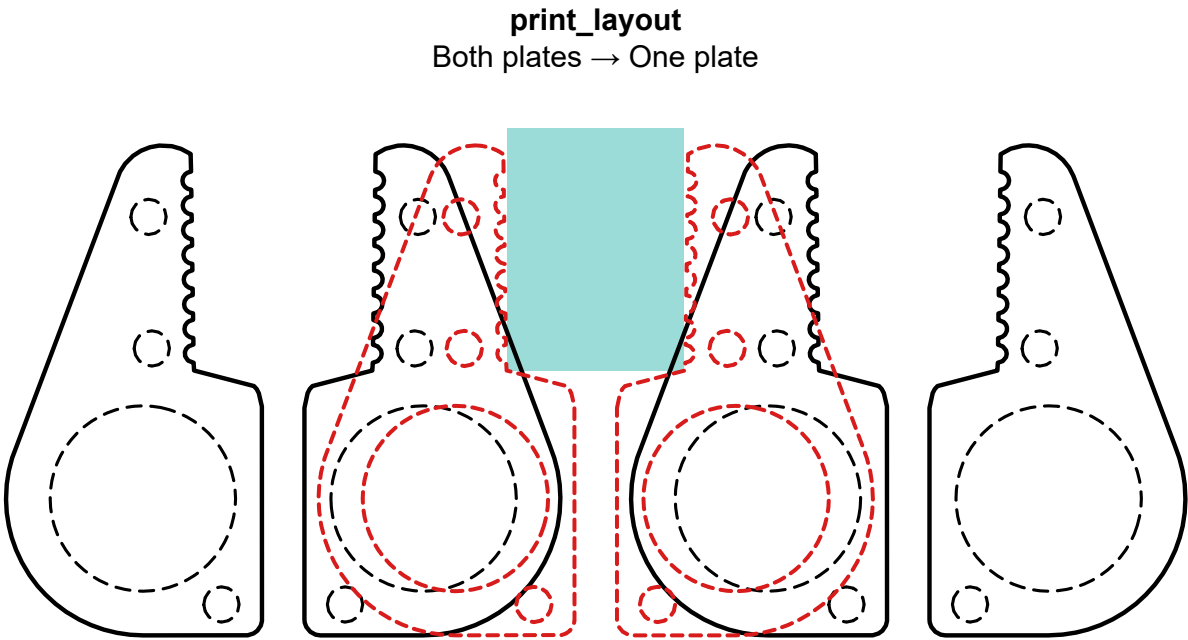
Moves: arms, strap slot, zip stations.

Can trip: STRAP WIDER THAN ARM SLOT WINDOW - NARROW THE STRAP.

print_layout

Two-sided puller · Step 4 - Print Layout

Print layout



black = the tool at its defaults; teal = the plug you measured; red dashed = what this dial moves

Puts both identical plates side by side in one file, or just one plate.

Default	Range	Step	Unit
Both plates	2 options	—	—

Options: Both plates, One plate.

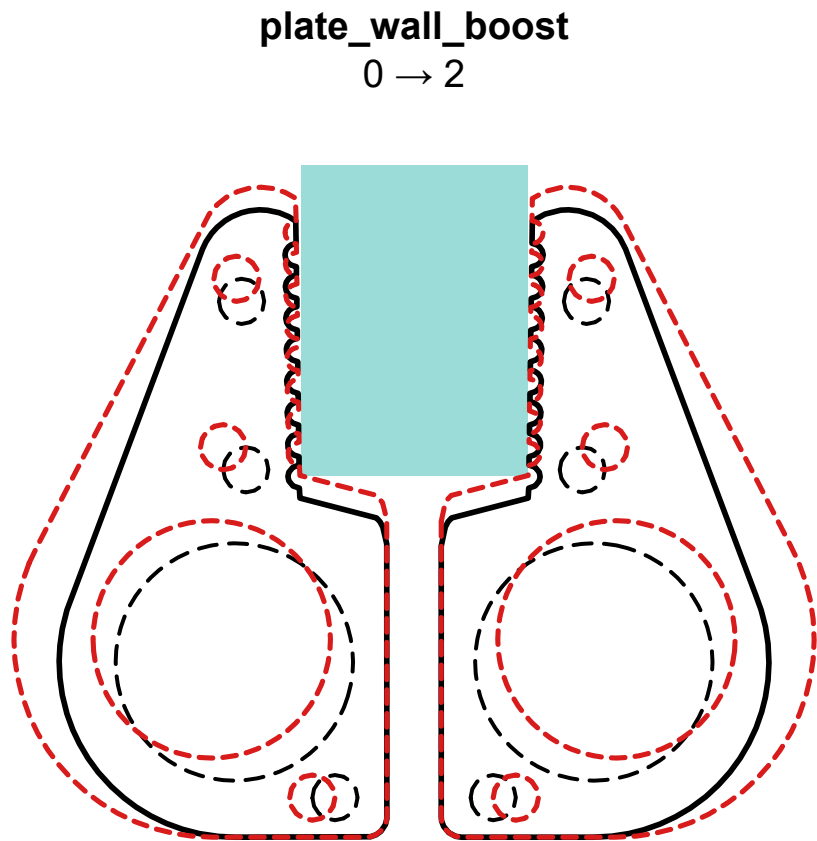
Before: Both plates. After: One plate.

Moves: arms, finger lobes, plate edge, zip stations.

plate_wall_boost

Two-sided puller · Advanced - Two-Sided Puller

Extra wall on the plate



black = the tool at its defaults;
teal = the plug you measured;
red dashed = what this dial moves

Thickens every wall around the finger holes, the cord channel, the zip-tie holes and the strap slot at once, so the plate outline grows and the openings shift to keep their walls.

Default	Range	Step	Unit
0	0 to 5	0.25	mm

Before: 0. After: 2.

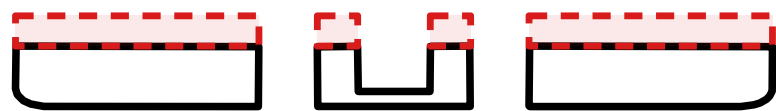
Moves: arms, finger lobes, zip stations.

plate_thickness

Two-sided puller · Advanced - Two-Sided Puller

Plate thickness

plate_thickness
4 → 6
(section-y: 3)



black = the tool at its defaults;
teal = the plug you measured;
red dashed = what this dial moves

Thickens each plate, so the finished pair is twice as thick.

Default	Range	Step	Unit
4	2 to 8	0.25	mm

Before: 4. After: 6. Section at y = 3 mm.

Moves: cord channel, finger lobes, zip stations.

Can trip: PLATE THINNER THAN 2MM - TOO FLIMSY.

plate_grip_bite

Two-sided puller · Advanced - Two-Sided Puller

Grip bite

plate_grip_bite

-1 → -2

(plug_sides: Flat sides, section-y: 45)



black = the tool at its defaults;
teal = the plug you measured;
red dashed = what this dial moves

Squeezes the plug harder or softer, because the arms close in by this much per side against the plug's width.

Default	Range	Step	Unit
-1	-2 to 2	0.1	mm

Before: -1. After: -2. Context: plug_sides = Flat sides. Section at y = 45 mm.

Only acts when plug_sides is Flat sides or a plug preset is chosen.

Moves: arms, teeth.

Can trip: NO GRIP BITE - PLUG WONT BE HELD.

plate_cradle_depth

Two-sided puller · Advanced - Two-Sided Puller

Cradle depth

plate_cradle_depth

2.5 → 0

(plug_sides: Rounded sides, section-y: 45)



black = the tool at its defaults;
teal = the plug you measured;
red dashed = what this dial moves

Slopes each arm's gripping edge from the mating face down to the outer face, so two plates form a cradle that centers a round plug.

Default	Range	Step	Unit
2.5	0 to 3.5	0.25	mm

Before: 2.5. After: 0. Context: plug_sides = Rounded sides. Section at y = 45 mm.

Only acts when plug_sides is Rounded sides.

Moves: arms, teeth.

Can trip: CRADLE SHALLOWER THAN ASKED - PLUG NARROW.

plate_grip_clearance

Two-sided puller · Advanced - Two-Sided Puller

Grip clearance

plate_grip_clearance

0.5 → 2

(plug_sides: Rounded sides, section-y: 45)



black = the tool at its defaults;
teal = the plug you measured;
red dashed = what this dial moves

Opens the gap between the arms where the plates meet, so a hard round plug can drop in flat before the cradle holds it.

Default	Range	Step	Unit
0.5	0 to 2	0.1	mm

Before: 0.5. After: 2. Context: plug_sides = Rounded sides. Section at y = 45 mm.

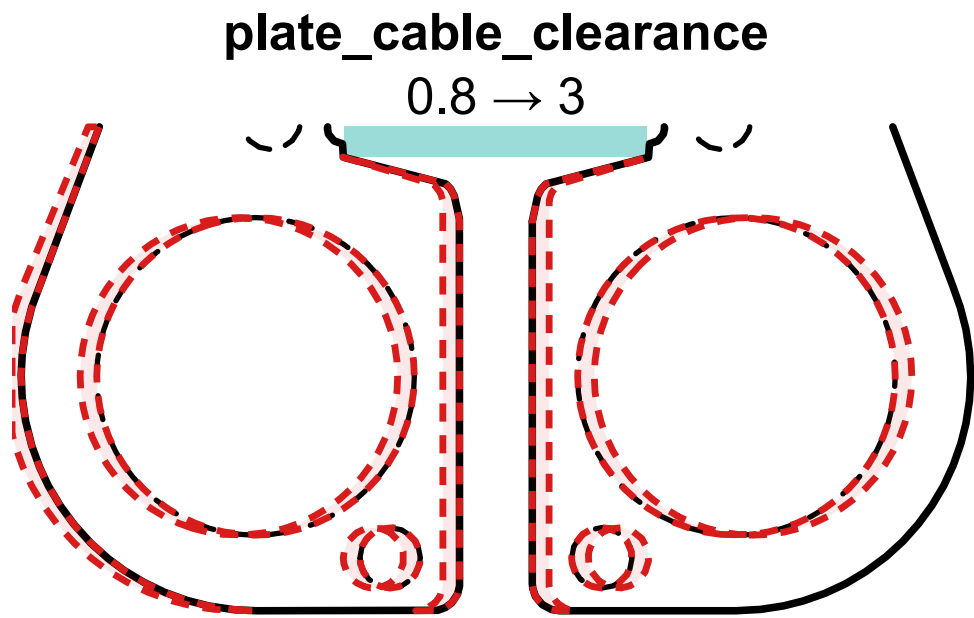
Only acts when plug_sides is Rounded sides.

Moves: arms, teeth.

plate_cable_clearance

Two-sided puller · Advanced - Two-Sided Puller

Cord channel clearance



black = the tool at its defaults;
teal = the plug you measured;
red dashed = what this dial moves

Widens the cord channel between the finger holes beyond the cord's thickness.

Default	Range	Step	Unit
0.8	0 to 5	0.1	mm

Before: 0.8. After: 3.

Moves: arms, cord channel, finger lobes, zip stations.

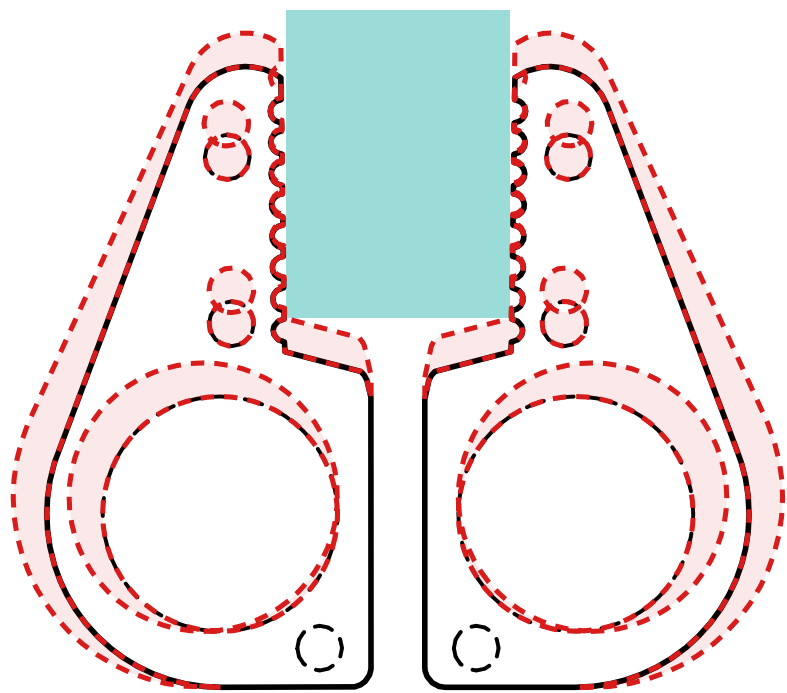
Can trip: CORD TOO THICK FOR CABLE CHANNEL; PLUG NARROWER THAN THE CORD CHANNEL - ARMS CANNOT TOUCH IT.

plate_finger_fit

Two-sided puller · Advanced - Two-Sided Puller

Finger hole fit

plate_finger_fit
1 → 4



black = the tool at its defaults;
teal = the plug you measured;
red dashed = what this dial moves

Widens both finger holes beyond your finger width.

Default	Range	Step	Unit
1	0 to 8	0.25	mm

Before: 1. After: 4.

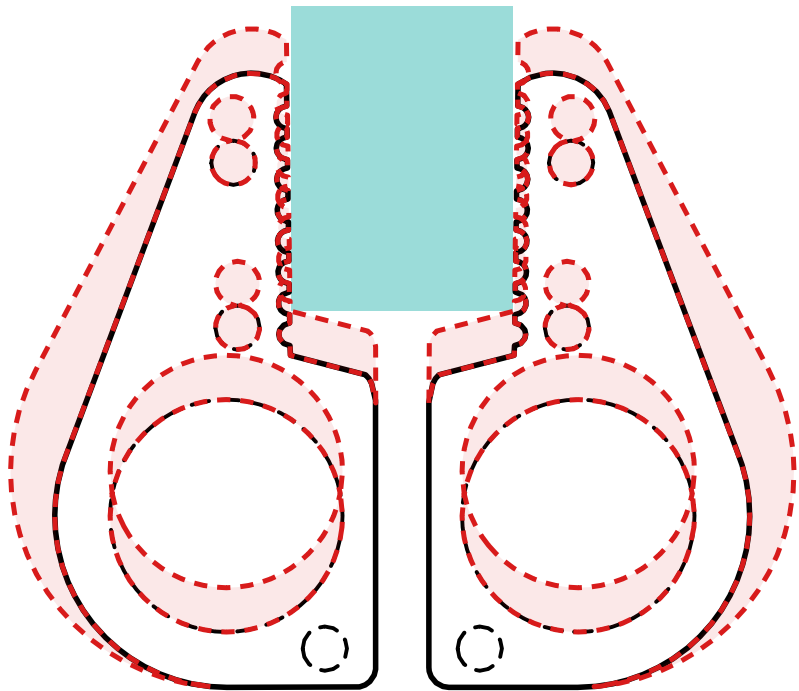
Moves: arms, finger lobes, zip stations.

plate_finger_wall

Two-sided puller · Advanced - Two-Sided Puller

Finger lobe wall

plate_finger_wall
5 → 9



black = the tool at its defaults;
teal = the plug you measured;
red dashed = what this dial moves

Thickens the wall around each finger hole, so the rounded lobes grow.

Default	Range	Step	Unit
5	3 to 12	0.25	mm

Before: 5. After: 9.

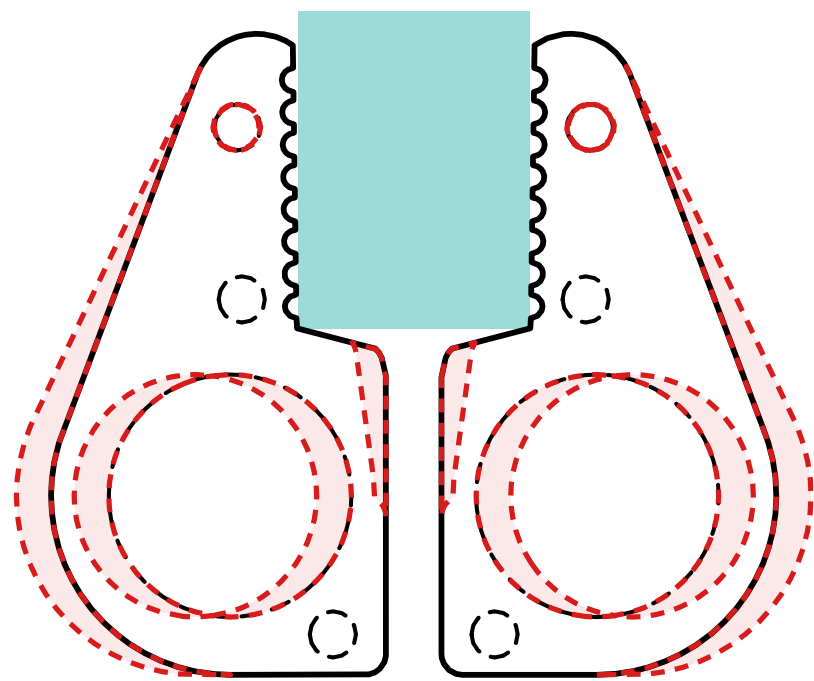
Moves: arms, finger lobes, zip stations.

plate_finger_inner_wall

Two-sided puller · Advanced - Two-Sided Puller

Wall beside the cord channel

plate_finger_inner_wall
3 → 6



black = the tool at its defaults;
teal = the plug you measured;
red dashed = what this dial moves

Thickens the wall between the cord channel and each finger hole, pushing the finger holes outward.

Default	Range	Step	Unit
3	1 to 8	0.25	mm

Before: 3. After: 6.

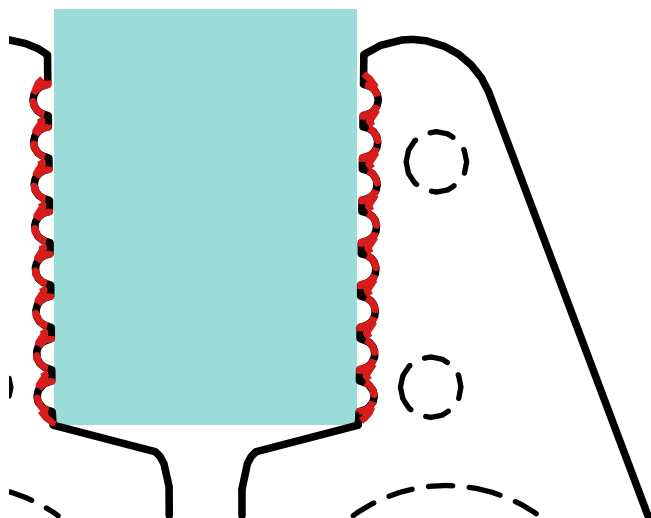
Moves: arms, finger lobes, zip stations.

plate_tooth_diameter

Two-sided puller · Advanced - Two-Sided Puller

Tooth size

plate_tooth_diameter
2 → 4



black = the tool at its defaults;
teal = the plug you measured;
red dashed = what this dial moves

Cuts bigger or smaller scallops into the arms' gripping edges.

Default	Range	Step	Unit
2	0 to 4	0.1	mm

Before: 2. After: 4.

Moves: teeth.

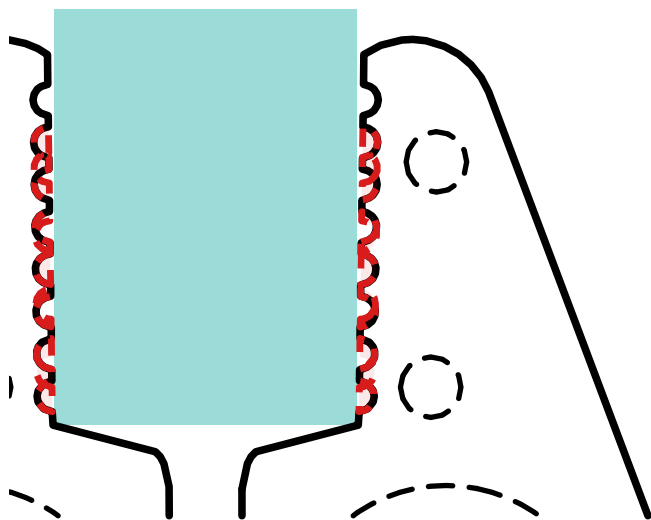
plate_tooth_pitch

Two-sided puller · Advanced - Two-Sided Puller

Tooth spacing

plate_tooth_pitch

2.8 → 4.5



black = the tool at its defaults;
teal = the plug you measured;
red dashed = what this dial moves

Spaces the teeth farther apart or closer together along the gripping edges.

Default	Range	Step	Unit
2.8	0.5 to 5	0.1	mm

Before: 2.8. After: 4.5.

Moves: teeth.

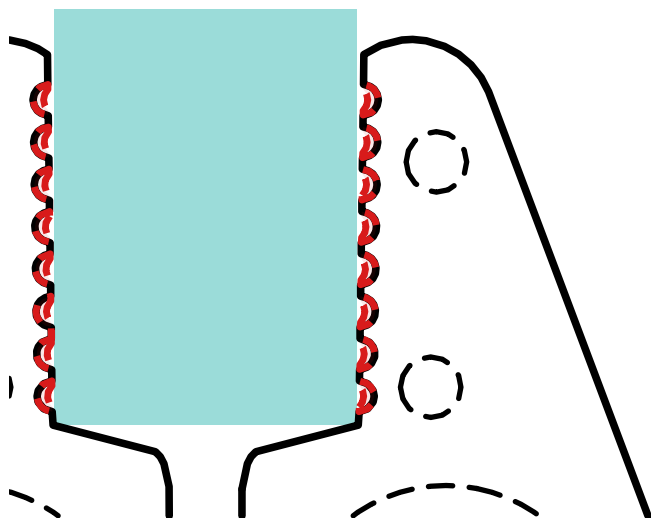
plate_tooth_depth

Two-sided puller · Advanced - Two-Sided Puller

Tooth depth

plate_tooth_depth

1 → 0.3



black = the tool at its defaults;
teal = the plug you measured;
red dashed = what this dial moves

Cuts each tooth deeper or shallower into the gripping edge.

Default	Range	Step	Unit
1	0 to 1.5	0.05	mm

Before: 1. After: 0.3.

Moves: teeth.

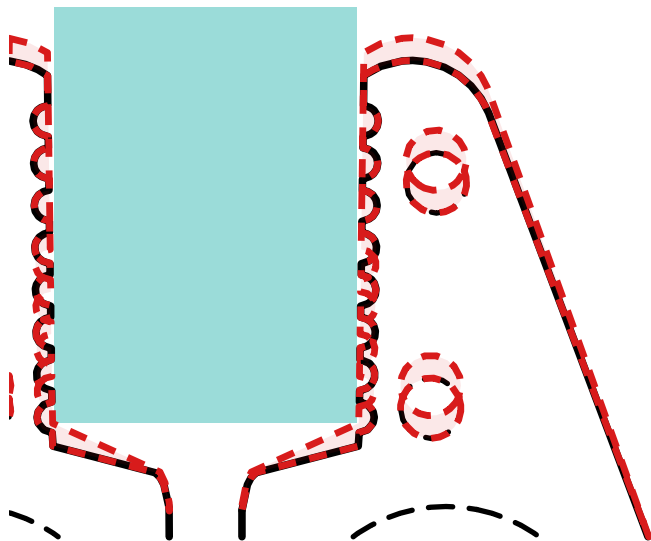
plate_grip_zone_start

Two-sided puller · Advanced - Two-Sided Puller

Toothed zone start

plate_grip_zone_start

4 → 15



black = the tool at its defaults;
teal = the plug you measured;
red dashed = what this dial moves

Moves where the teeth begin, measured back from the arm tips, and the arms lengthen when the zone needs the room.

Default	Range	Step	Unit
4	0 to 25	0.5	mm

Before: 4. After: 15.

Moves: arms, teeth, zip stations.

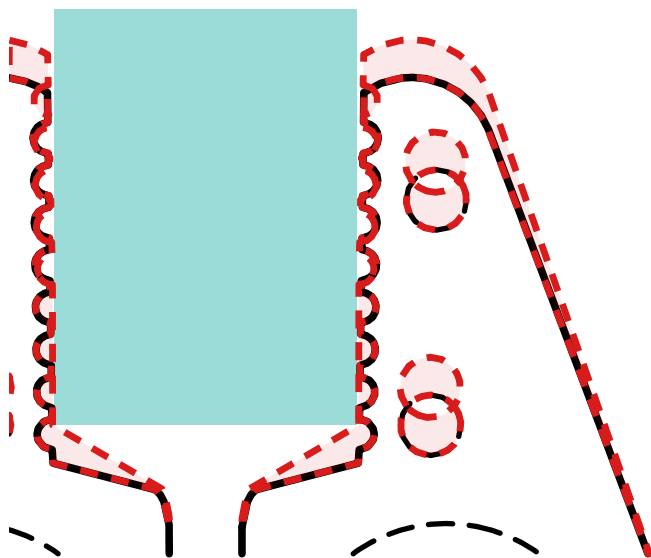
plate_grip_zone_length

Two-sided puller · Advanced - Two-Sided Puller

Toothed zone length

plate_grip_zone_length

0 → 12



black = the tool at its defaults;
teal = the plug you measured;
red dashed = what this dial moves

Sets how far the teeth run along each arm instead of covering the whole plug body.

Default	Range	Step	Unit
0	0 to 60	1	mm

Before: 0. After: 12.

Moves: arms, zip stations.

plate_tip_flare

Two-sided puller · Advanced - Two-Sided Puller

Tip flare

plate_tip_flare

0.7 → 3



black = the tool at its defaults;
teal = the plug you measured;
red dashed = what this dial moves

Opens the gap wider right at the arm tips so the plug head can enter before the teeth bite.

Default	Range	Step	Unit
0.7	0 to 4	0.1	mm

Before: 0.7. After: 3.

Moves: arms, cord channel, finger lobes, zip stations.

plate_arm_tip_width

Two-sided puller · Advanced - Two-Sided Puller

Arm tip width

plate_arm_tip_width

11 → 16



black = the tool at its defaults;
teal = the plug you measured;
red dashed = what this dial moves

Widens the rounded tip of each arm, so the arms taper less.

Default	Range	Step	Unit
11	5 to 16	0.5	mm

Before: 11. After: 16.

Moves: arms, zip stations.

plate_edge_rounding

Two-sided puller · Advanced - Two-Sided Puller

Outer edge rounding

plate_edge_rounding
1.2 → 0
(section-y: 3)



black = the tool at its defaults;
teal = the plug you measured;
red dashed = what this dial moves

Rounds or squares the outer face's edge all around the plate, while the plug-contact face stays square.

Default	Range	Step	Unit
1.2	0 to 2	0.1	mm

Before: 1.2. After: 0. Section at y = 3 mm.

Moves: finger lobes.

plate_strip_thickness

Two-sided puller · Advanced - Two-Sided Puller

Cable strip thickness

plate_strip_thickness

1 → 0

(section-y: 3)



black = the tool at its defaults;
teal = the plug you measured;
red dashed = what this dial moves

Thickens or removes the thin strip that bridges the cord channel on the outer face and ties the two arms together.

Default	Range	Step	Unit
1	0 to 4	0.25	mm

Before: 1. After: 0. Section at y = 3 mm.

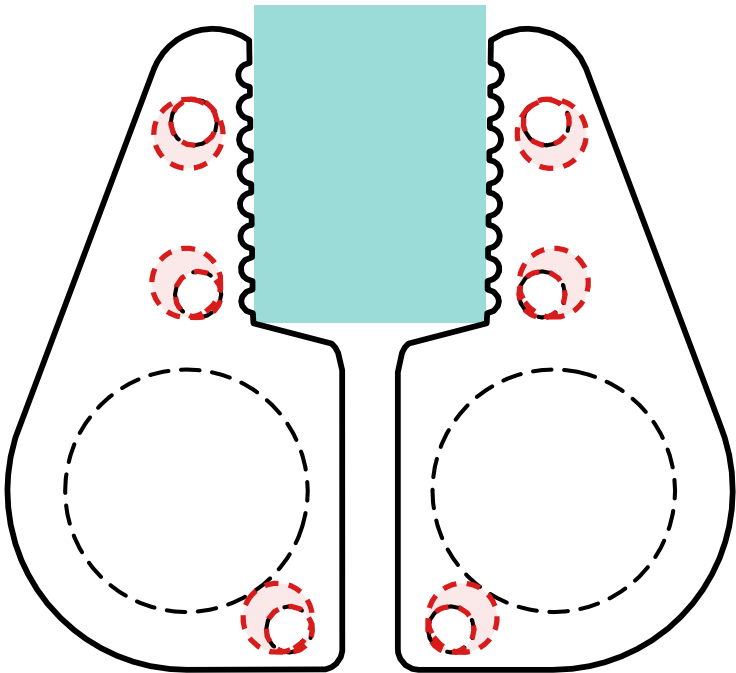
Moves: cord channel.

plate_zip_hole_diameter

Two-sided puller · Advanced - Two-Sided Puller

Zip-tie hole diameter

plate_zip_hole_diameter
4 → 6



black = the tool at its defaults;
teal = the plug you measured;
red dashed = what this dial moves

Widens the three zip-tie holes in each arm, and the stations shift to keep their walls.

Default	Range	Step	Unit
4	0 to 8	0.1	mm

Before: 4. After: 6.

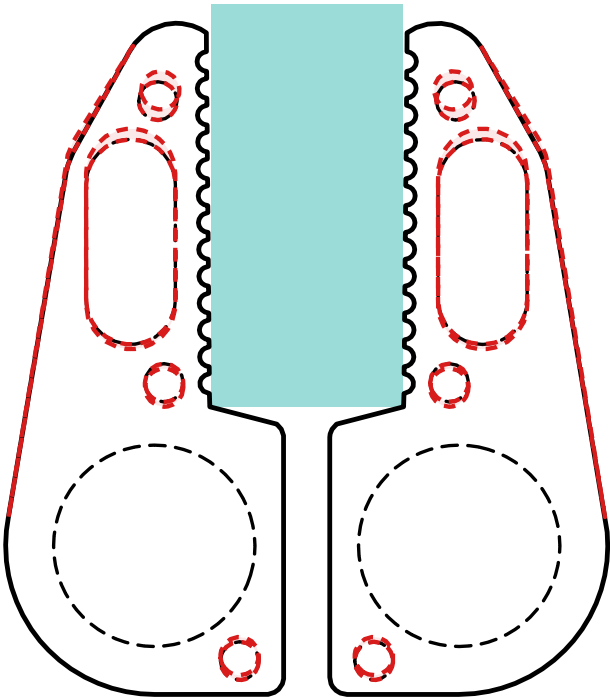
Moves: arms, finger lobes, strap slot, zip stations.

plate_zip_placement

Two-sided puller · Advanced - Two-Sided Puller

Zip-tie station placement

plate_zip_placement
Auto → Manual
(measure_plug_length: 40)



black = the tool at its defaults;
teal = the plug you measured;
red dashed = what this dial moves

Switches the three zip-tie stations in each arm from the automatic positions to the three position dials.

Default	Range	Step	Unit
Auto	2 options	—	—

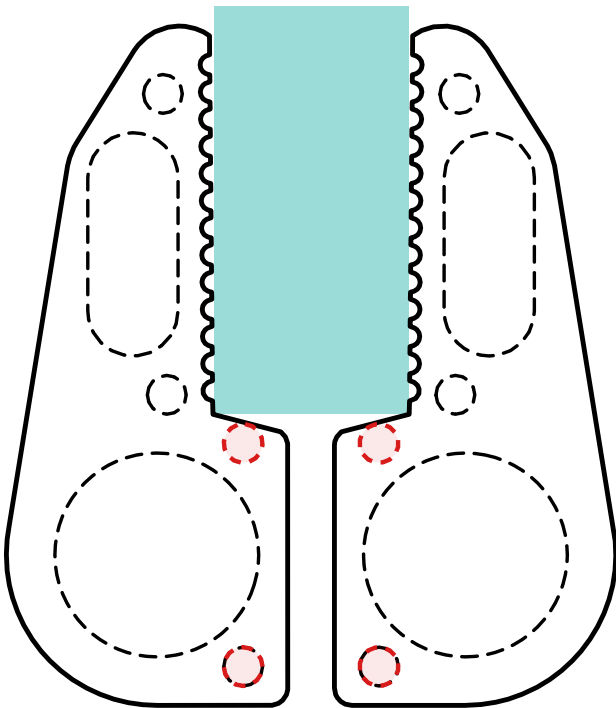
Options: Auto, Manual.
Before: Auto. After: Manual. Context: measure_plug_length = 40.
Drawn on a 40 mm plug so the default manual positions sit on the arm.
Moves: arms, strap slot, zip stations.

plate_zip_pos_1

Two-sided puller · Advanced - Two-Sided Puller

Zip-tie station 1 position

plate_zip_pos_1
4 → 27
(measure_plug_length: 40, plate_zip_placement: Manual)



black = the tool at its defaults;
teal = the plug you measured;
red dashed = what this dial moves

Slides the rear zip-tie station along each arm, measured from the cord end.

Default	Range	Step	Unit
4	0 to 80	0.5	mm

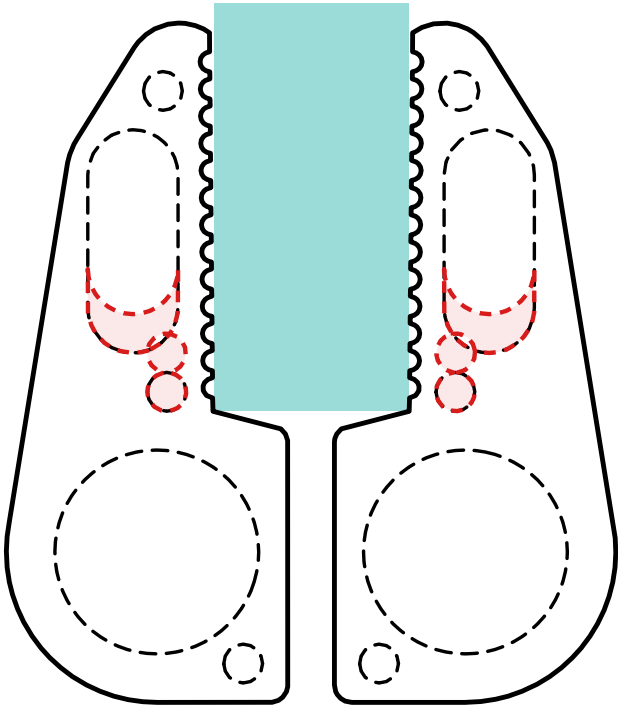
Before: 4. After: 27. Context: measure_plug_length = 40, plate_zip_placement = Manual.
Only acts when plate_zip_placement is Manual; drawn on a 40 mm plug so every station sits on the arm.
Moves: finger lobes, zip stations.
Can trip: ZIP STATION OFF THE ARM; ZIP STATIONS OVERLAP EACH OTHER; STRAP WIDER THAN ARM SLOT WINDOW - NARROW THE STRAP.

plate_zip_pos_2

Two-sided puller · Advanced - Two-Sided Puller

Zip-tie station 2 position

plate_zip_pos_2
32 → 36
(measure_plug_length: 40, plate_zip_placement: Manual)



black = the tool at its defaults;
teal = the plug you measured;
red dashed = what this dial moves

Slides the middle zip-tie station along each arm, and the strap slot's window moves with it.

Default	Range	Step	Unit
32	0 to 80	0.5	mm

Before: 32. After: 36. Context: measure_plug_length = 40, plate_zip_placement = Manual.

Only acts when plate_zip_placement is Manual; drawn on a 40 mm plug so every station sits on the arm.

Moves: strap slot, zip stations.

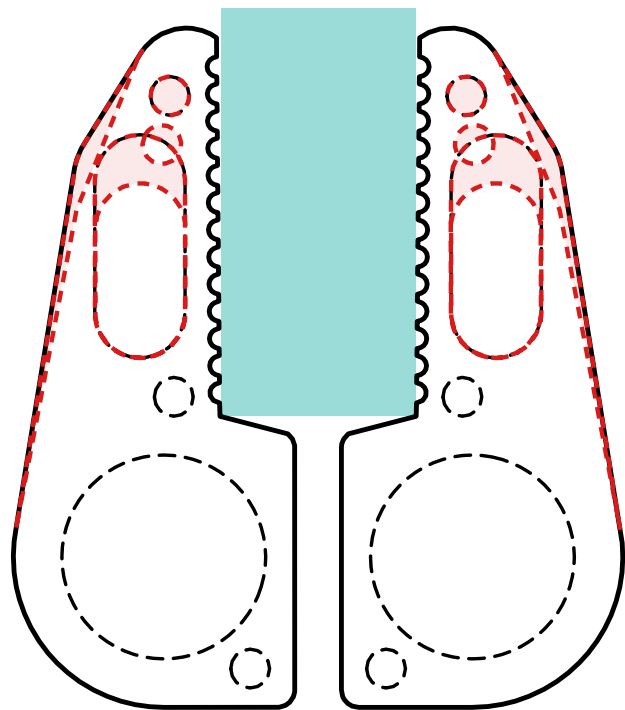
Can trip: ZIP STATION OFF THE ARM; ZIP STATIONS OVERLAP EACH OTHER; ZIP STATION HITS VELCRO SLOT; STRAP WIDER THAN ARM SLOT WINDOW - NARROW THE STRAP.

plate_zip_pos_3

Two-sided puller · Advanced - Two-Sided Puller

Zip-tie station 3 position

plate_zip_pos_3
63 → 58
(measure_plug_length: 40, plate_zip_placement: Manual)



black = the tool at its defaults;
teal = the plug you measured;
red dashed = what this dial moves

Slides the tip zip-tie station along each arm, and the strap slot's window moves with it.

Default	Range	Step	Unit
63	0 to 80	0.5	mm

Before: 63. After: 58. Context: measure_plug_length = 40, plate_zip_placement = Manual.

Only acts when plate_zip_placement is Manual; drawn on a 40 mm plug so every station sits on the arm.

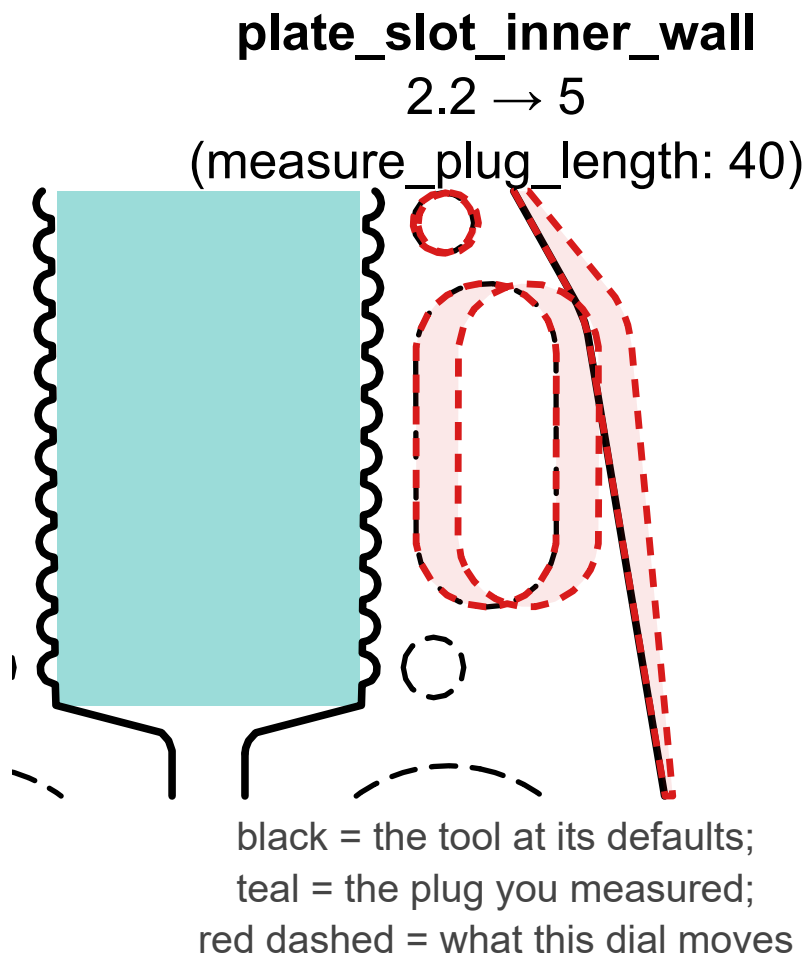
Moves: arms, strap slot, zip stations.

Can trip: ZIP STATION OFF THE ARM; ZIP STATIONS OVERLAP EACH OTHER; ZIP STATION HITS VELCRO SLOT; STRAP WIDER THAN ARM SLOT WINDOW - NARROW THE STRAP.

plate_slot_inner_wall

Two-sided puller · Advanced - Two-Sided Puller

Wall between the teeth and the slot



Thickens the wall between the toothed edge and the strap slot, moving the slot outward.

Default	Range	Step	Unit
2.2	1 to 8	0.1	mm

Before: 2.2. After: 5. Context: measure_plug_length = 40.

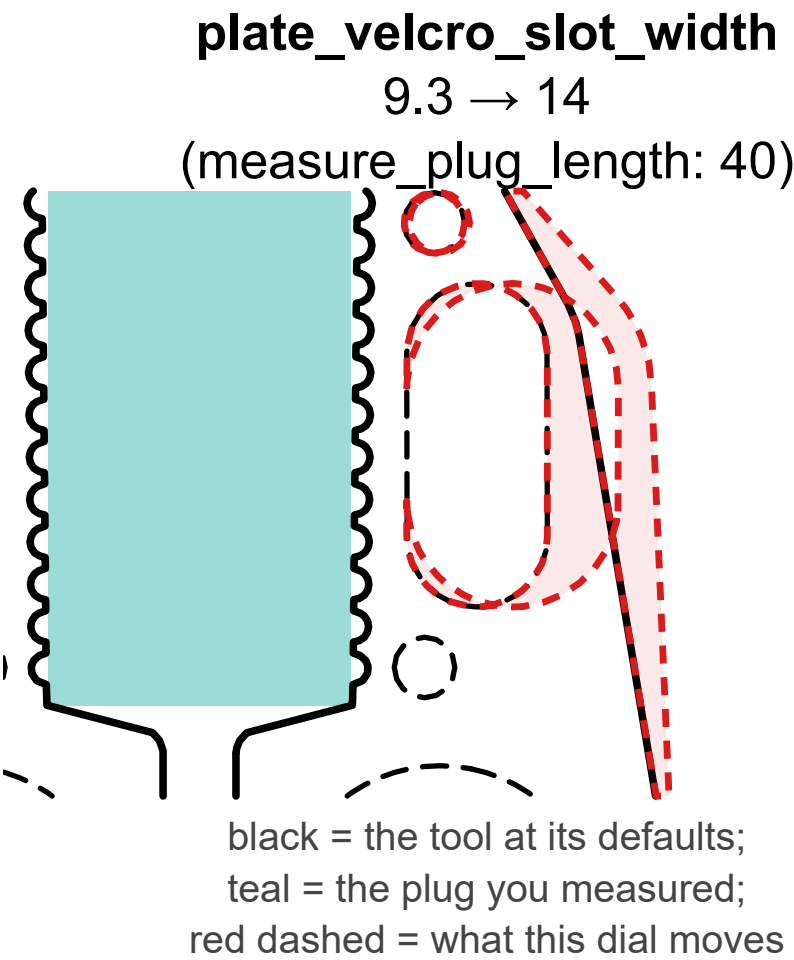
Drawn on a 40 mm plug: on the default 25.5 mm plug the arms are too short for a strap slot.

Moves: arms, strap slot, zip stations.

plate_velcro_slot_width

Two-sided puller · Advanced - Two-Sided Puller

Strap slot width



Widens the strap slot in each arm, and the arm bulges outward to keep its wall.

Default	Range	Step	Unit
9.3	0 to 20	0.25	mm

Before: 9.3. After: 14. Context: measure_plug_length = 40.

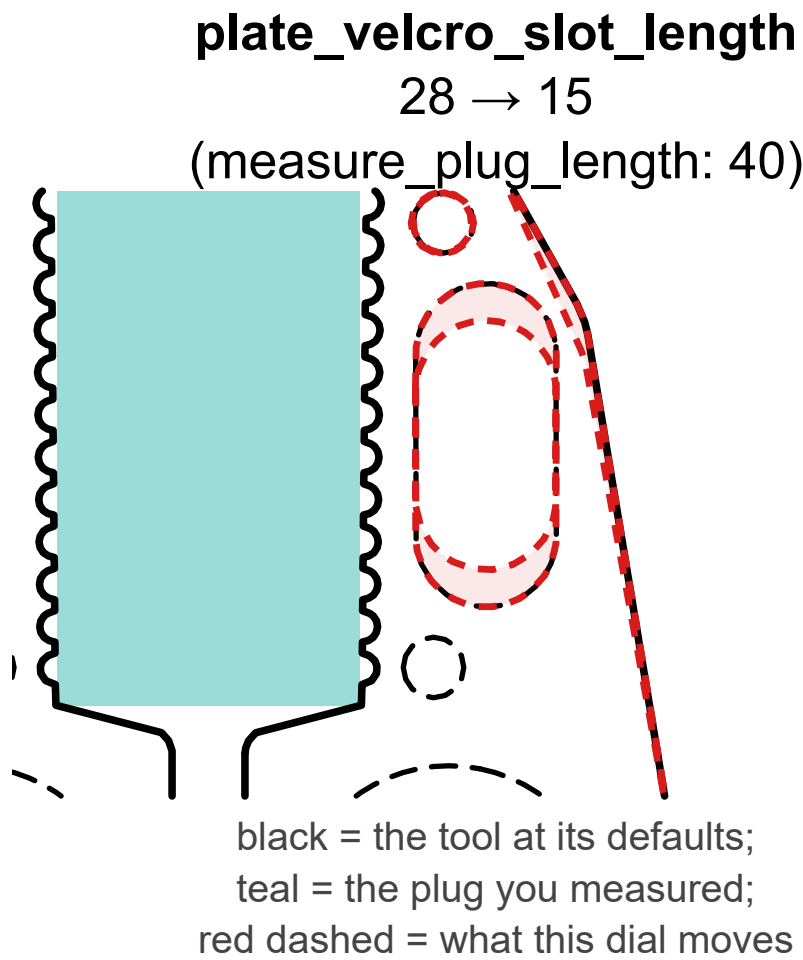
Drawn on a 40 mm plug: on the default 25.5 mm plug the arms are too short for a strap slot.

Moves: arms, strap slot, zip stations.

plate_velcro_slot_length

Two-sided puller · Advanced - Two-Sided Puller

Strap slot length



Lengthens or shortens the strap slot along each arm, within the window between the zip-tie stations.

Default	Range	Step	Unit
28	5 to 60	1	mm

Before: 28. After: 15. Context: measure_plug_length = 40.

Drawn on a 40 mm plug: on the default 25.5 mm plug the arms are too short for a strap slot.

Moves: arms, strap slot, zip stations.

Can trip: STRAP WIDER THAN ARM SLOT WINDOW - NARROW THE STRAP.

quality

Two-sided puller · Advanced - Render Quality

Render quality

quality

This dial changes no shape.

Changes no shape: it only sets the number of segments in every circle and arc.

Sets how many flat segments draw each curve and changes no dimension.

Default	Range	Step	Unit
64	24 to 128	8	segments

This dial changes no shape. Changes no shape: it only sets the number of segments in every circle and arc.